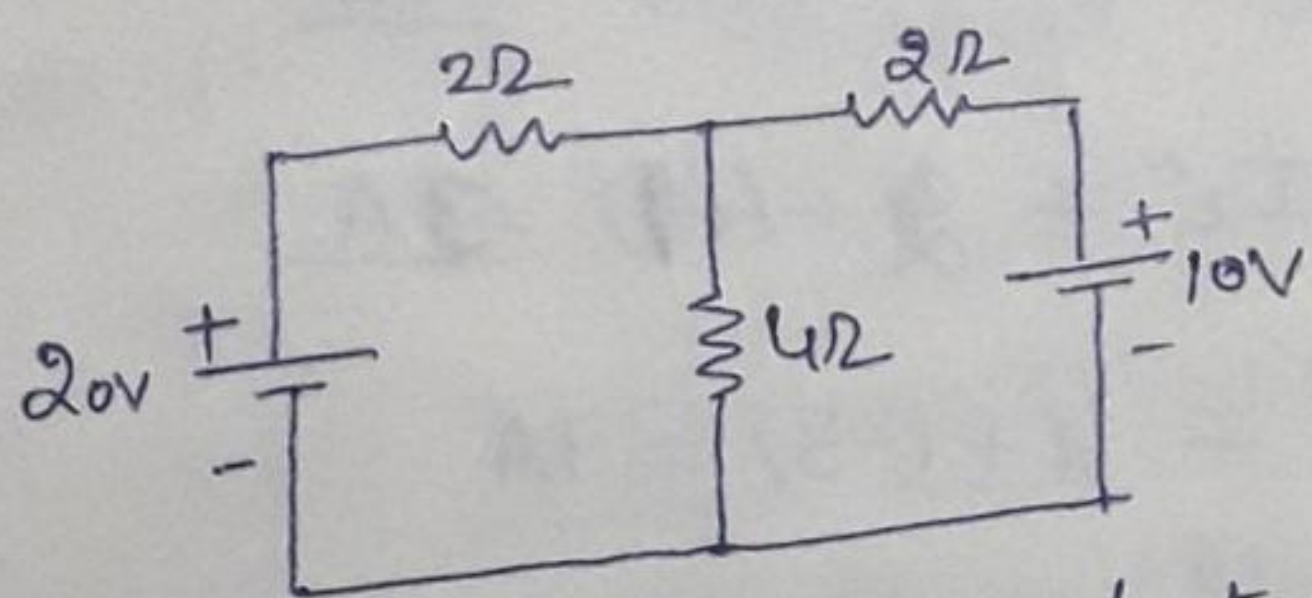


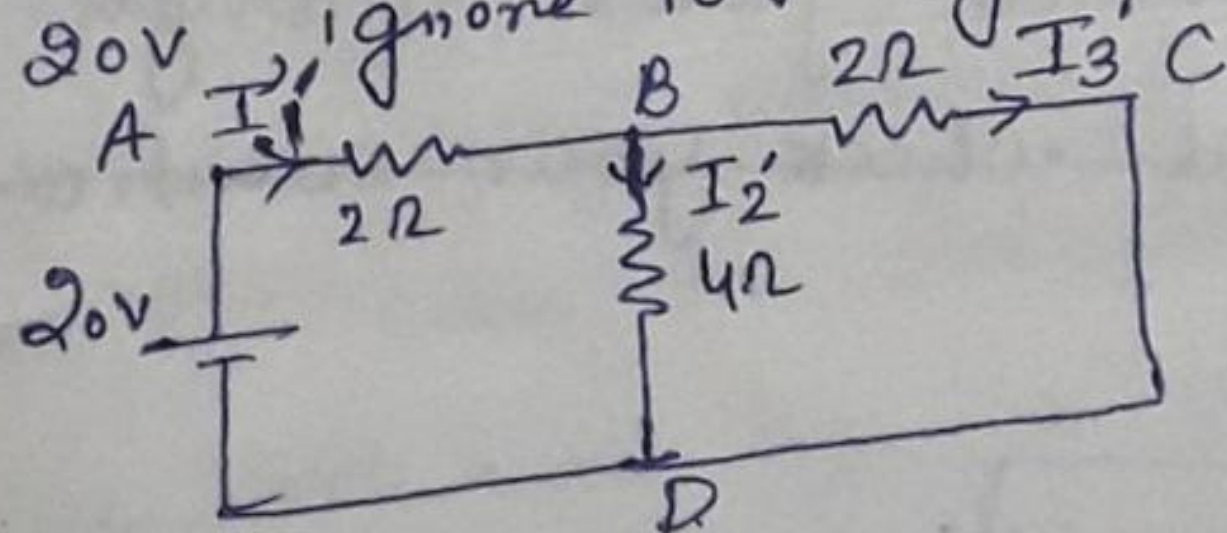
Superposition Theorem:- In a linear resistive network ^①

containing two or more voltage sources, the current through any element may be determined by adding together algebraically the currents produced by each source acting alone, when all other voltage sources are replaced by their internal resistances. If there are current sources present they are removed and network terminals to which they are connected are left open.

Q1:- Calculate the current in each branch by superposition theorem:-



1) Take 20V source and ignore 10V by short circuiting it.

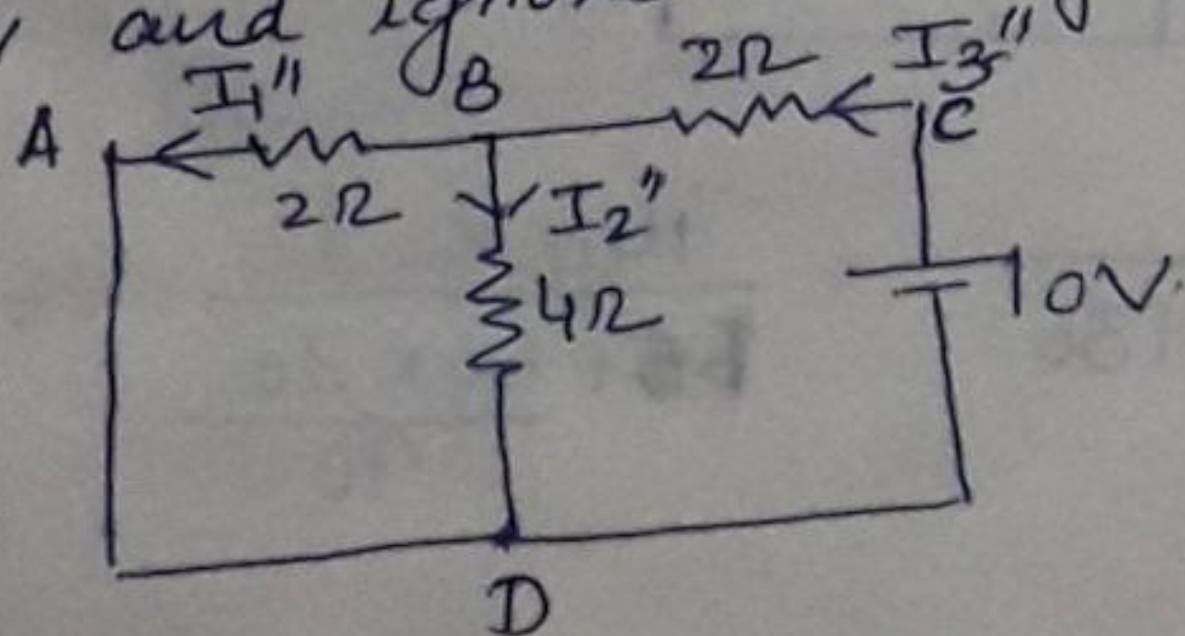


$$I_1' = \frac{20V}{R_1} = \frac{20}{(2 + 2 \parallel 4)} \quad 20 \parallel \left(2 + \frac{1}{\frac{1}{2} + \frac{1}{4}} \right) = \frac{20 \parallel 10}{3} \Omega = 6A$$

$$I_3' = I_1' \times \frac{R_{BD}}{R_{BC} + R_{BD}} = 6 \times \frac{4}{2+4} = 4A$$

$$I_2' = I_1' \times \frac{R_{BC}}{R_{BC} + R_{BD}} = 6 \times \frac{2}{6} = 2A$$

2) Take 10V source and ignore 20V by short circuiting it.



②

$$I_3'' = \frac{10}{R''} = \frac{10}{2+4 \parallel 2} = \frac{10}{10/3} = 3A$$

$$I_2'' = I_3'' \times \frac{R_{BA}}{R_{BA} + R_{BD}} = 3 \times \frac{2}{2+4} = 1A$$

$$I_1' = I_3'' \times \frac{R_{BD}}{R_{BA} + R_{BD}} = 3 \times \frac{4}{2+4} = 2A$$

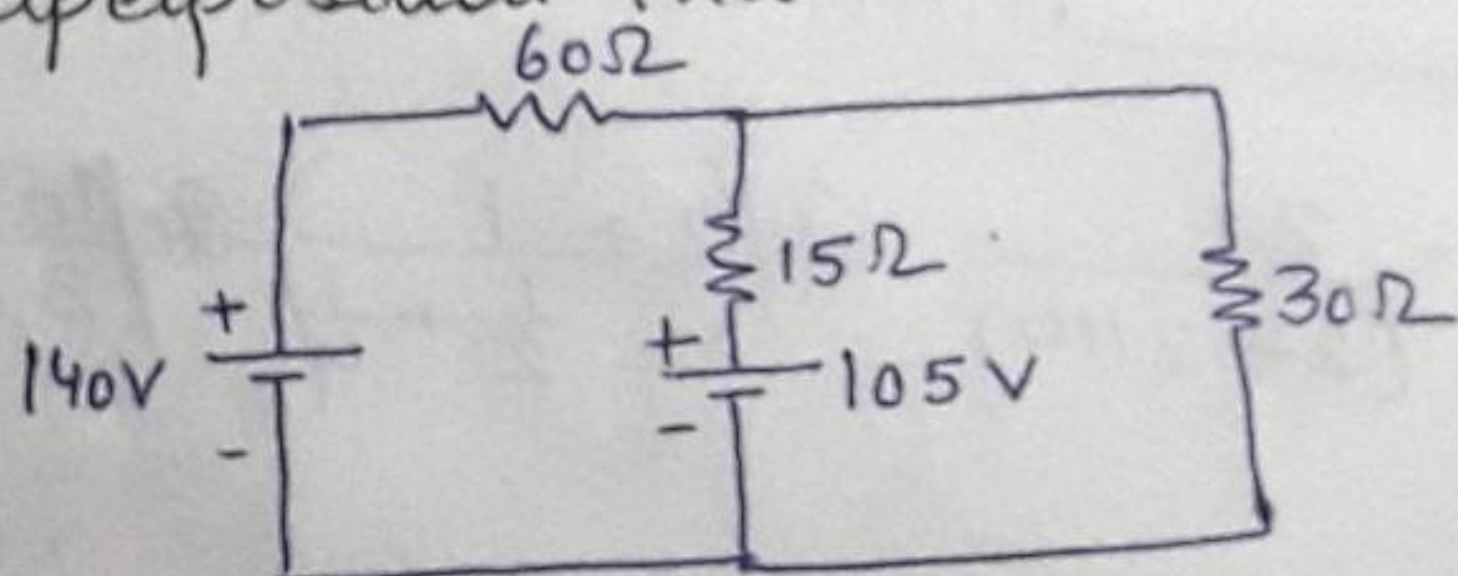
Current in branch AB is equal to algebraic sum of I_1' and I_1'' .

$$\text{i.e. } I_1 = I_1' + I_1'' = 6 + (-2) = 4A$$

$$\text{Similarly } I_2 = I_2' + I_2'' = 2 + (-1) = 1A$$

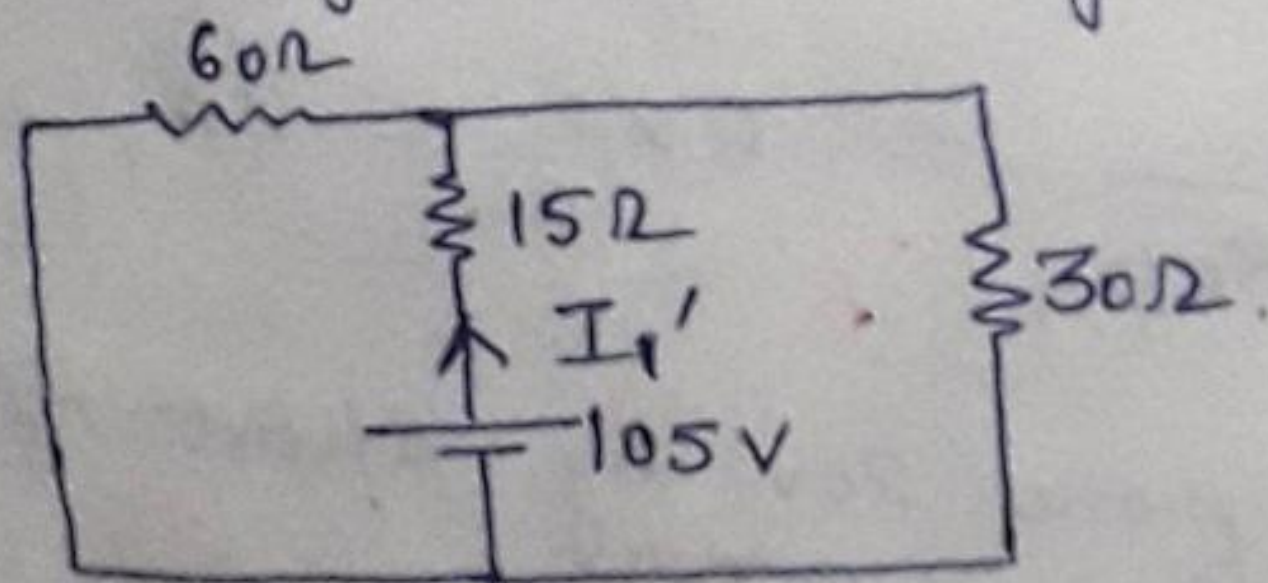
$$\text{and } I_3 = I_3' + I_3'' = 4 + (-3) = 1A$$

Q2:- Find the current which flows through 15Ω resistor by using the mesh method. Check your solution by using superposition theorem.



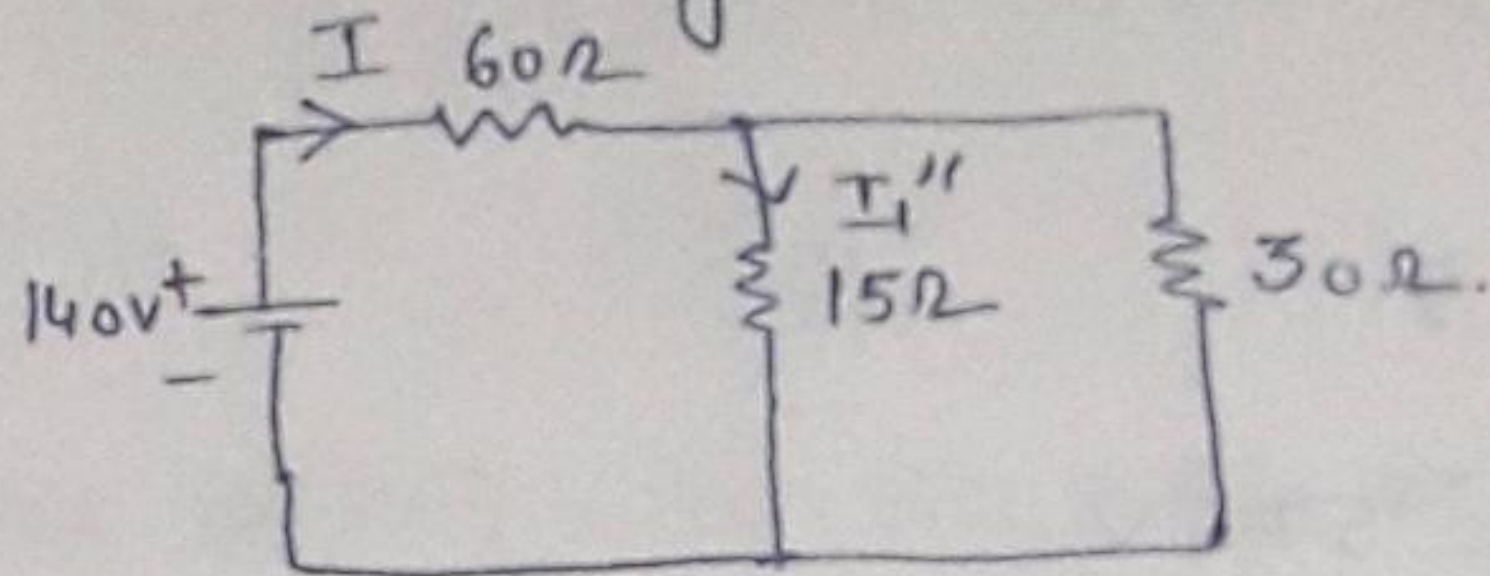
Current through $15\Omega = ?$

1) Take 105V and ignore 140V by shortcircuiting it.



$$I_1' = \frac{105}{15 + 60 \parallel 30} = \frac{105}{15 + \frac{60 \times 30}{90}} = 3A \text{ upward}$$

2) Take 140V and ignore 105V by shortcircuiting it. ③



$$I = \frac{140}{60 + 15 \parallel 30} = 2A$$

$$\text{and } I_1'' = \frac{2}{15 + 30} \times 30 = 1.333A \text{ downward.}$$

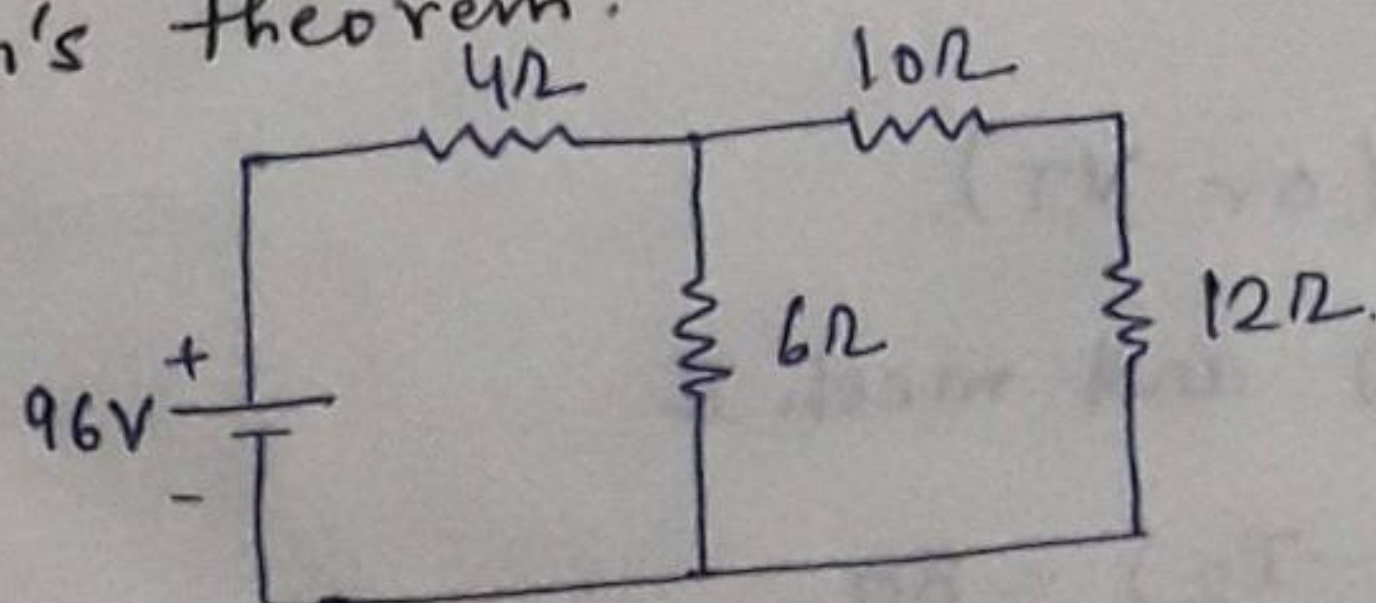
So, current flowing through 15Ω resistance = $3 - 1.333 = 1.667A$

THEVENIN'S THEOREM

The thevenin's theorem can be stated as follows:

The current in any passive circuit element, which may be called R_L , can be determined by replacing the rest of the network by a voltage source of open circuit voltage (V_{OC} or V_T) and internal resistance R_{in} (or R_T) and then connecting the given load resistance R_L across this ^{Thenerin's} equivalent source.

Q1:- For the circuit shown, find the current in 12Ω resistor using thevenin's theorem.

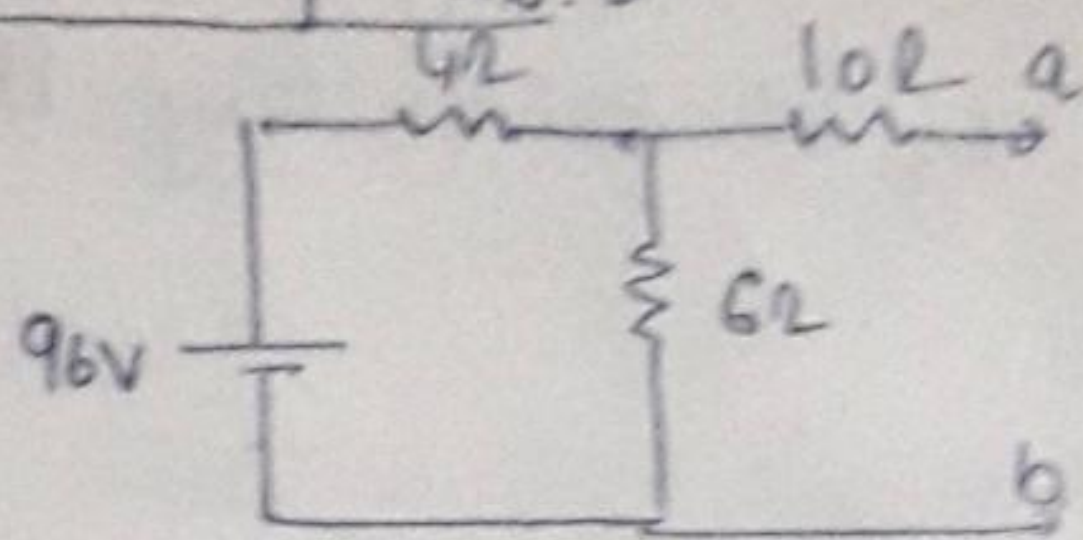


1) Remove the load which is between a and b.

2) Calculate V_{ab} .

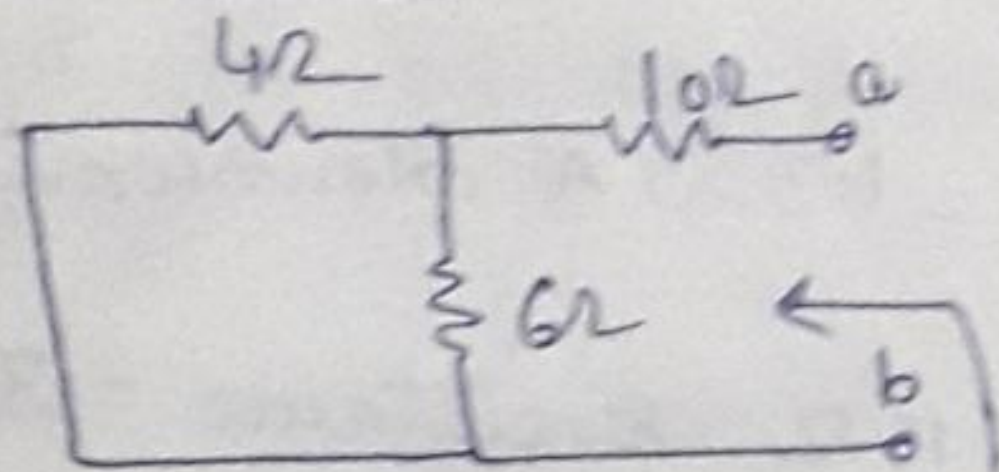
3) Calculate R_{th} .

④ Calculation of V_{ab} :-



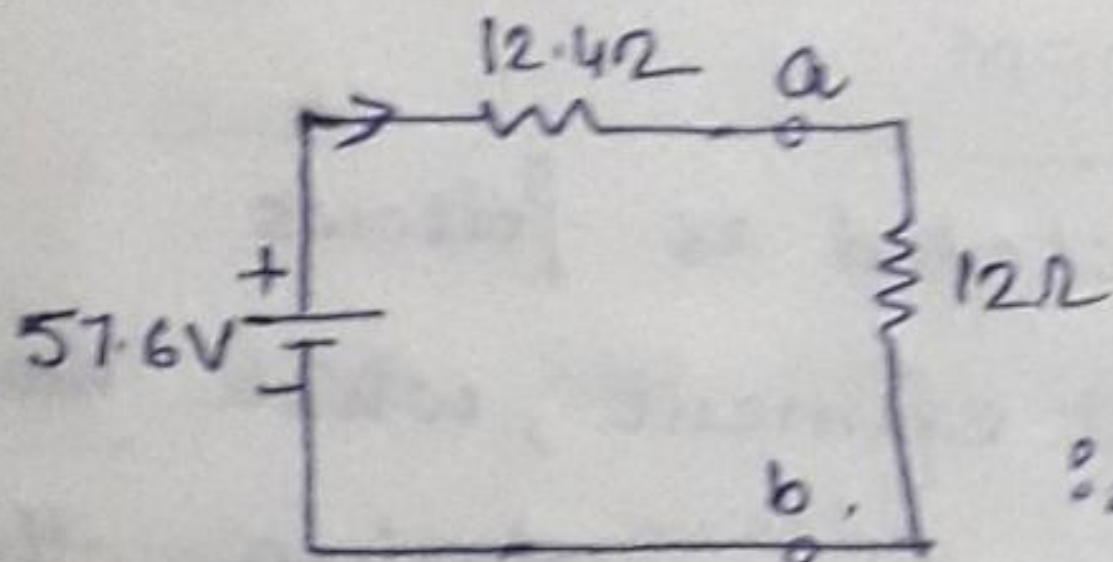
$$V_{ab} = \frac{96}{4+6} \times 6 = \underline{57.6V}$$

Calculation of R_{th} :-



$$R_{th} = 12.4\Omega$$

④ Make the equivalent circuit. by connecting the load across a and b.

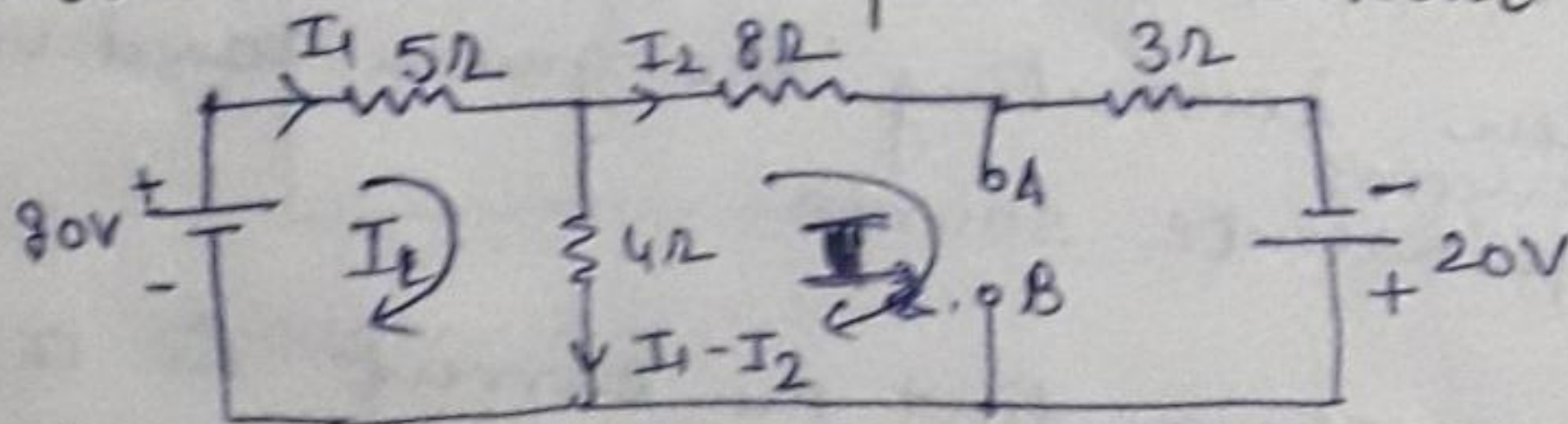


⑤ Note the values to be found.

$$\therefore \text{Current through } 12\Omega = I_L = \frac{57.6}{12.4 + 12}$$

$$I_L = 2.36A$$

Q2 :- Obtain Thevenin's equivalent circuit at AB as shown.



1) Remove the load.

2) Calculate V_{ab} (or V_T).

We have in mesh (I) and mesh II

$$5I_1 + 4(I_1 - I_2) = 80$$

$$\text{or } 9I_1 - 4I_2 = 80 \quad \text{--- (i)}$$

$$\text{and } 11I_2 - 4(I_1 - I_2) = 20$$

$$\text{or } 4I_1 - 15I_2 = -20 \quad \text{--- (ii)}$$

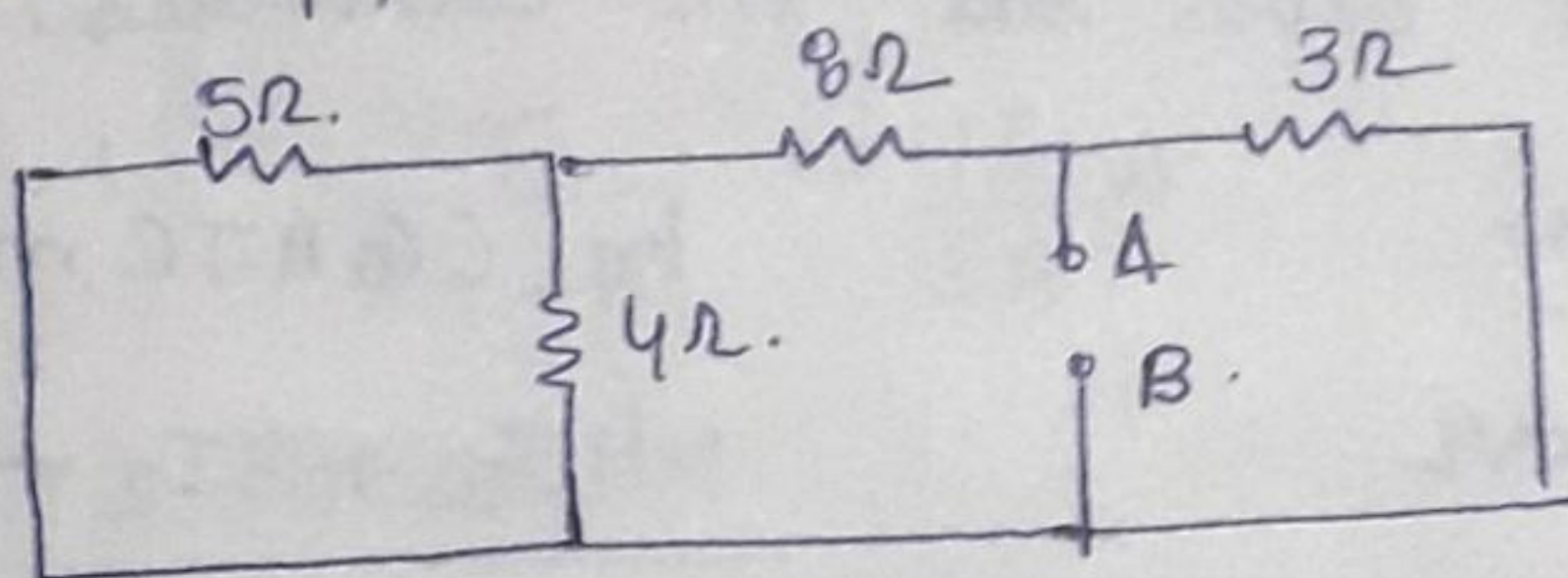
$$I_1 = \frac{1280}{119} \text{ A}$$

$$\text{and } I_2 = \frac{500}{119} \text{ A}$$

$$V_{AB} \text{ (or } V_T) = 80 - 5I_1 - 8I_2 \text{ or } -8I_2 + 4(I_1 - I_2).$$

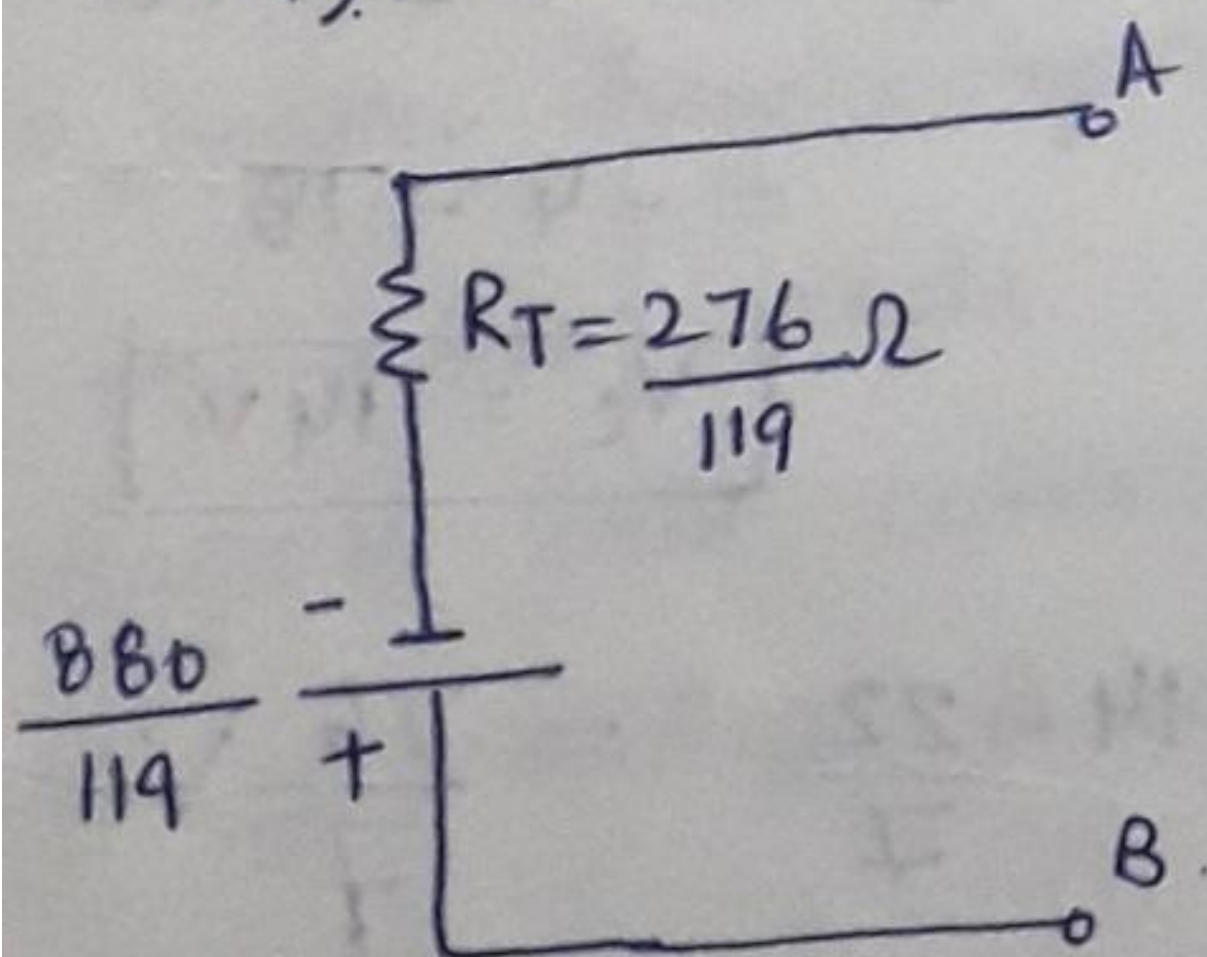
$$= 80 - 5 \times \frac{1280}{119} - 8 \times \frac{500}{119} = -\frac{880}{119} \text{ A.}$$

3) Calculate R_T .



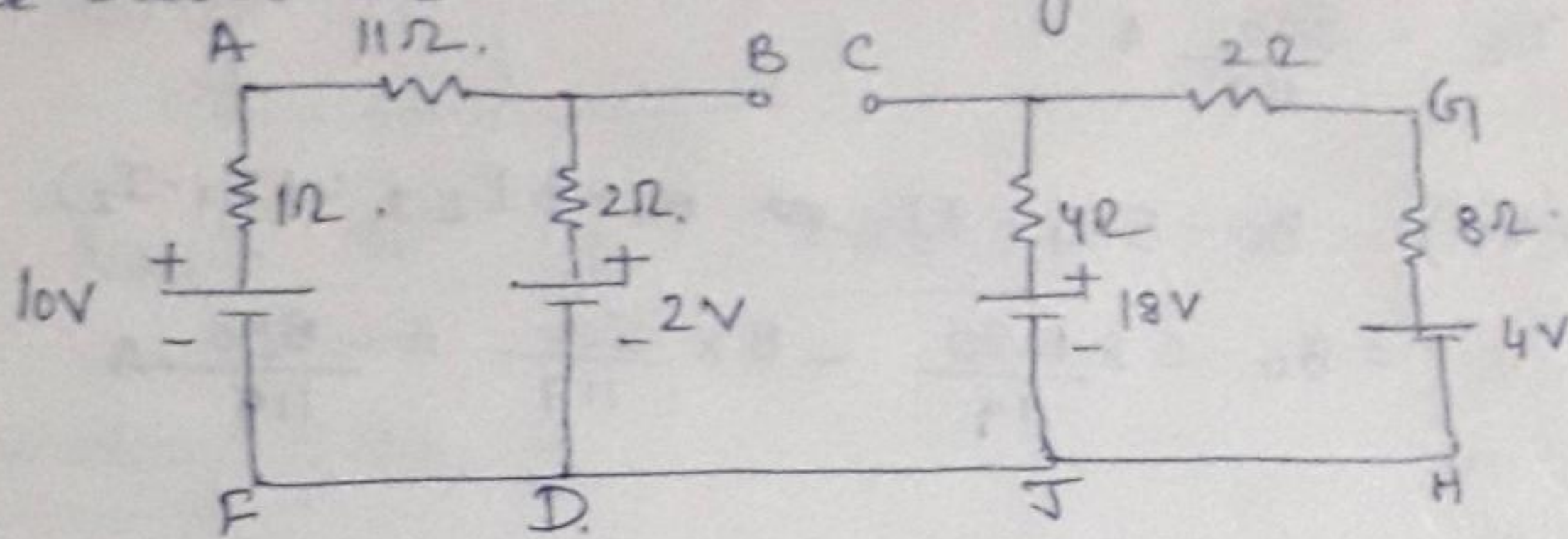
$$R_T = 3 \parallel [8 + 4 \parallel 5] = \frac{276}{119} \Omega$$

4) Make the thevenin's equivalent circuit as follows-



⑥

Q3:- Find thevenin's equivalent circuit at terminals A-B. Hence determine current through $R=1\Omega$.



1) Remove the load across ~~AB~~ BC terminals.

2) Calculation of V_{th}

In ABDEFA, apply KVL

$$12I_1 + 2I_1 + 2 - 10 = 0$$

$$I_1 = \frac{8}{14} A$$

$$V_B = \frac{8}{14} \times 2 + 2 = \frac{22}{7} V$$

In CGHJC, mesh, apply

$$4I_2 + 2I_2 + 8I_2 + 4 - 18 = 0$$

$$14I_2 = 14$$

$$I_2 = 1 A$$

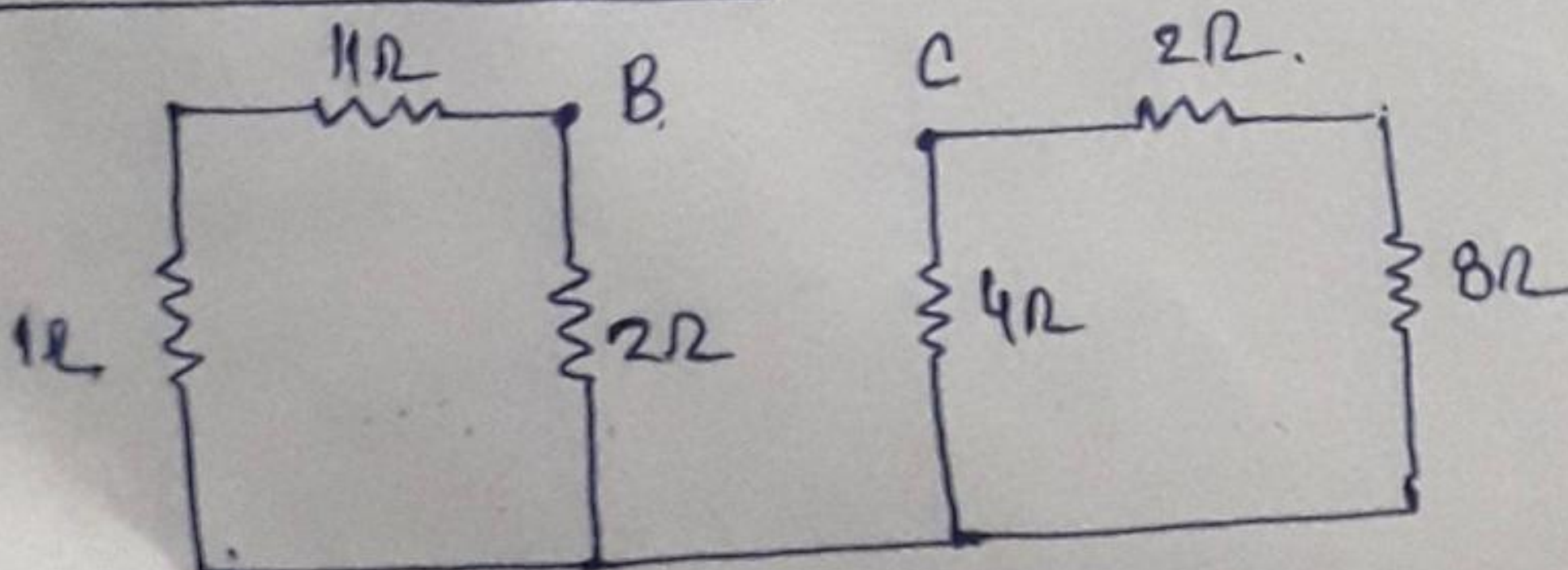
$$V_C = -4I_2 + 18$$

$$= -4 + 18$$

$$V_C = 14 V$$

$$\text{Therefore, } V_{th} = V_C - V_B = 14 - \frac{22}{7} = \frac{76}{7} V$$

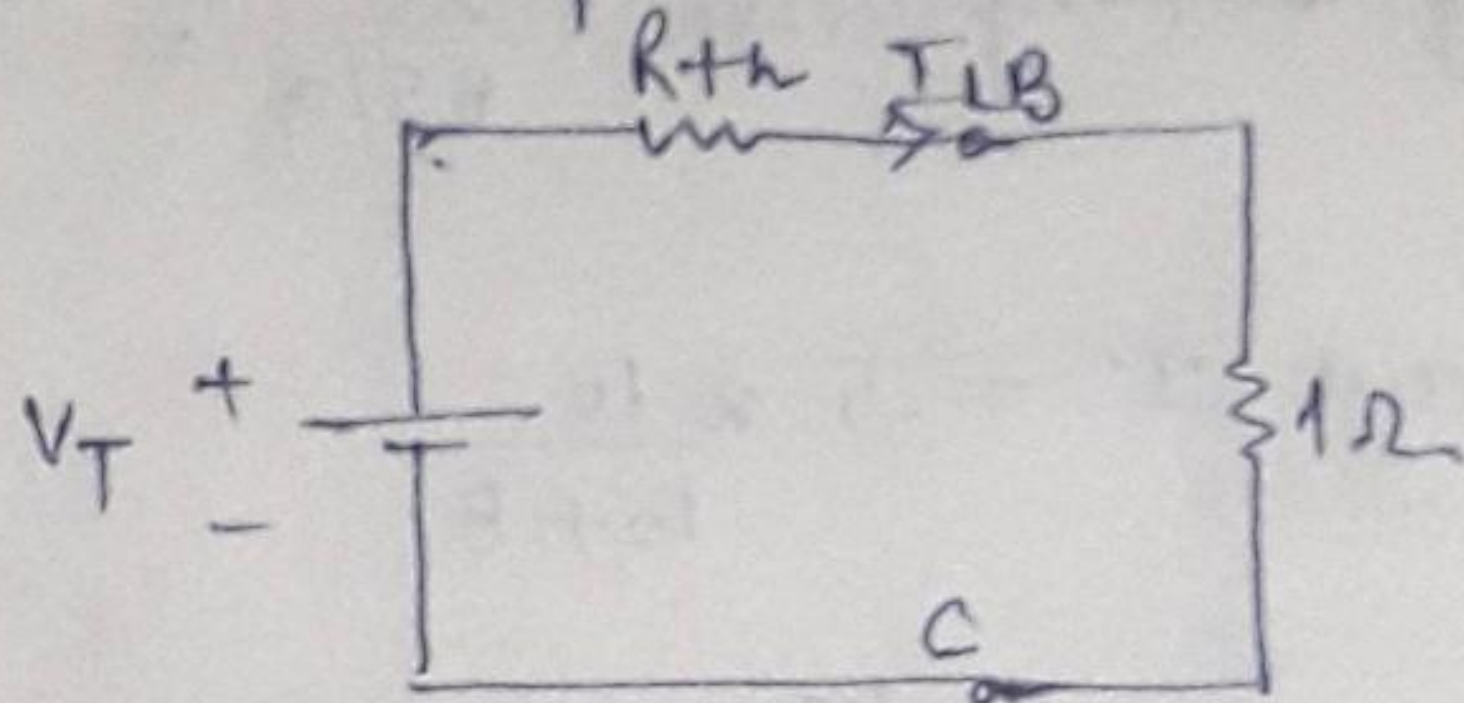
3) Calculation of R_{th} :-



$$R_{th} = [(1+1) \parallel 2] + [(2+8) \parallel 4]$$

$$\therefore R_{th} = \frac{32}{7} \Omega$$

(iii) Now the equivalent circuit is -



(iv) Note the values of load current

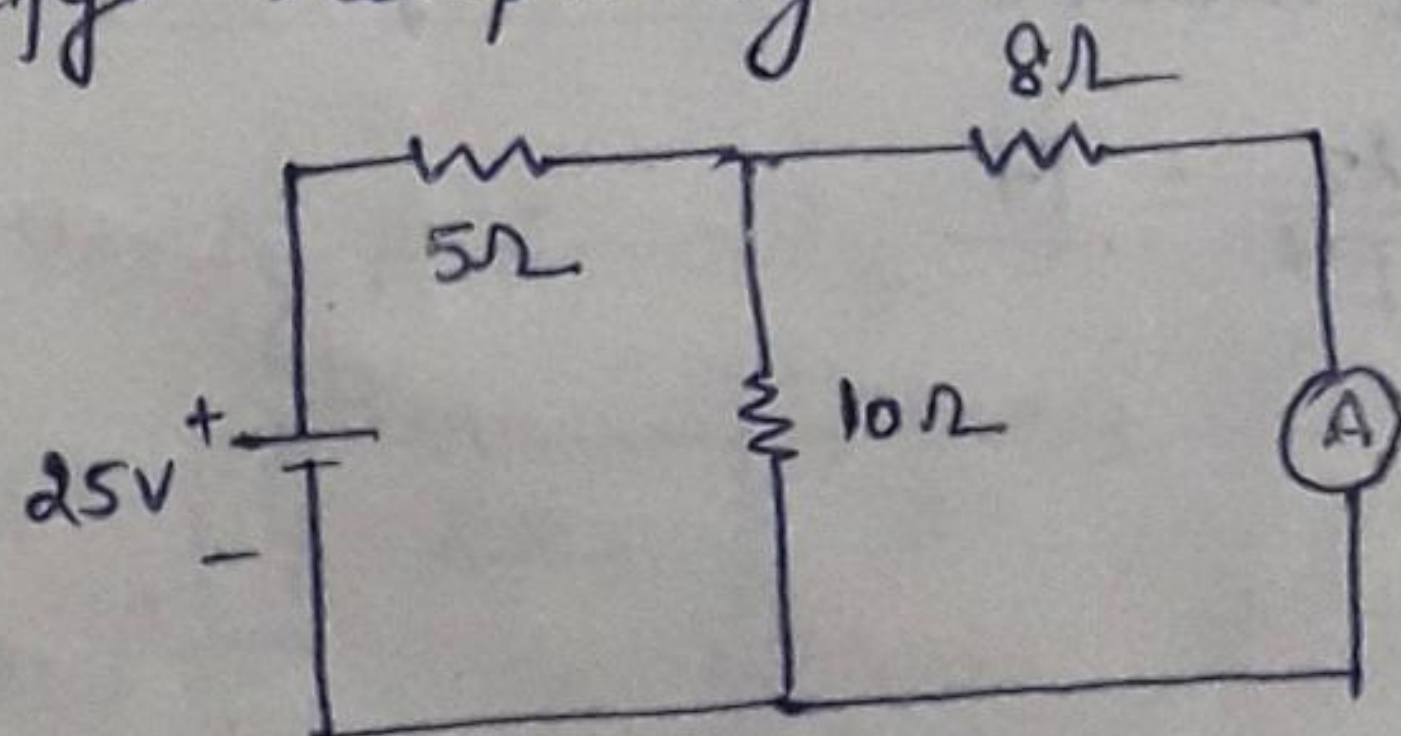
$$I_L = \frac{V_T}{R_T + R} = \frac{76}{39} \text{ A}$$

RECIPROCITY THEOREM

According to this theorem if the source voltage and zero-resistance ammeter, are interchanged, the magnitude of current through the ammeter will be the same, no matter how complicated the network.

In other words, in a linear passive network, supply voltage V and current I are mutually transferable. The ratio of V and I is called transfer resistance.

Q1: Verify reciprocity theorem.



Steps:-

- 1). Apply V_1 , note I_1'
- 2). Apply V_2 from load side and note I''

$$\text{Equivalent resistance : } R' = 5 + 10 \parallel 8 = 5 + \frac{10 \times 8}{18} = \frac{85}{9} \Omega$$

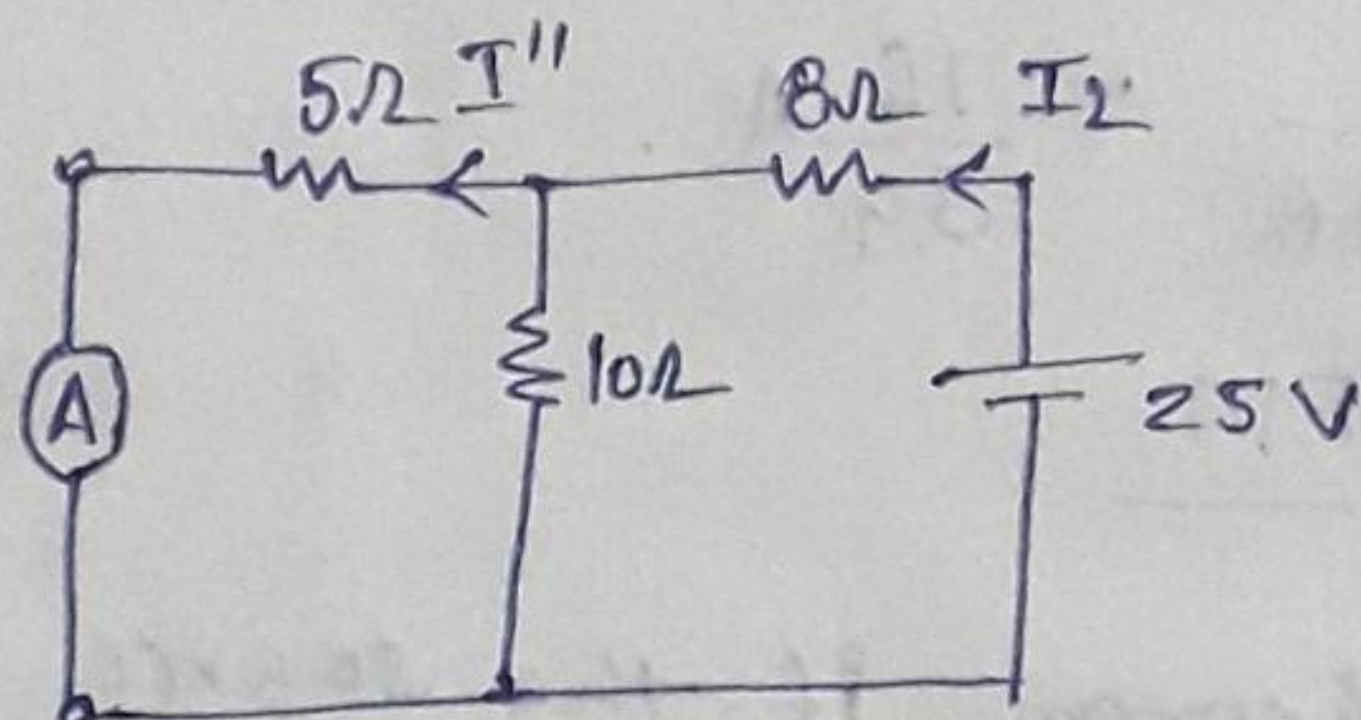
(8)

Current supplied by battery, $I_1 = \frac{25}{85/9} = \frac{45}{17} \text{ A}$

$$\text{Ammeter current } I' = I_1 \times \frac{10}{10+8}$$

$$= \frac{45}{17} \times \frac{10}{18} = \frac{25}{17} \text{ A}$$

2) Apply $V_2 = 25 \text{ V}$ from load side and note I'' ,
ie.



Equivalent resistance of network, $R'' = 8 + \frac{1}{\frac{1}{10} + \frac{1}{5}}$

Current supplied by battery, $I_2 = \frac{25}{34/3} = \frac{75}{34} \text{ A}$

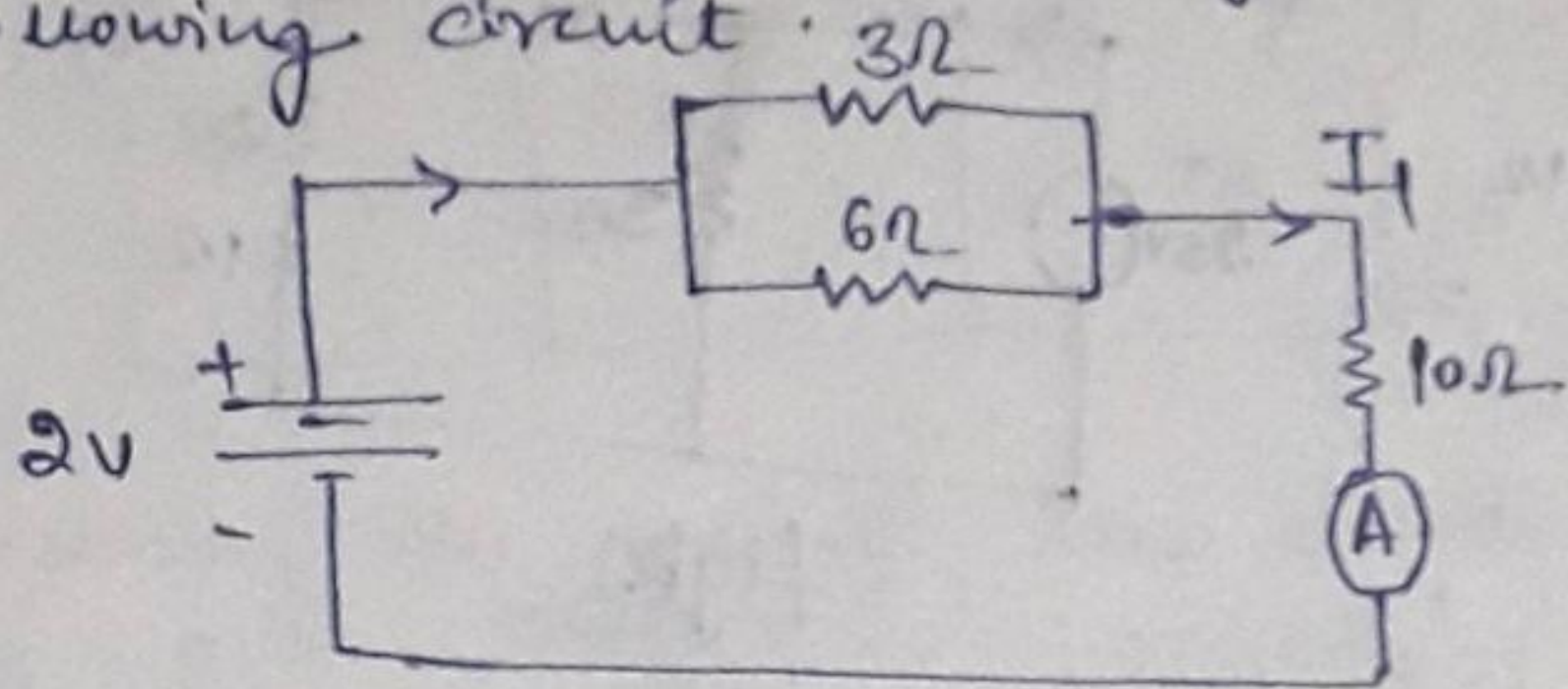
$$\text{Ammeter current } I'' = I_2 \times \frac{10}{10+5} = \frac{75}{34} \times$$

$$= \frac{25}{17}$$

Hence, reciprocity theorem is verified as

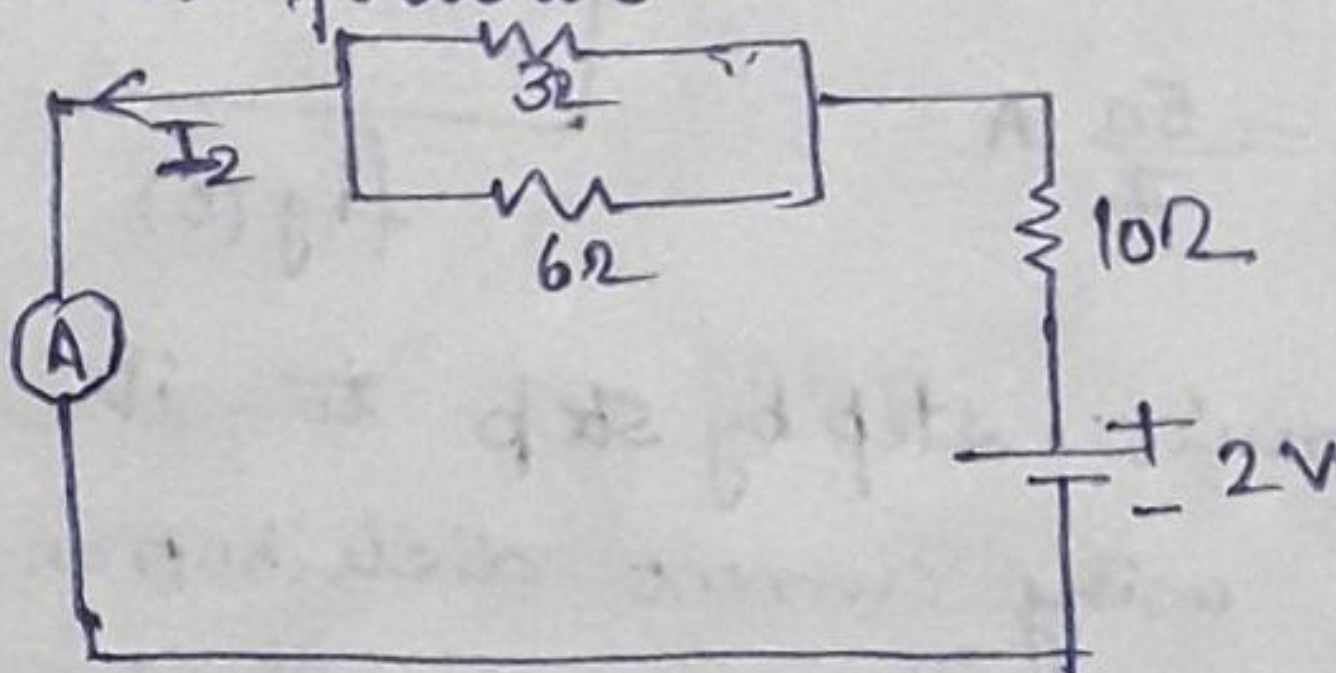
$$\frac{V_1}{I'} = \frac{V_2}{I''} = 17$$

Q2 Show that reciprocity theorem holds good for following circuit.



$$I_1 = \frac{2}{R'} = \frac{2}{(3 \parallel 6) + 10} = \frac{2}{12} = \frac{1}{6} \text{ A}$$

Now interchange source voltage and the ammeter the circuit is as follows -



$$I_2 = \frac{2}{R''} = \frac{2}{12} = \frac{1}{6} \text{ A}$$

Hence reciprocity theorem is verified as

$$\frac{V_1}{I_1} = \frac{V_2}{I_2} = 2 \times 6 = 12$$

Q3:- Verify Reciprocity theorem of the network.

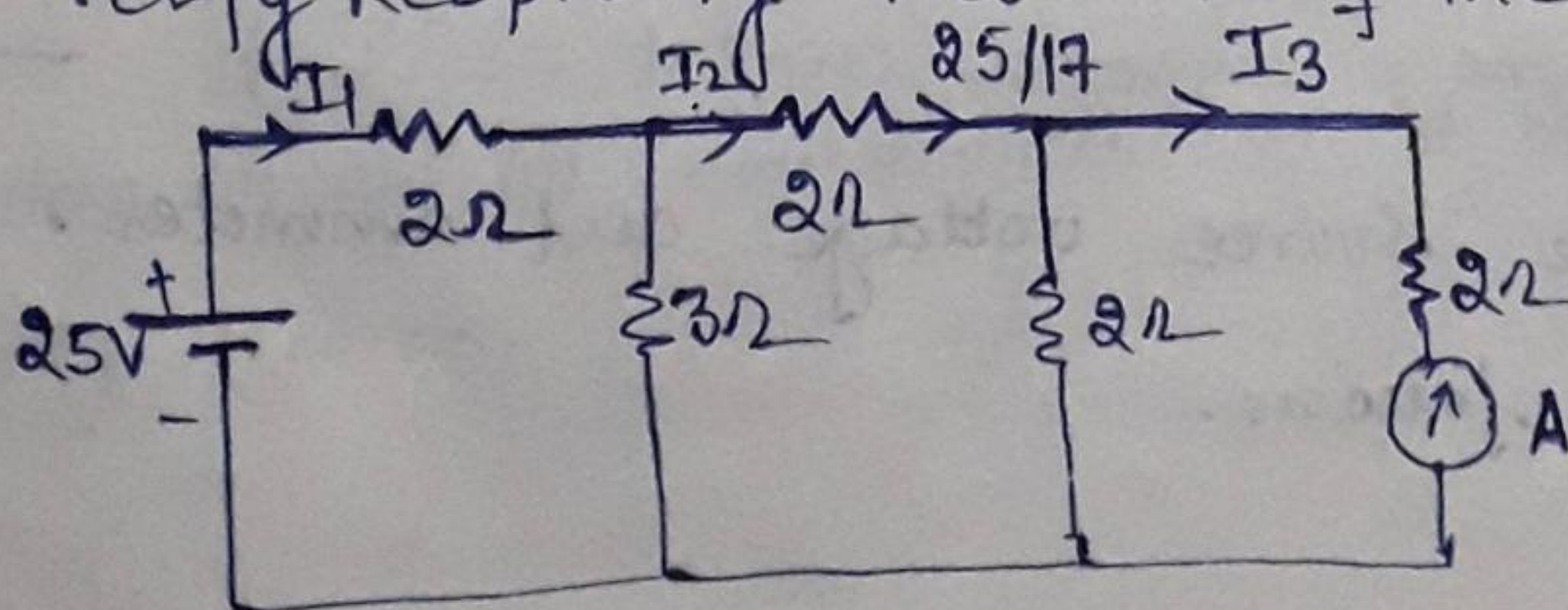
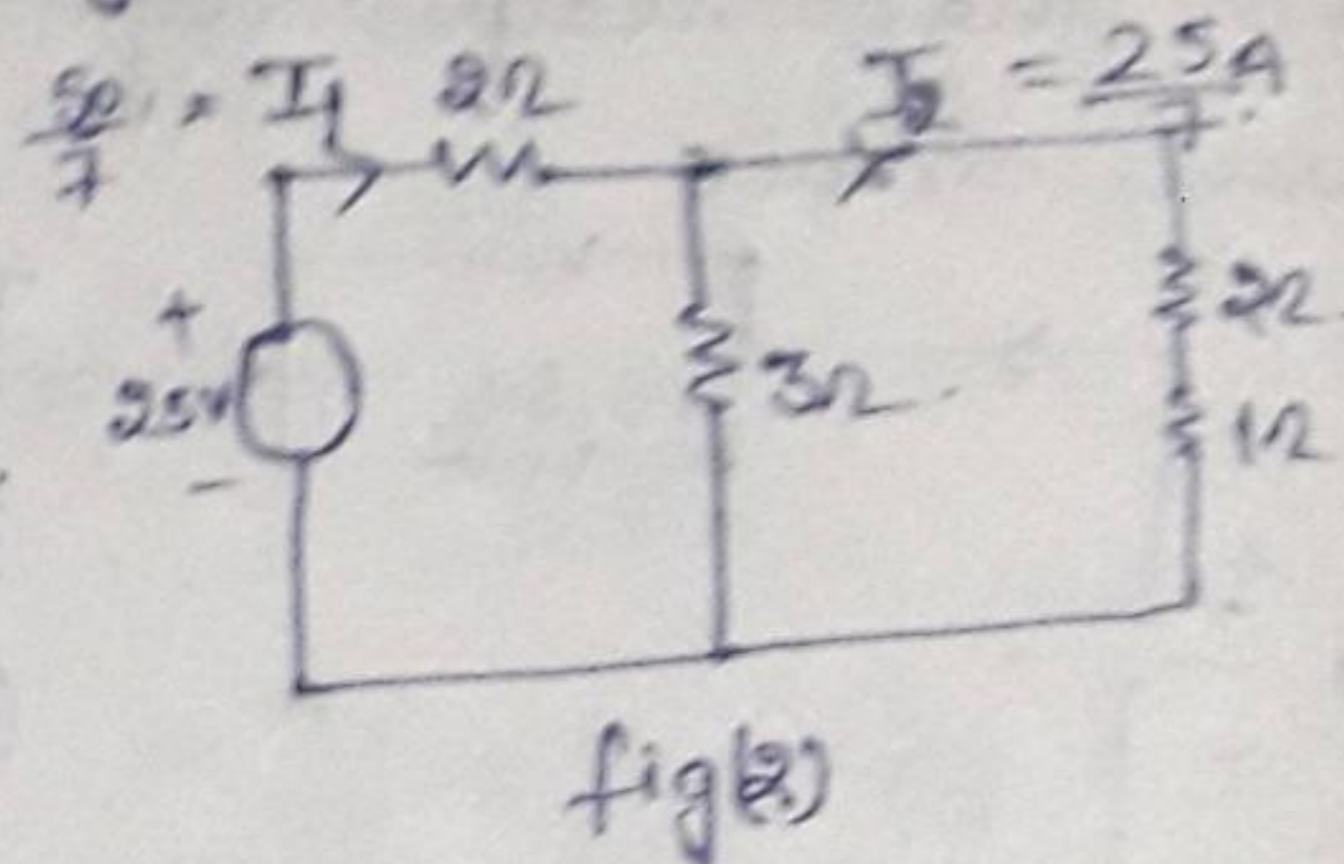
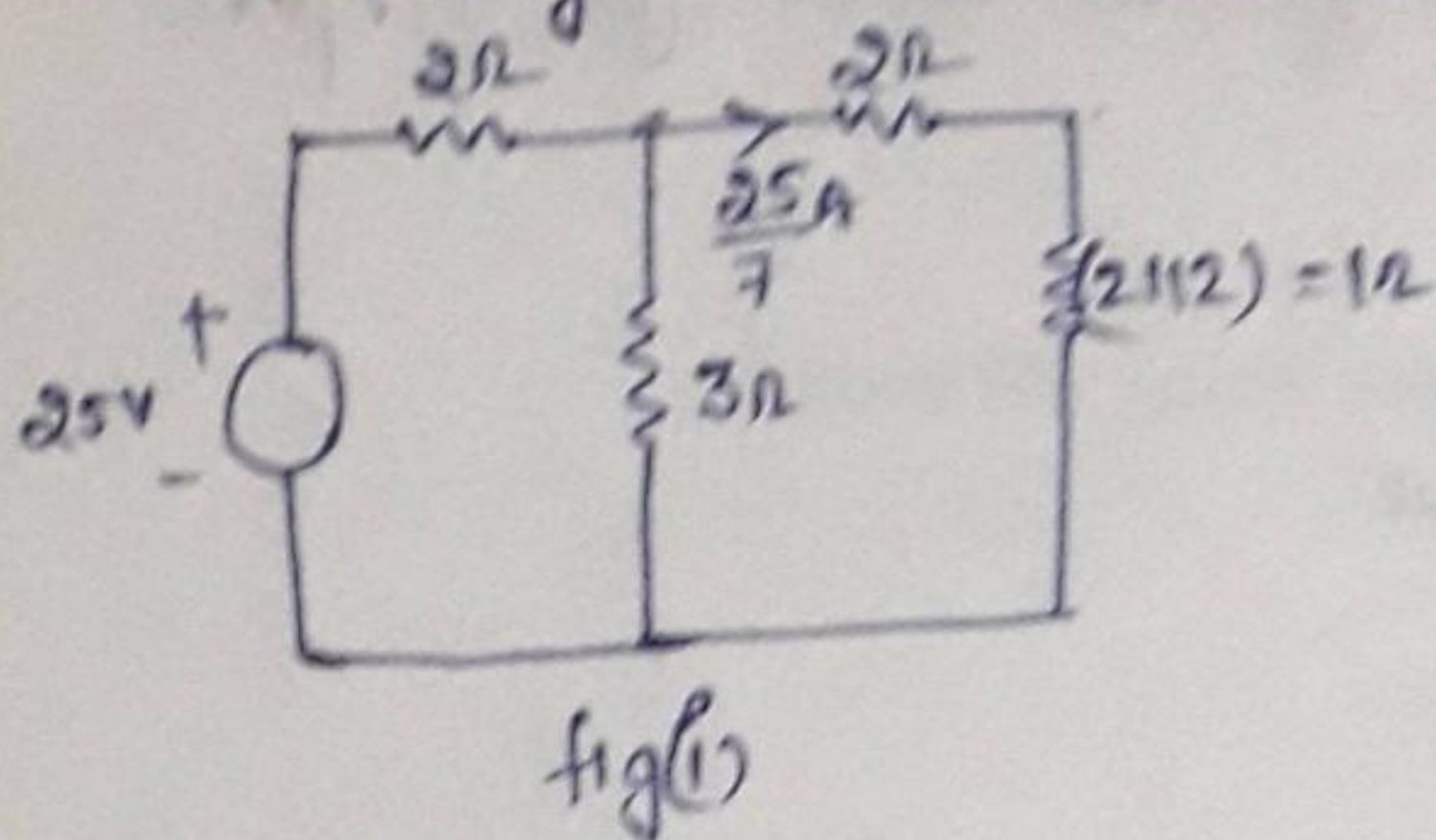


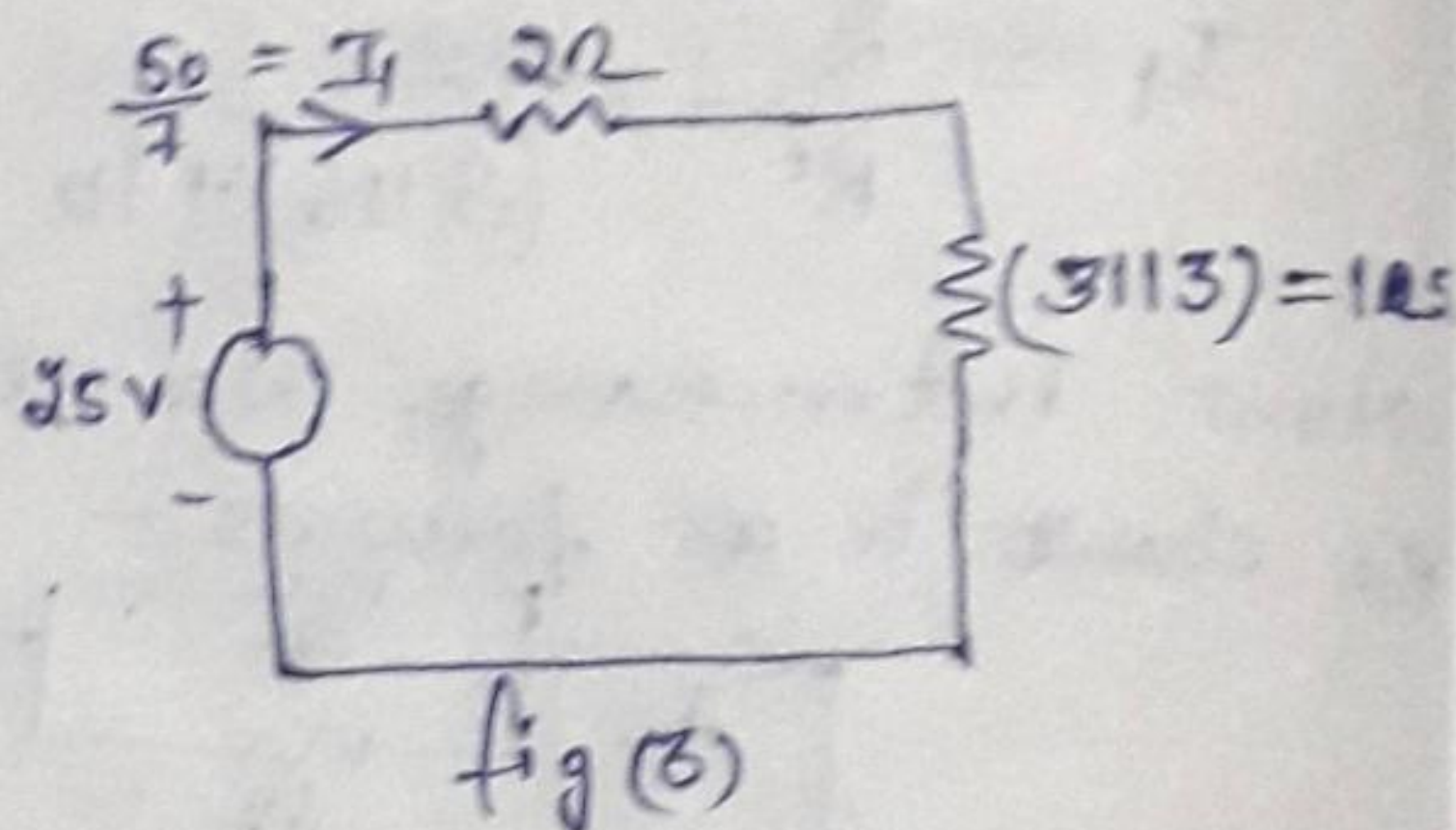
fig A.

(1) Reducing the network step by step, we have -



∴ From figure (3), we have current supplied by battery.

$$I_1 = \frac{25}{2 + 1.5} = \frac{25}{3.5} = \frac{50}{7} \text{ A}$$



By restoring the circuit step by step to its original form in reverse order by using current distribution rule in parallel circuits.

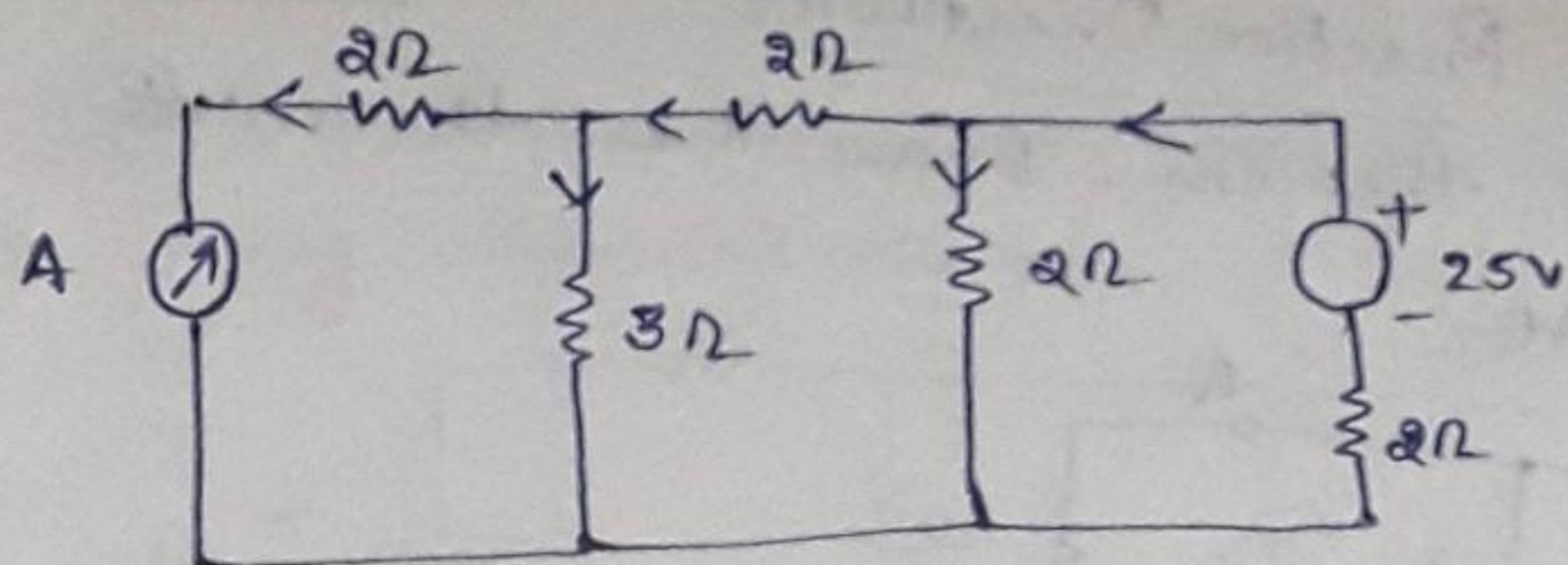
From fig(2) $I_2 = I_1 \times \frac{3}{3+3} = \frac{I_1}{2} = \frac{25}{7} \text{ A}$

From fig(1) current I_2 is dividing equally into parallel branches of 2Ω each.

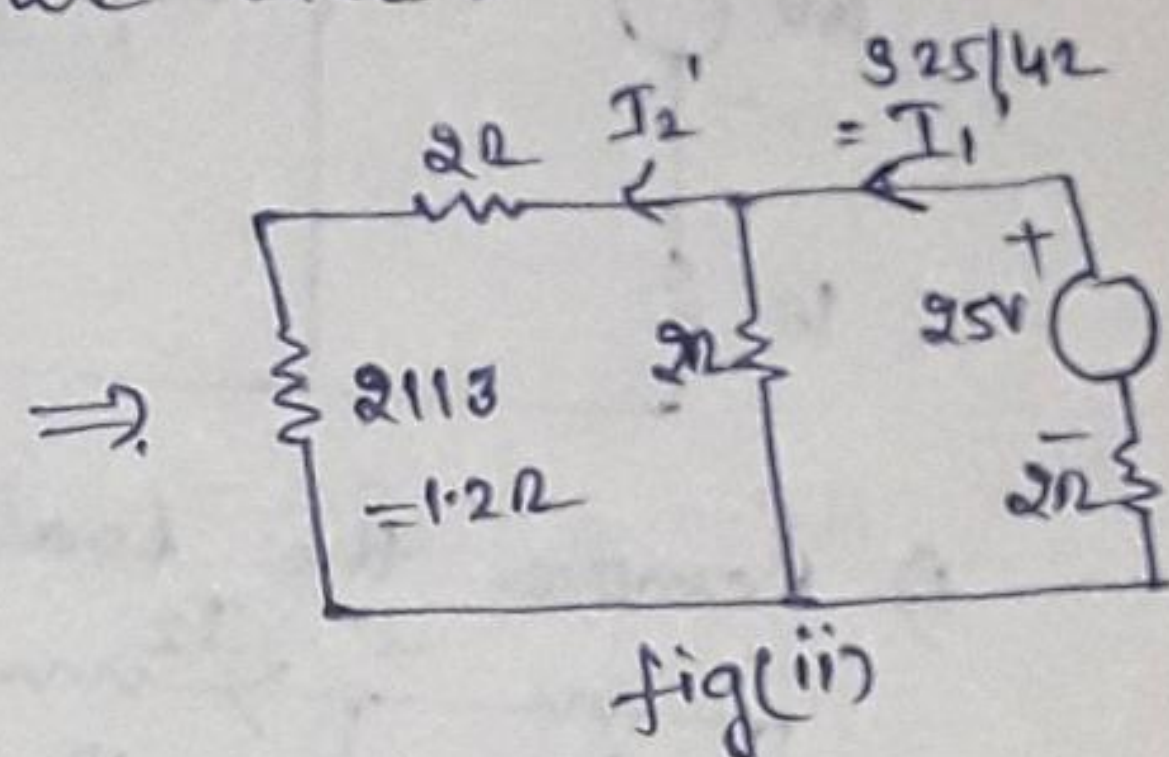
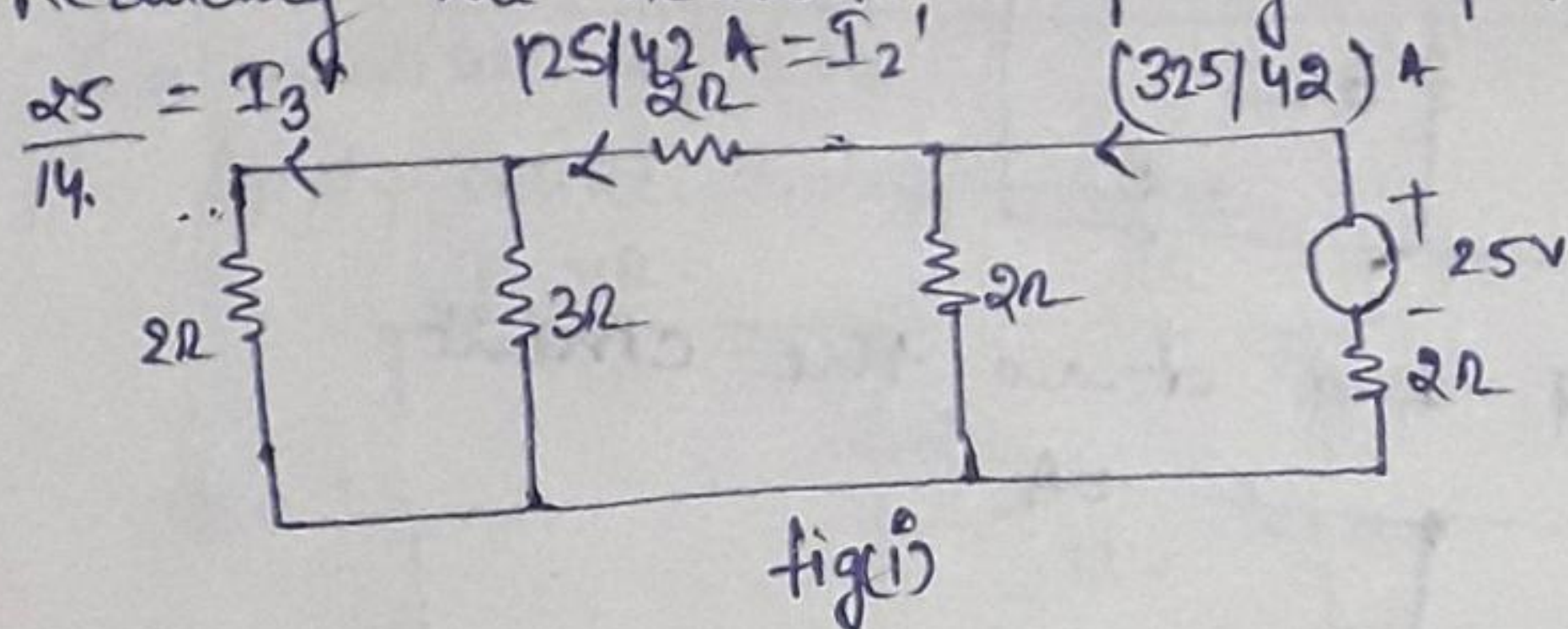
$$\therefore I_3 = I_2 \times \frac{2}{4} = \frac{25}{7} \times \frac{1}{2} = \frac{25}{14} \text{ A}$$

Therefore, ammeter current = $\frac{25}{14} \text{ A}$ — (i)

Now interchange source voltage and ammeter. the circuit is as follows -



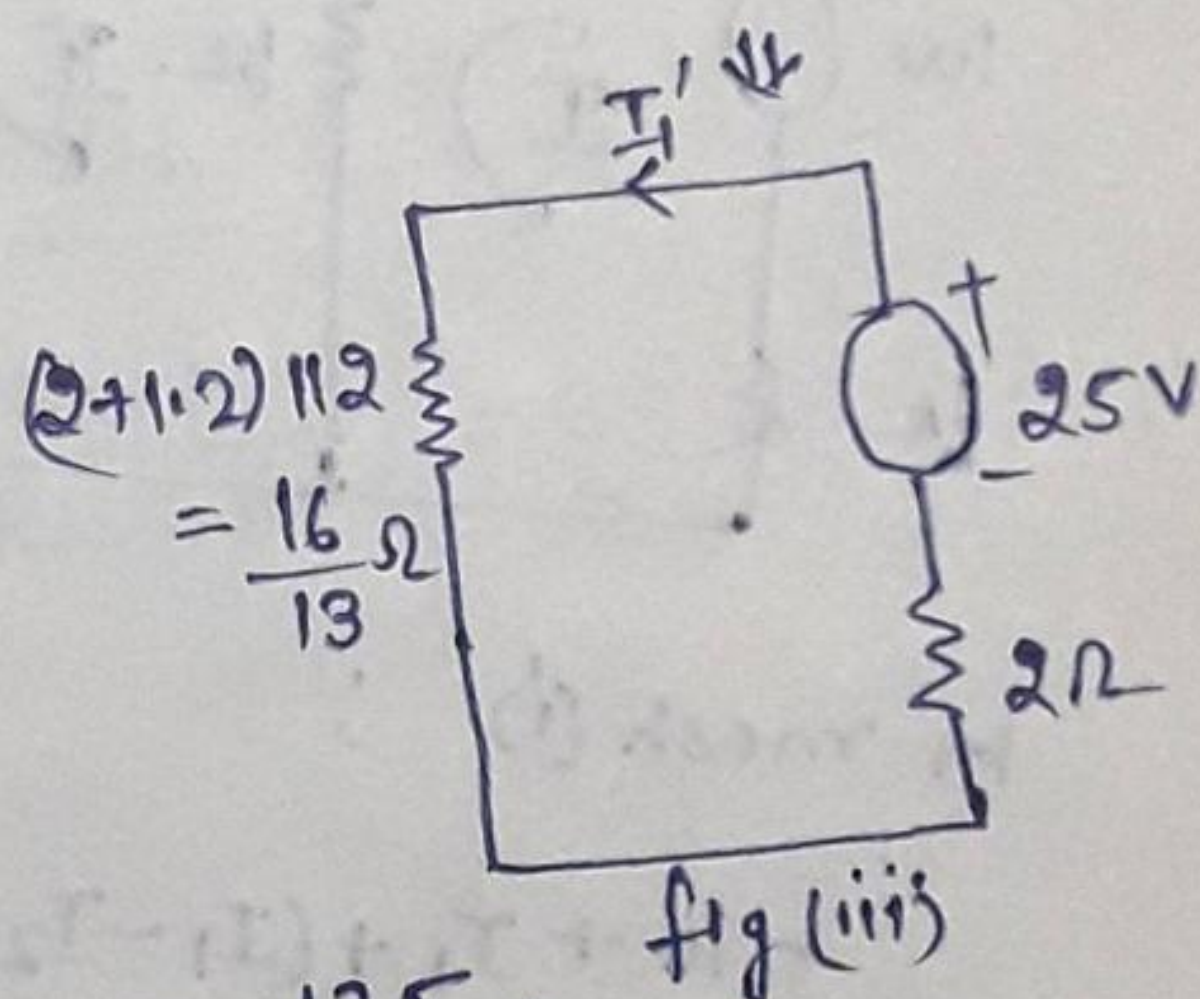
Reducing the network step by step, we have -



From fig(iii)

Current supplied by battery

$$I_1' = \frac{25}{2 + \frac{16}{13}} = \frac{25 \times 13}{2 \times 13 + 16} = \frac{325}{42} \text{ A}$$



From fig(ii), we have,

$$I_2' = I_1' \times \frac{2}{2 + (1.2 + 2)} = \frac{325}{42} \times \frac{2}{5.2} = \frac{125}{42} \text{ A}$$

Current in ammeter $I_m' = \text{Current in } 2\Omega \text{ (in fig(i))}$

$$\frac{1}{3} I_m' = I_2' \times \frac{3}{2+3} = \frac{125}{42} \times \frac{3}{5} = \frac{25}{14} \text{ A}$$

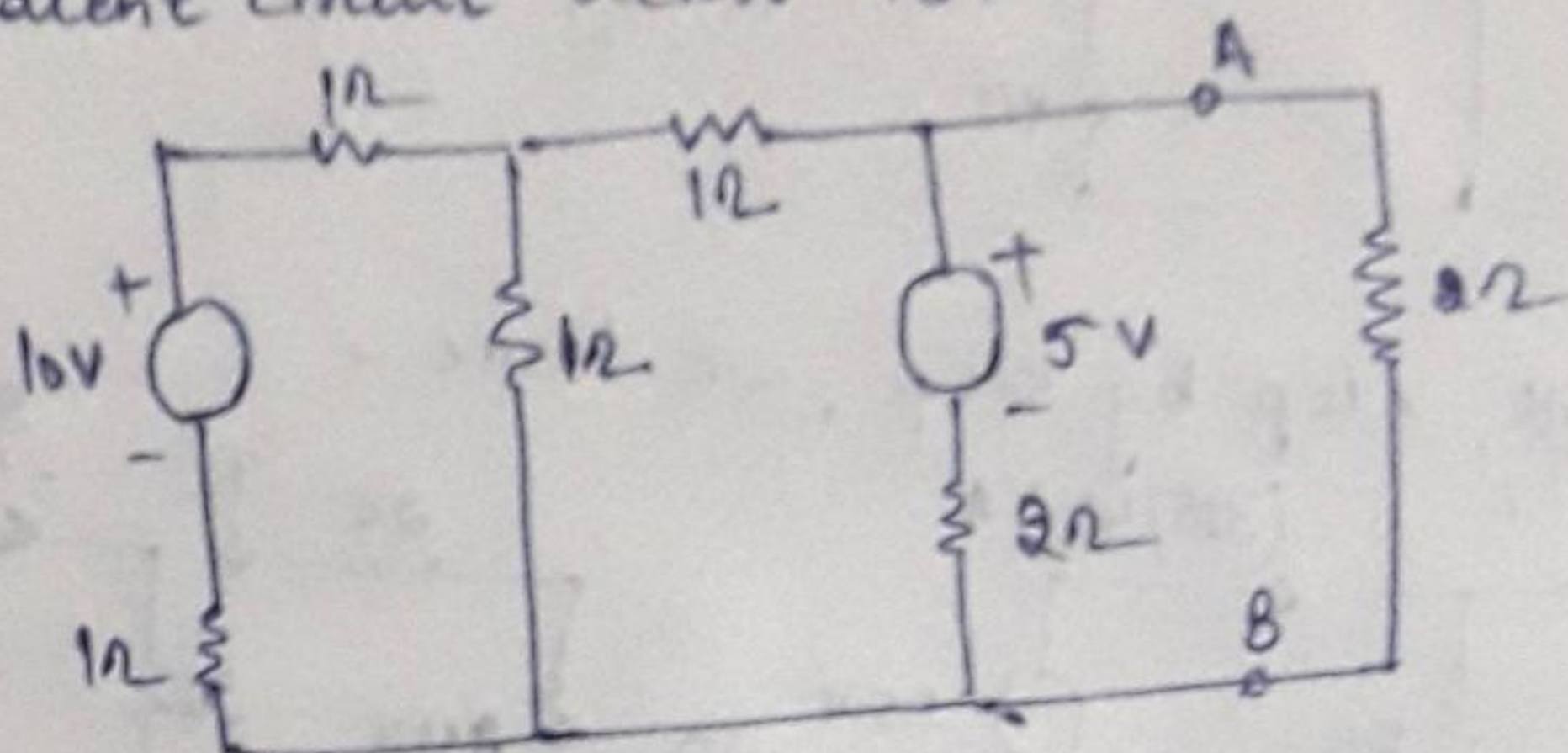
Expressions (i) and (ii) for ammeter current verify the reciprocity theorem.

⑫

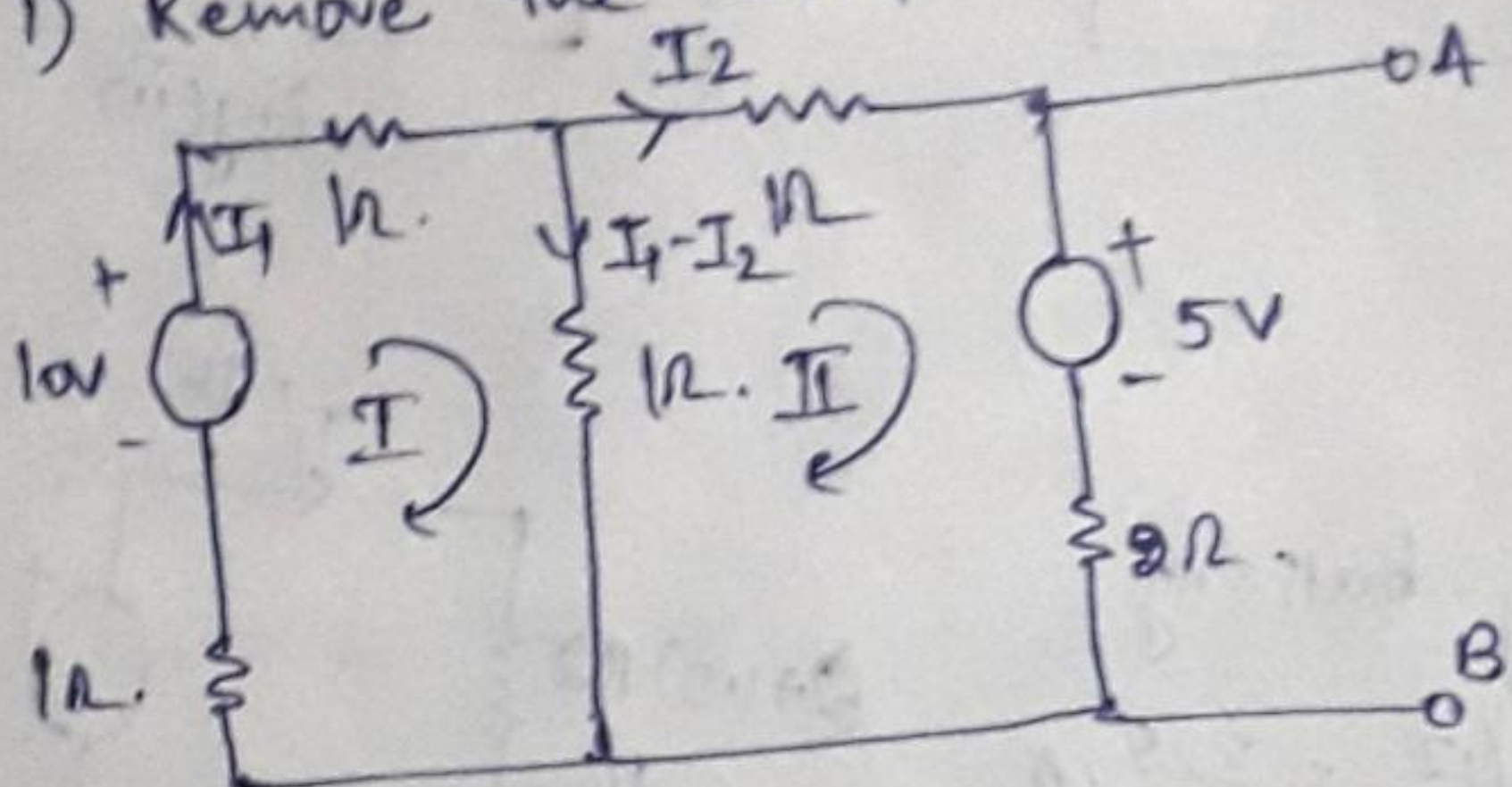
Practice Questions

02-09-

Q1: Explain thevenin's theorem. Draw the thevenin's equivalent circuit across AB.



1) Remove the load and draw the circuit



In mesh (i) ;

$$-10 + I_1 + (I_1 - I_2) + I_1 = 0$$

$$\text{or } 3I_1 - I_2 = 10 \quad \text{--- (1)}$$

In mesh (ii) $I_2 + 5 + 2I_2 - (I_1 - I_2) = 0$

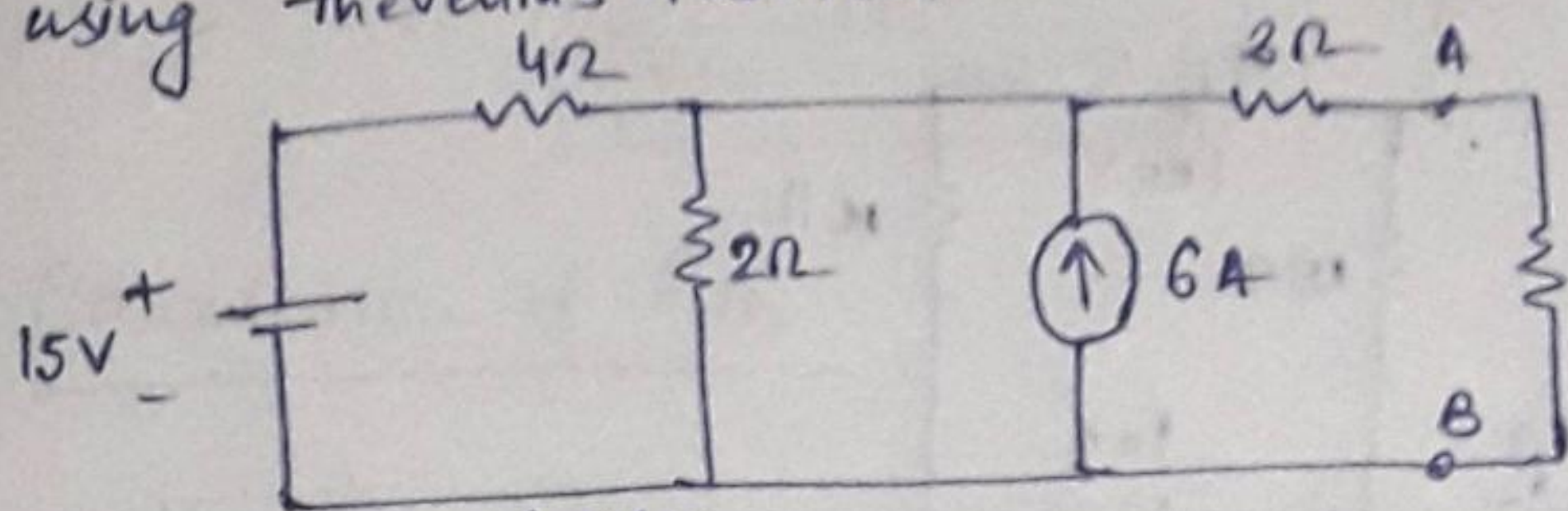
$$\text{or } I_1 - 4I_2 = 5 \quad \text{--- (2)}$$

$$\therefore I_1 = \frac{35}{11} \text{ A and } I_2 = \frac{-5}{11} \text{ A}$$

$$\therefore V_T = 5 + 2I_2 = 5 - \frac{5}{11} \times 2 = \frac{45}{11} \text{ V}$$

$$\text{Current through } 2\Omega \text{ resistance} = \frac{V_T}{R_T + R_L} = \frac{45}{32} \text{ A}$$

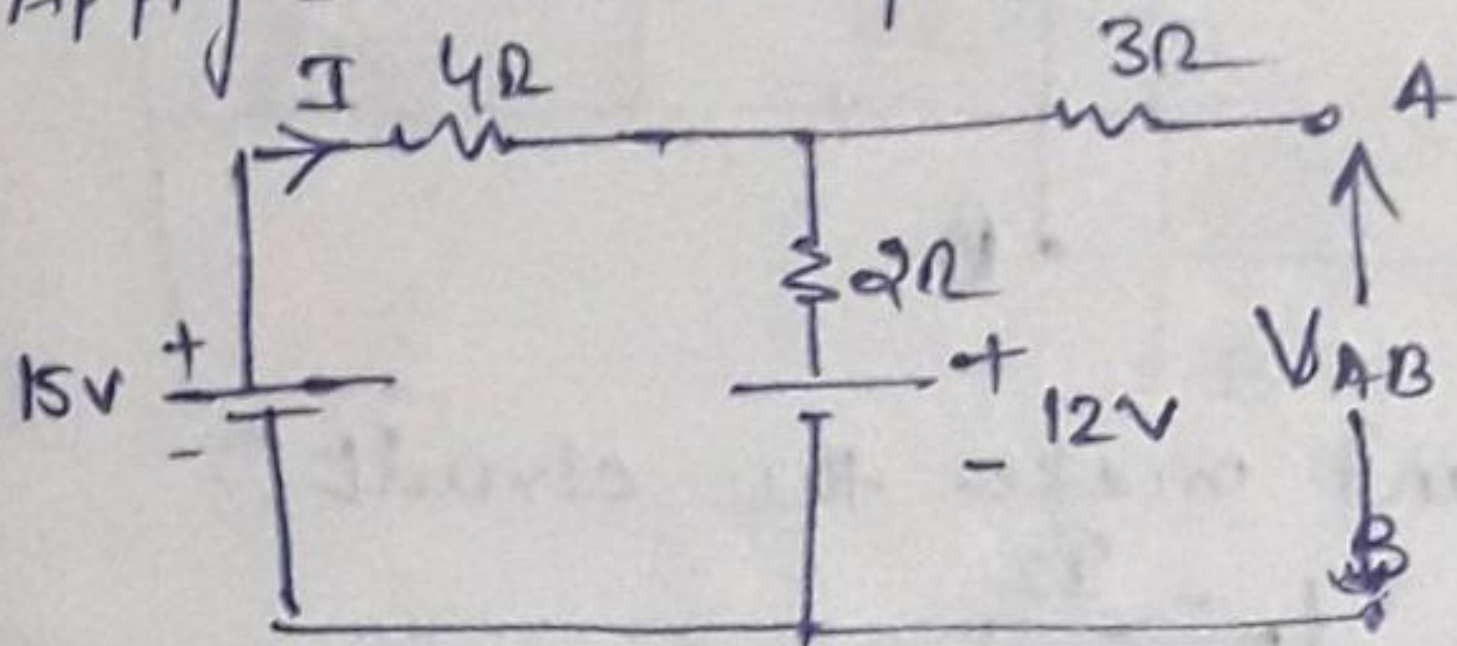
Q2 Determine current through 5Ω resistor in network using Thevenin's theorem.



(1) Remove the load and draw the circuit.

(2) Calculation of V_{th} .

Apply source transformation.

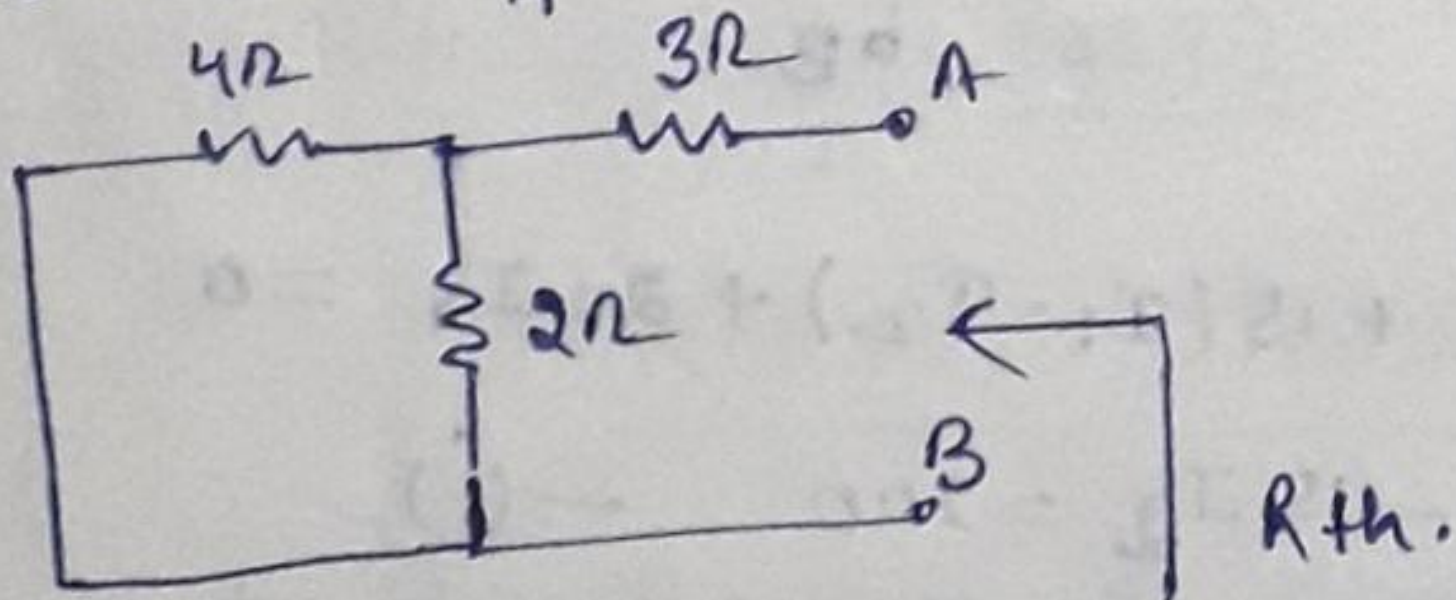


$$I = \frac{15 - 12}{4 + 2} = 0.5 \text{ A}$$

$$\therefore V_{AB} = 12 + 2I = 13 \text{ V} = V_{th}$$

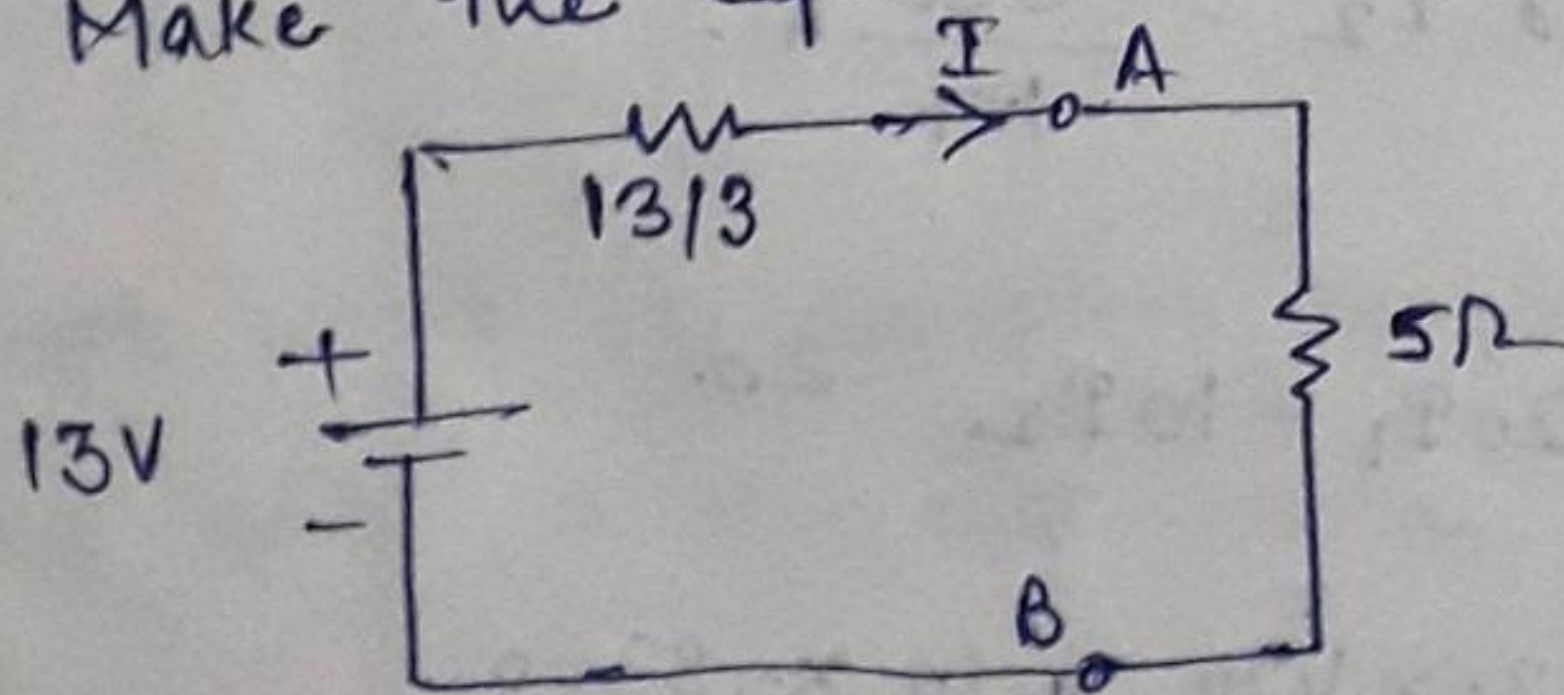
$$\text{Current flowing through } 5\Omega = \frac{V_{th}}{R_{th} + R_L} = \frac{13}{28} \text{ A}$$

(3) Calculation of R_{th} .



$$R_{th} = 3 + 4 \parallel 2 = \frac{13}{3} \Omega$$

(4) Make the equivalent circuit as shown

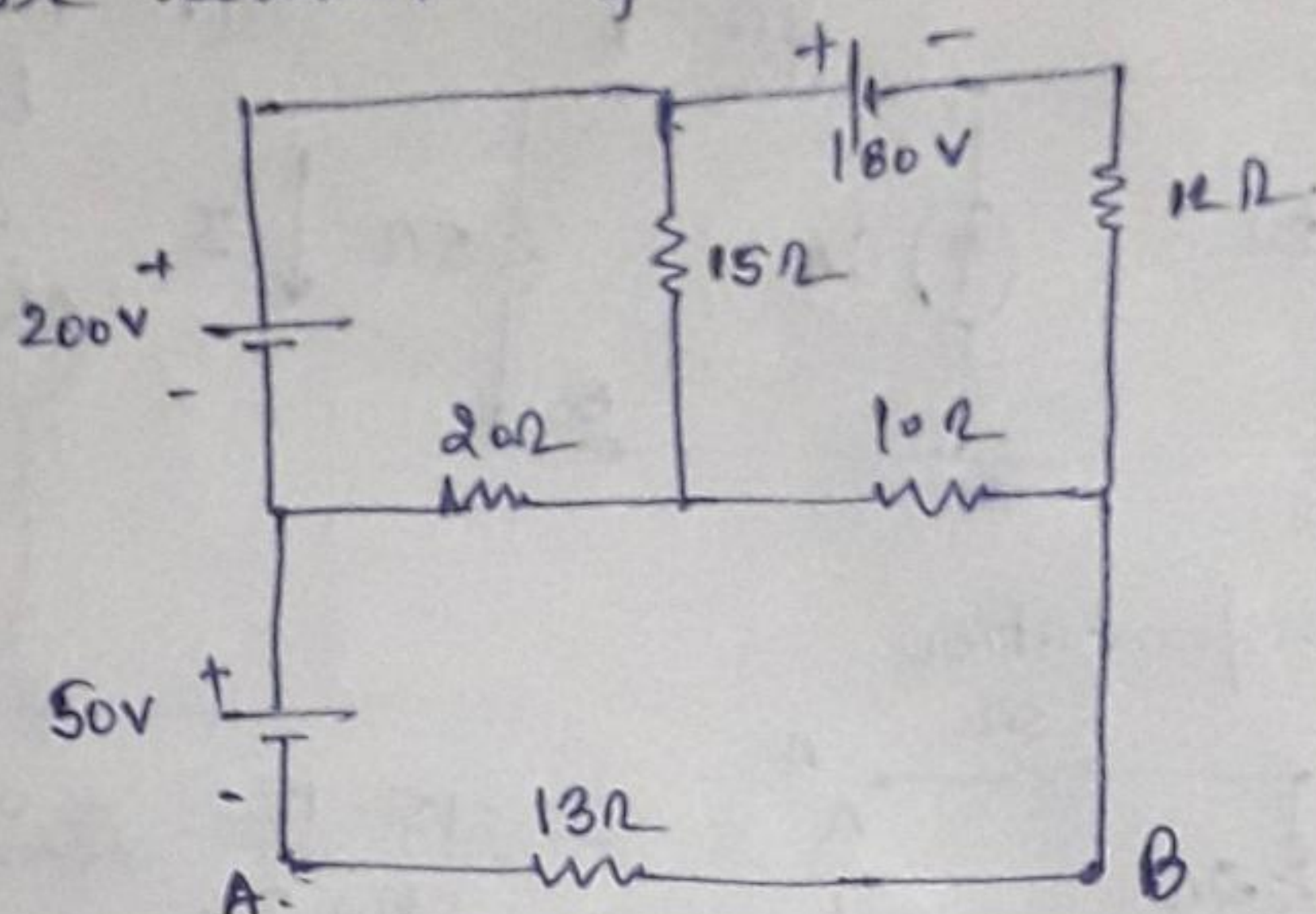


(5) Current flowing through 5Ω is

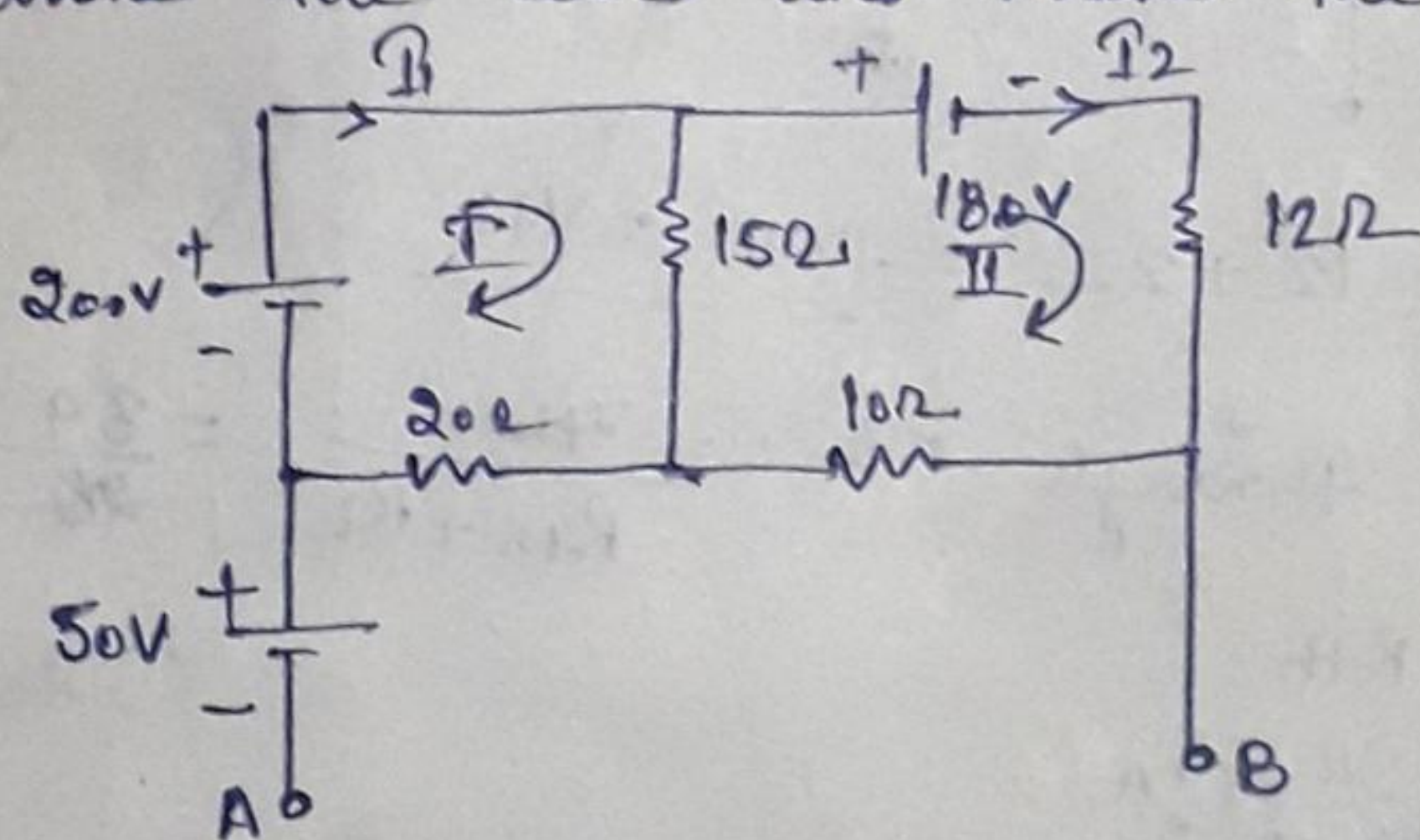
$$I_L = \frac{13}{\frac{13}{3} + 5} = \frac{39}{28} \text{ A}$$

(14)

Q3:- Applying Thevenin's theorem, calculate the current in 13Ω resistor of circuit.



1) Remove the load and make the circuit.



In mesh (i), $-200 + 15(I_1 - I_2) + 20I_1 = 0$

$$35I_1 - 15I_2 = 200 \quad \text{--- (i)}$$

In mesh (ii), $+180 + 12I_2 + 10I_2 - 15(I_1 - I_2) = 0$

$$15I_1 - 37I_2 = 180 \quad \text{--- (ii)}$$

$$\therefore I_1 = \frac{470}{107} \text{ A and } I_2 = -\frac{330}{107} \text{ A}$$

$$\therefore +V_{th} = 50 - 20I_1 - 10I_2 = 0$$

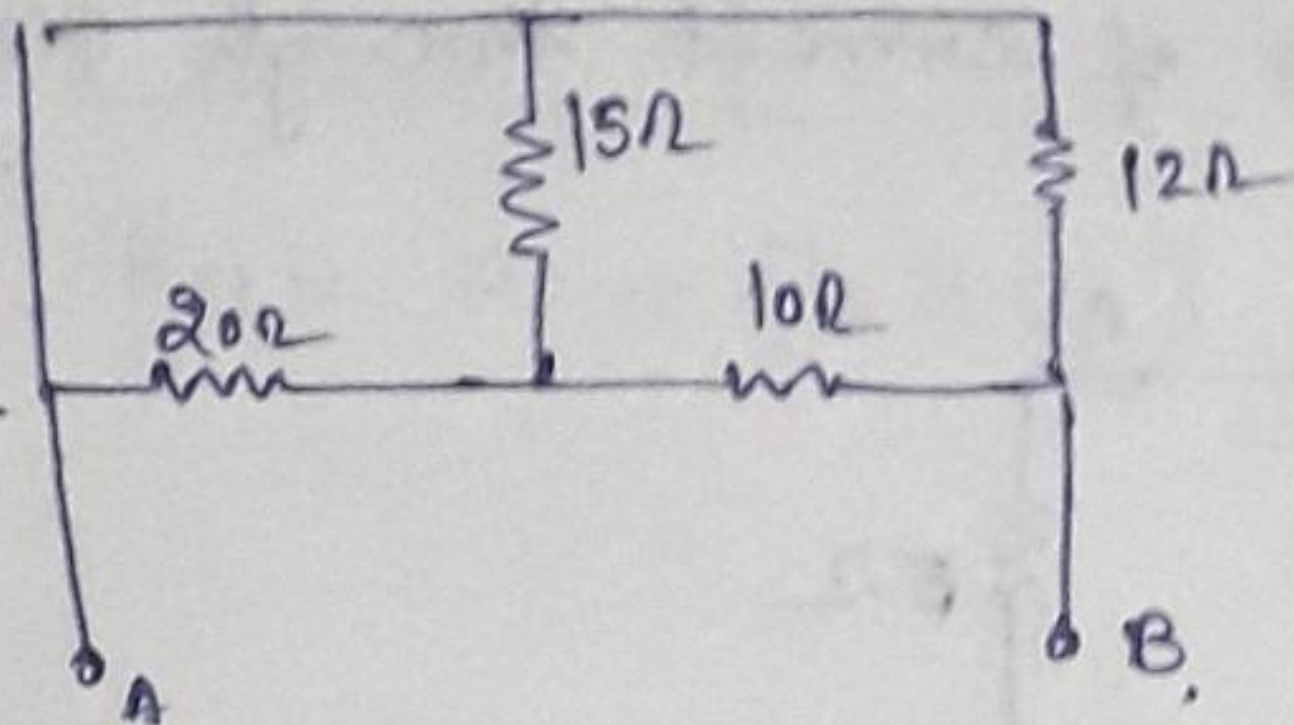
$$V_{th} = +50 + 20 \times \frac{470}{107} + 10 \times -\frac{330}{107}$$

$$V_{th} = 50 + \frac{9400 - 3300}{107}$$

(15)

$$\therefore V_{th} = 107 \text{ V}$$

3) Calculation of R_{th}



$$R_{th} = 12 \parallel [10 + 20 \parallel 15]$$

$$= 12 \parallel \left[10 + \frac{60}{7} \right]$$

$$= 12 \parallel \frac{130}{7}$$

$$\therefore R_{th} = \frac{780}{107} = 7.29 \Omega$$

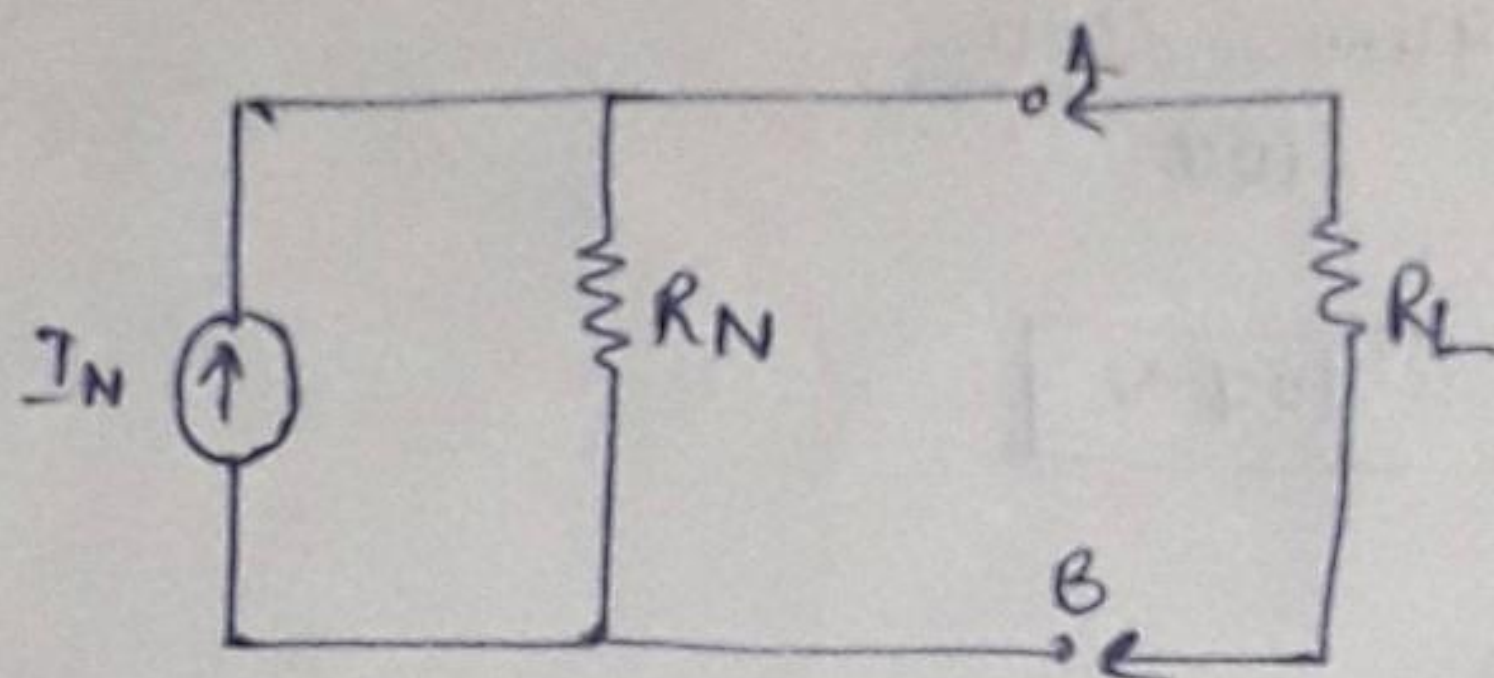
NORTON'S THEOREM

According to this theorem, a complex network is reduced into a simple circuit in which parallel combination of constant current source and looking back resistance feeds the load resistance.

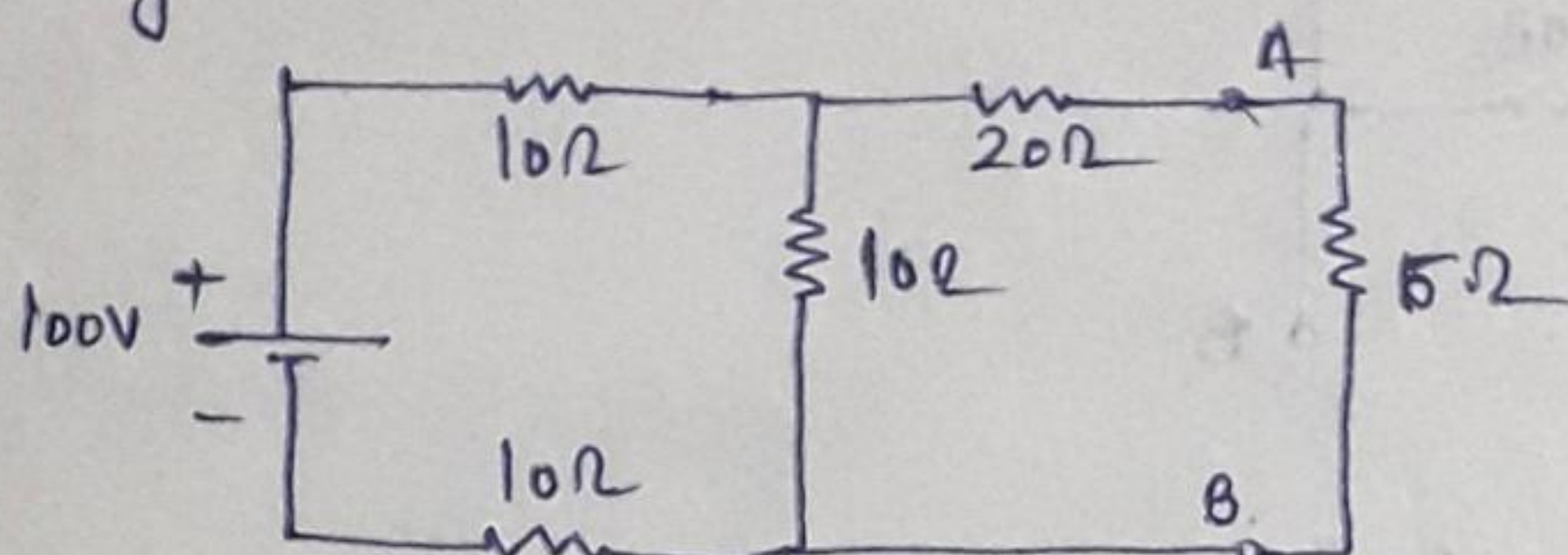
or

In simple words, a complex complicated network can be reduced into a current source in parallel with a resistance.

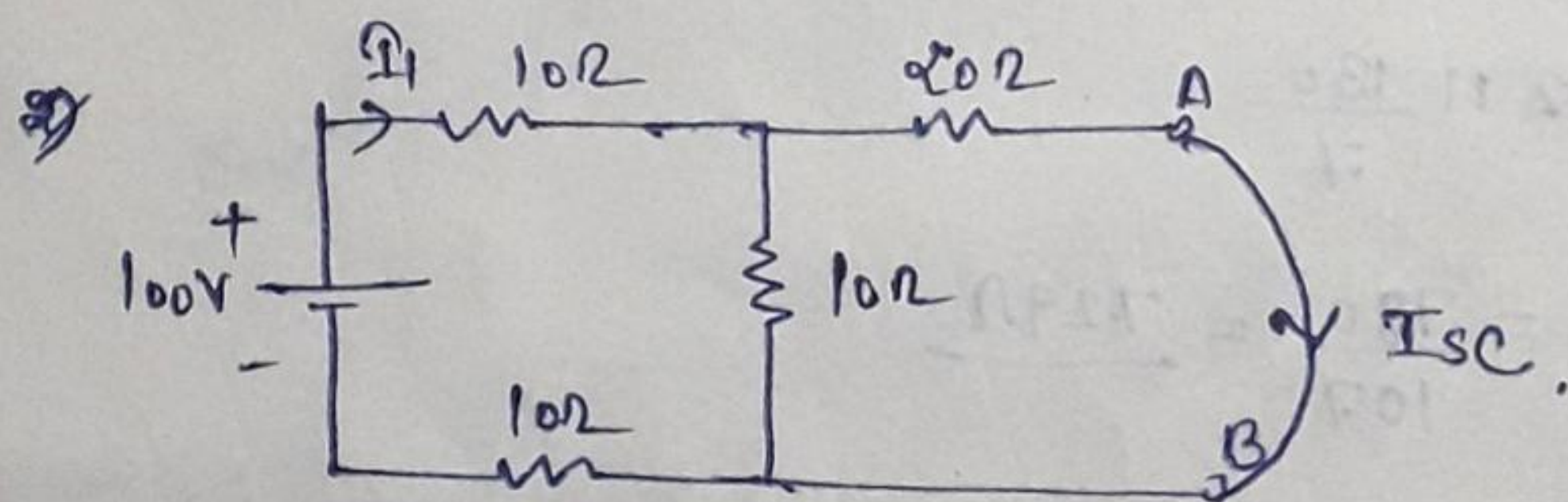
16



Q1:- Determine the value of current through 5Ω resistor using Norton's theorem.



1) Remove the load, and short circuit terminals A and B, draw the circuit.



2) Calculate I_{AB} or I_{sc} .

Current supplied by battery, $I_1 = \frac{100}{R'}$

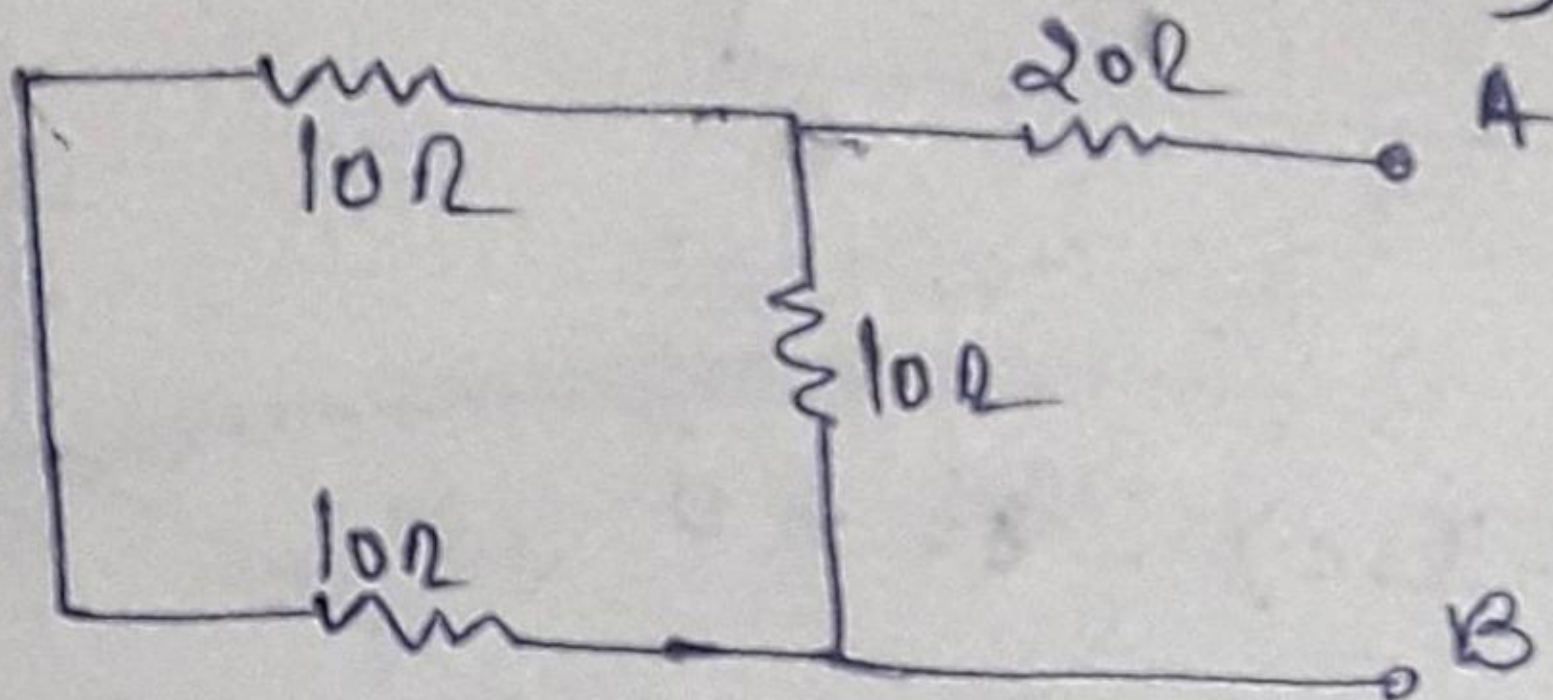
$$= \frac{100}{10 + (10 \parallel 20) + 10}$$

$$= \frac{100}{80/3}$$

$$\underline{I_1 = 3.75A}$$

$$\therefore I_{sc} = I_1 \times \frac{10}{10+20} = 3.75 \times \frac{10}{30} = \underline{1.25 \text{ A}} \quad (17)$$

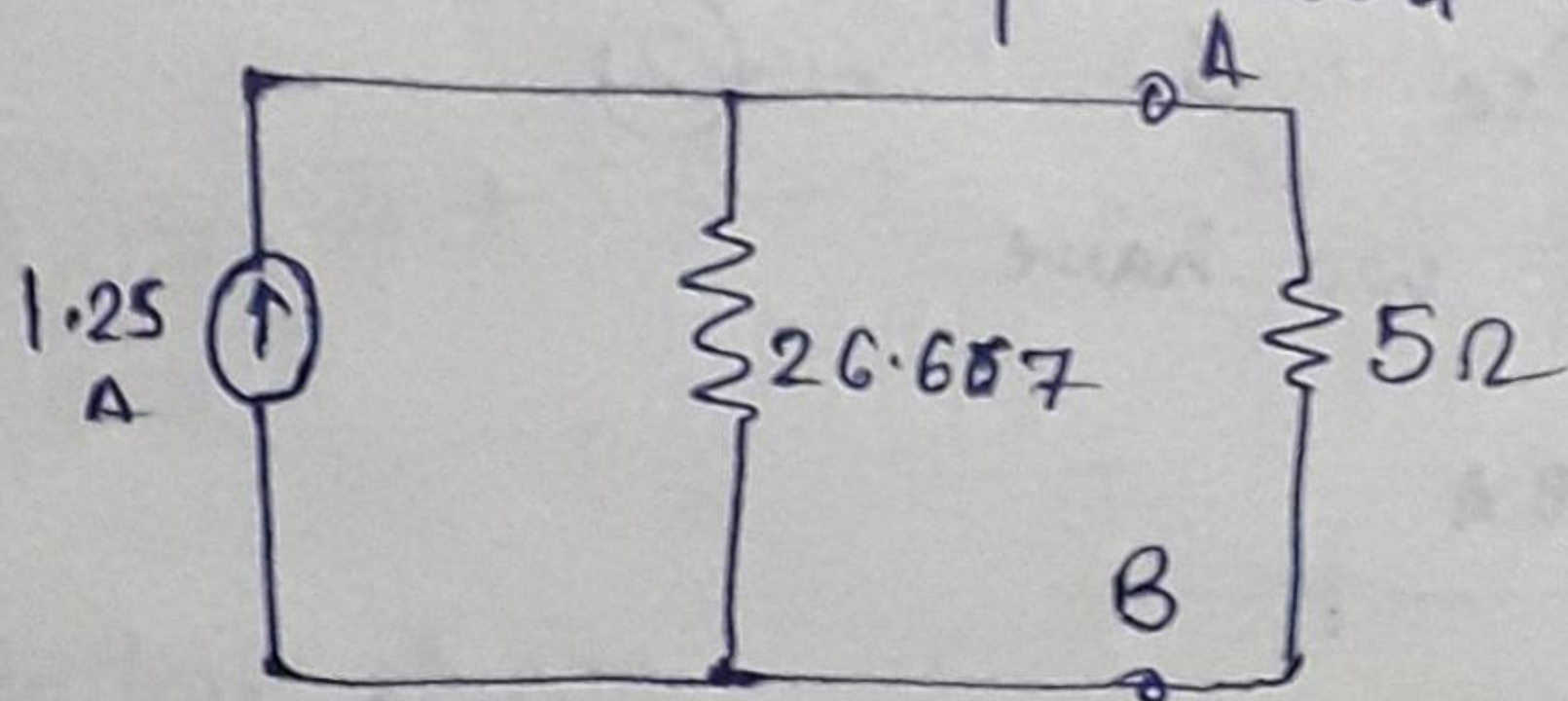
(3) Calculate norton's Resistance, R_N .



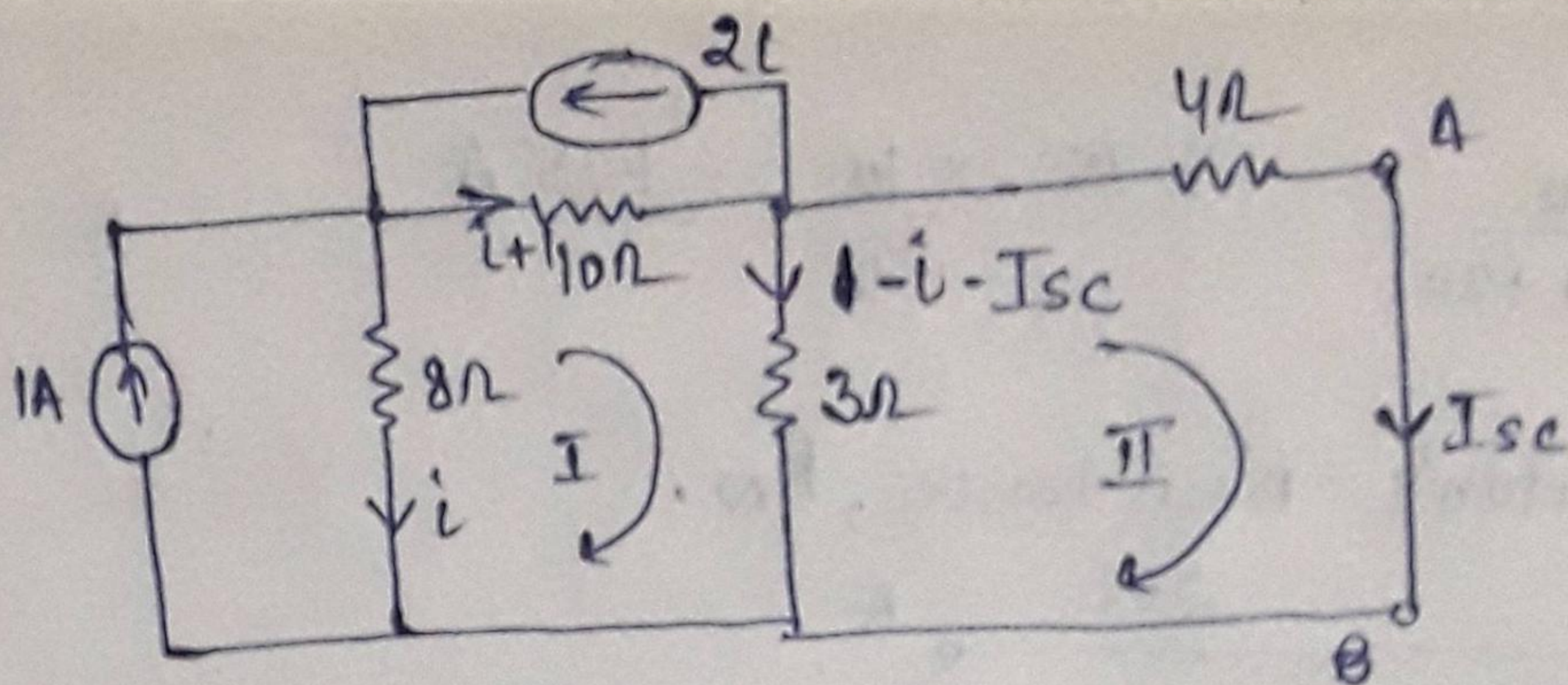
$$R_N = 20 + [10 \parallel 20] = 20 + \frac{200}{30}$$

$$\underline{R_N = 26.667 \Omega}$$

(4) Make the Norton's equivalent circuit,



(5) Note the value of current through 5Ω = $1.25 \times \frac{26.667}{26.667+5}$
 $= \underline{1.053 \text{ A}}$



In mesh (I)

$$10(i+1) + 3(1-i-I_{sc}) - 8i = 0$$

$$-i + 13 - 3I_{sc} = 0$$

$$i + 3I_{sc} = 13 \quad \text{--- (1)}$$

In mesh (II)

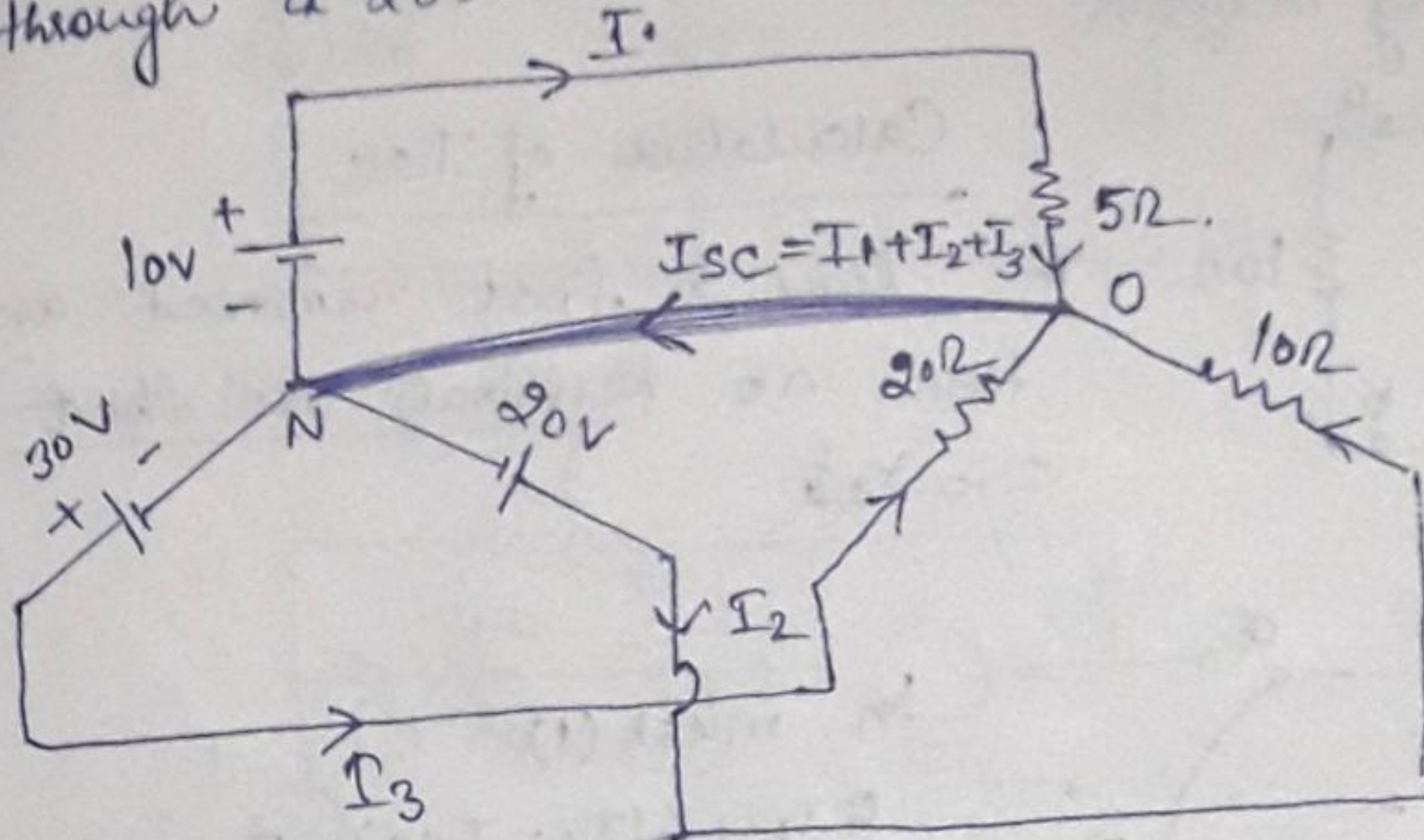
$$4I_{sc} - 3(1-i-I_{sc}) = 0$$

$$3i + 7I_{sc} = 3 \quad \text{--- (2)}$$

From (1) and (2), we have

$$\underline{I_{sc} = 18 \text{ A}}$$

Q3 Using Norton's theorem, find current which would flow through a 25Ω resistor connected between points N and O.

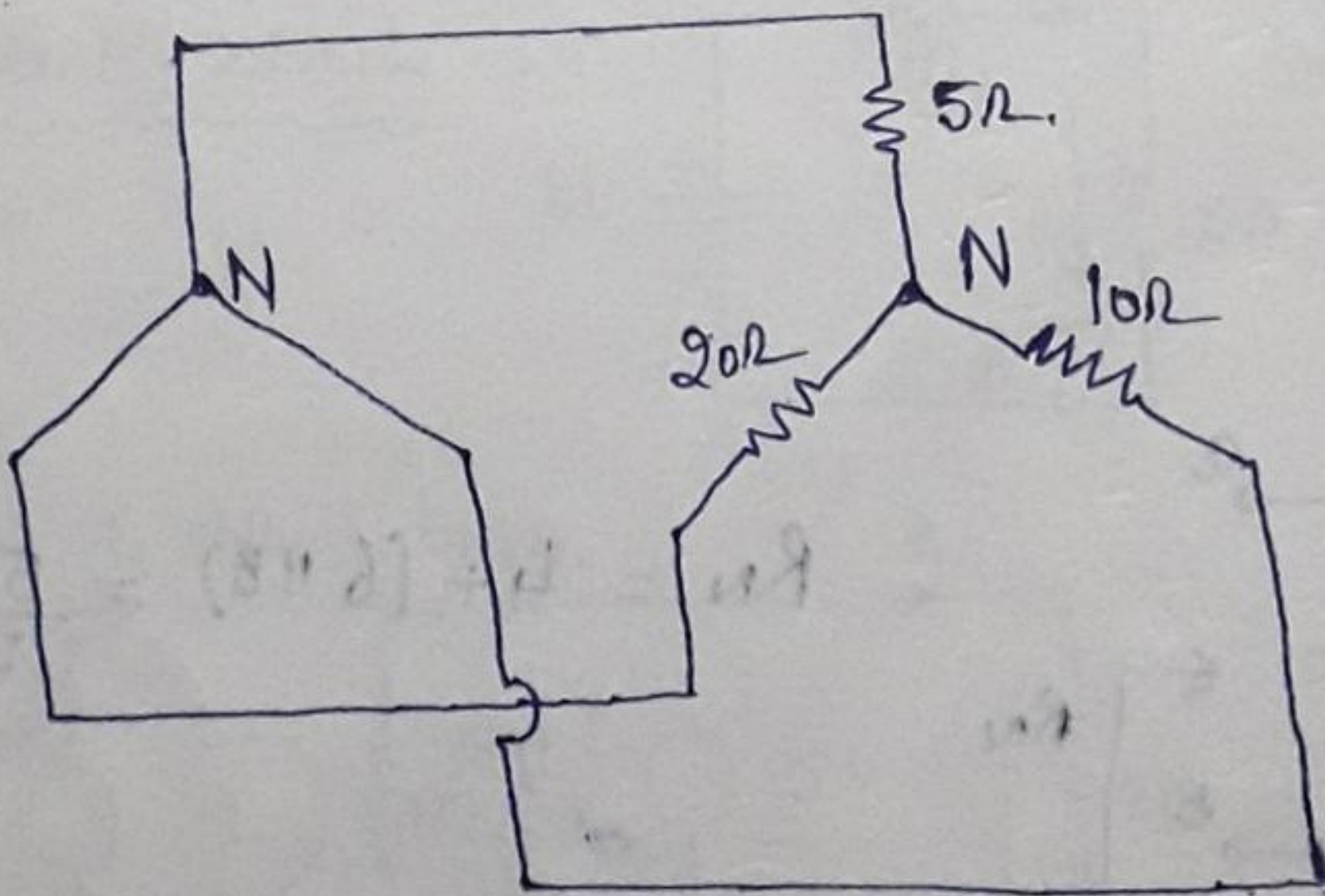


① Calculation of I_{sc}

$$I_{sc} = I_1 + I_2 + I_3 = \frac{10}{5} + \frac{20}{10} + \frac{30}{20}$$

$$I_{sc} = 2 + 2 + 1.5 = 5.5A$$

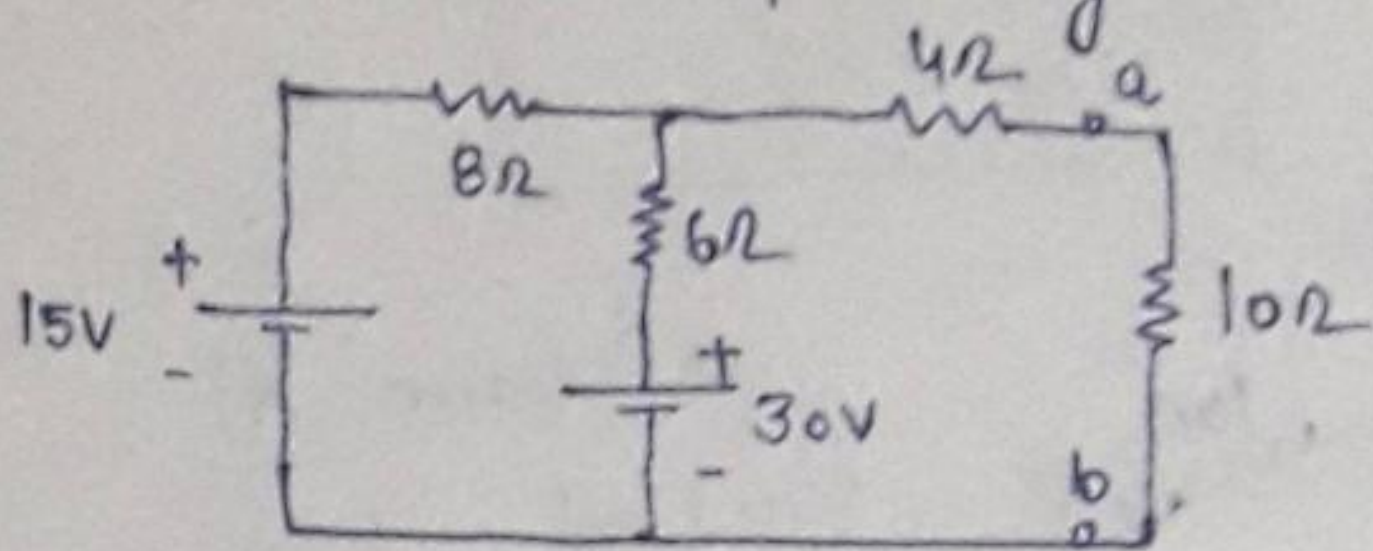
② Calculation of R_N



$$R_N = 5 \parallel 20 \parallel 10 = \frac{20}{7} \Omega$$

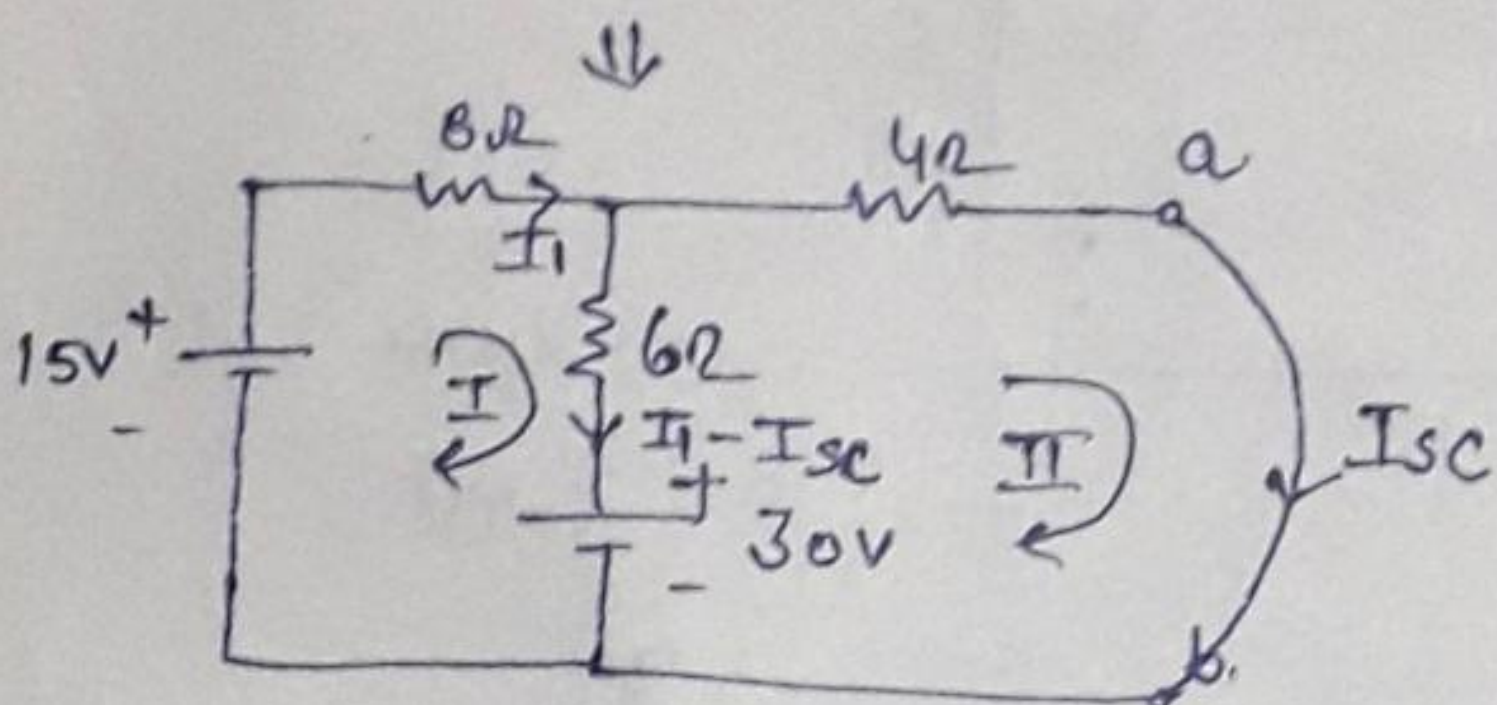
(20)

Q4 Determine current in 10Ω resistance using Norton's theorem in following network.



Calculation of I_{sc}

The load is first removed and then 'ab' terminals are short circuited.



In mesh (I)

$$8I_1 + 6(I_1 - I_{sc}) + 30 - 15 = 0$$

$$14I_1 - 6I_{sc} = -15 \quad \text{--- (1)}$$

In mesh (II)

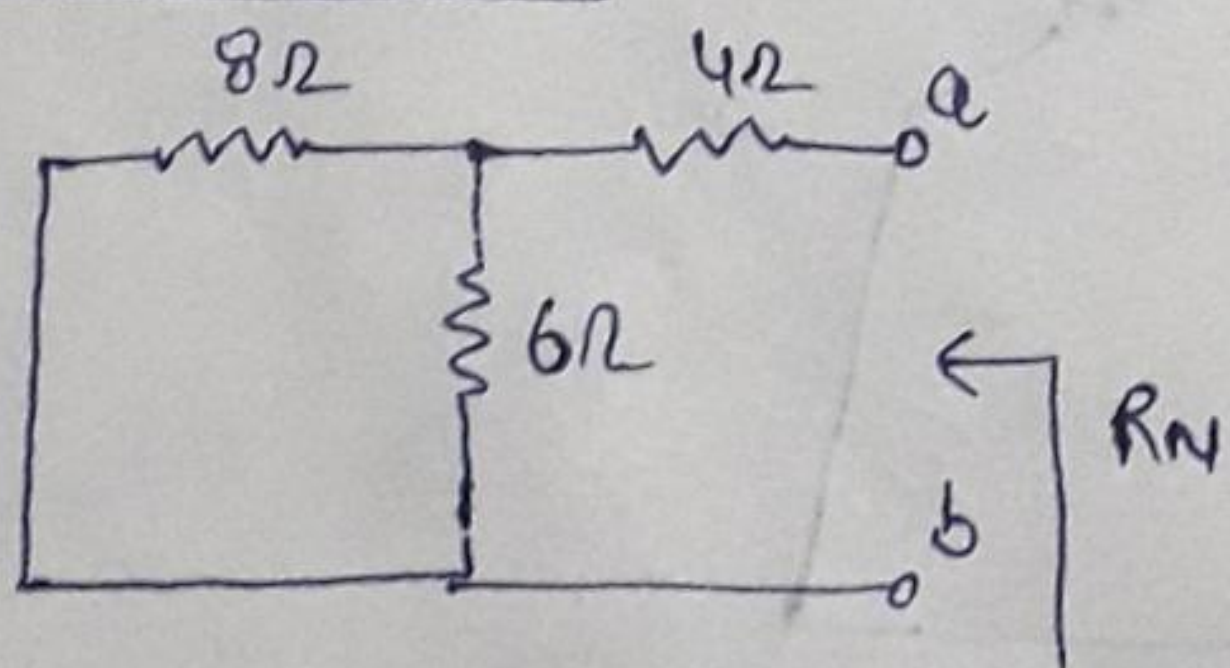
$$4I_{sc} - 30 - 6(I_1 - I_{sc}) = 0$$

$$-6I_1 + 10I_{sc} = 30$$

$$\text{or } 6I_1 - 10I_{sc} = -30 \quad \text{--- (2)}$$

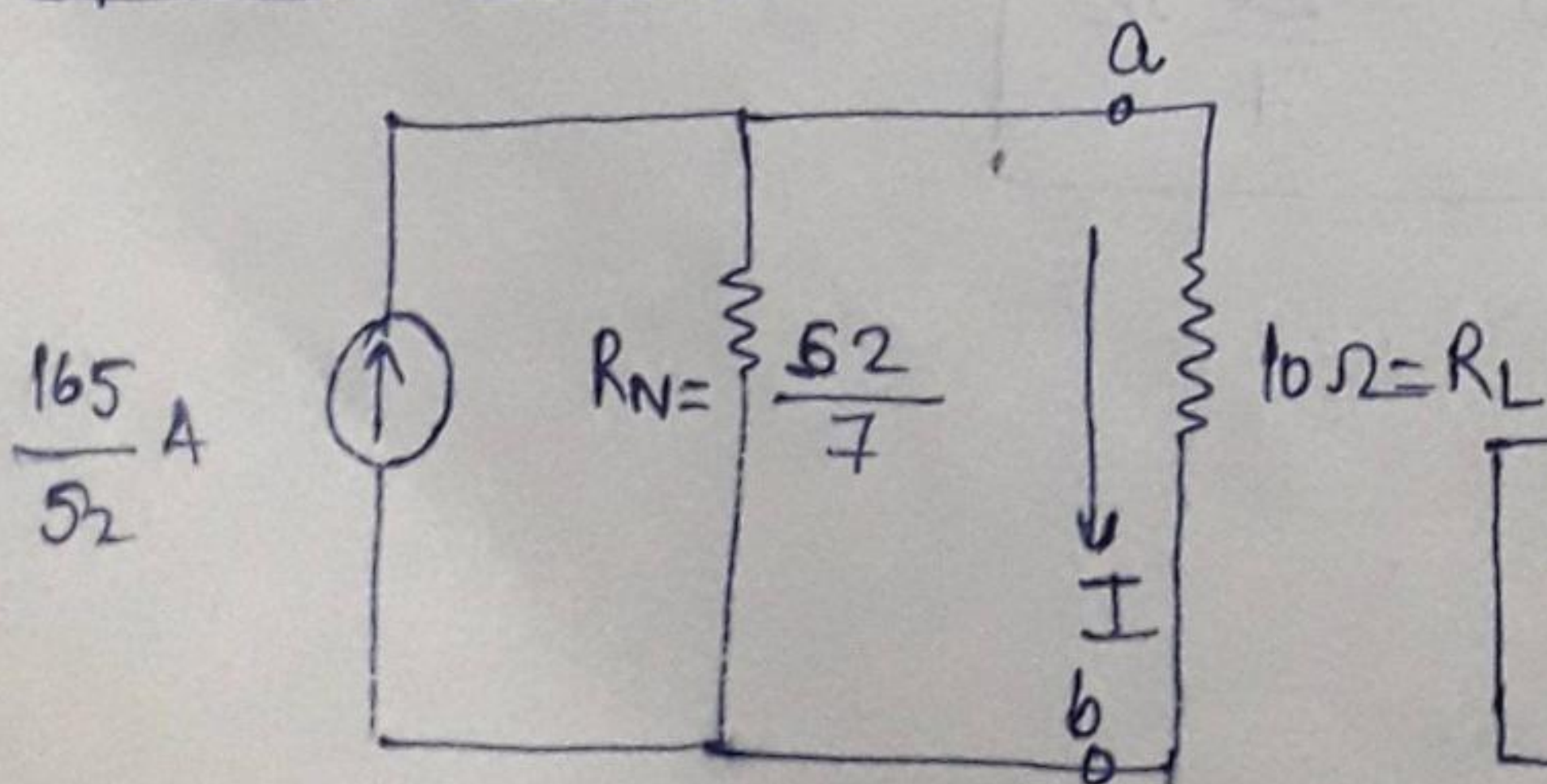
$$\therefore I_{sc} = \frac{165}{52} \text{ A}$$

Calculation of R_N



$$R_N = 4 + (6 \parallel 8) = \frac{52}{7} \Omega$$

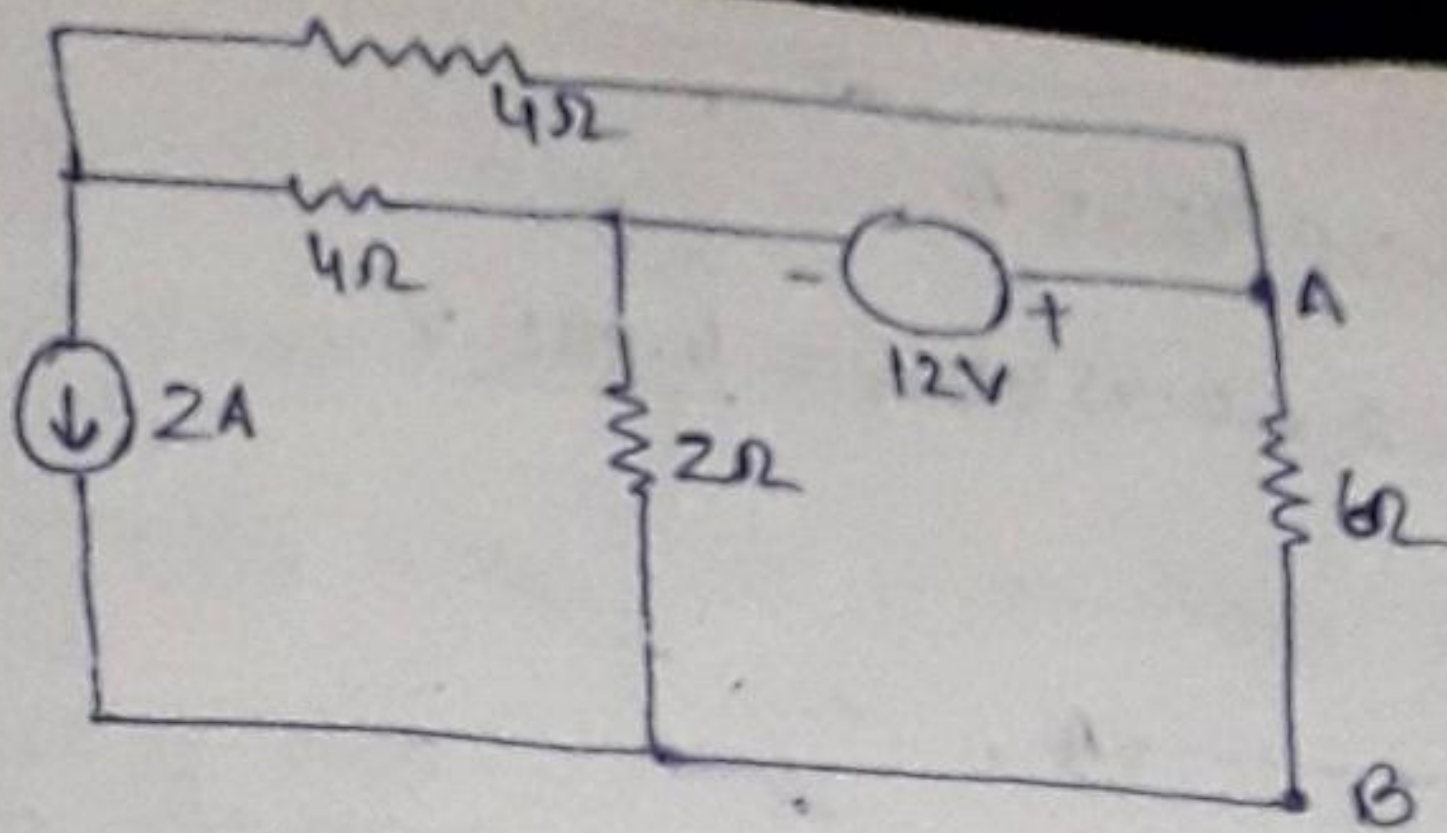
Equivalent circuit



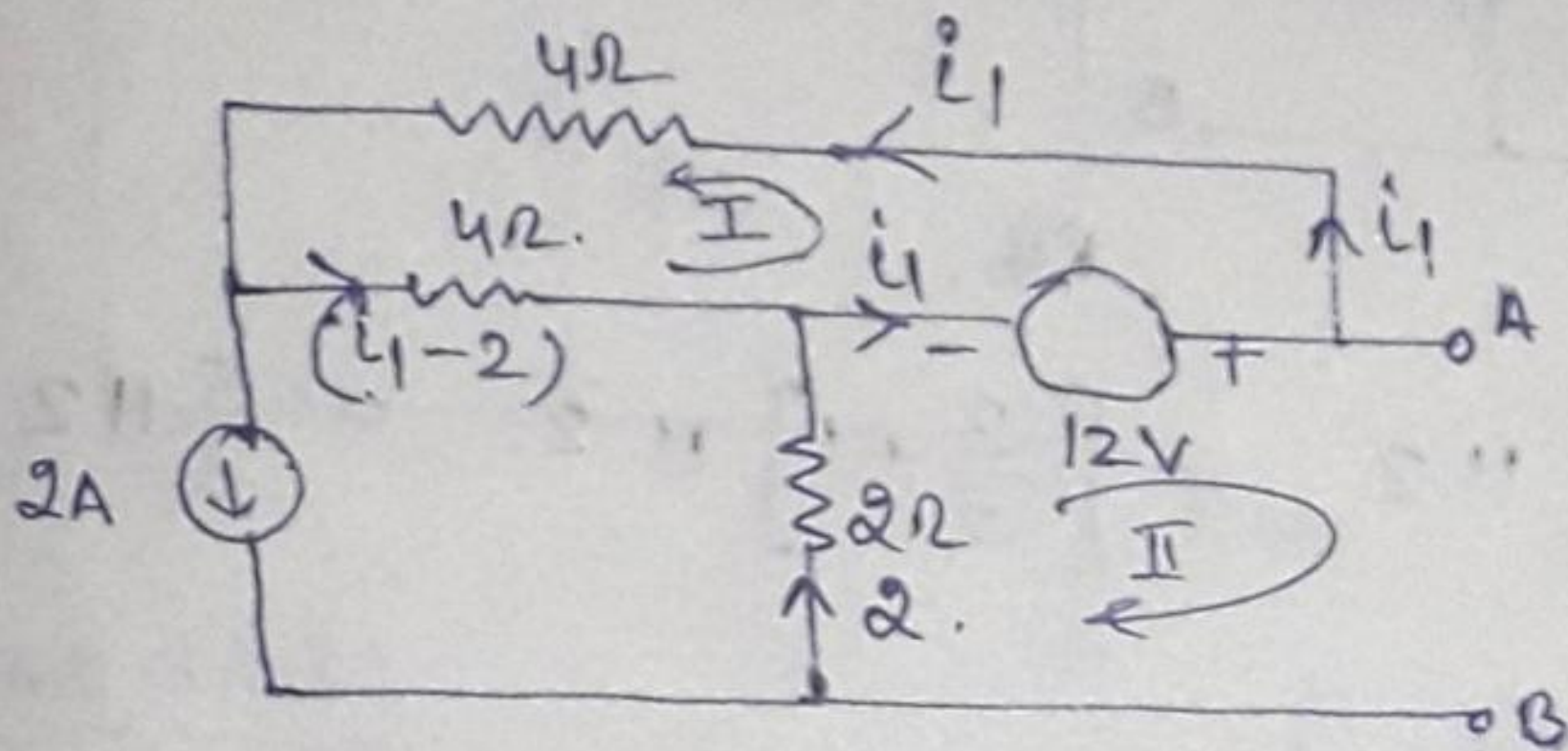
$$I = I_{sc} \times \frac{R_N}{R_N + R_L}$$

$$I = \frac{165}{122} \text{ A}$$

Q5



Calculation of V_{th} :-



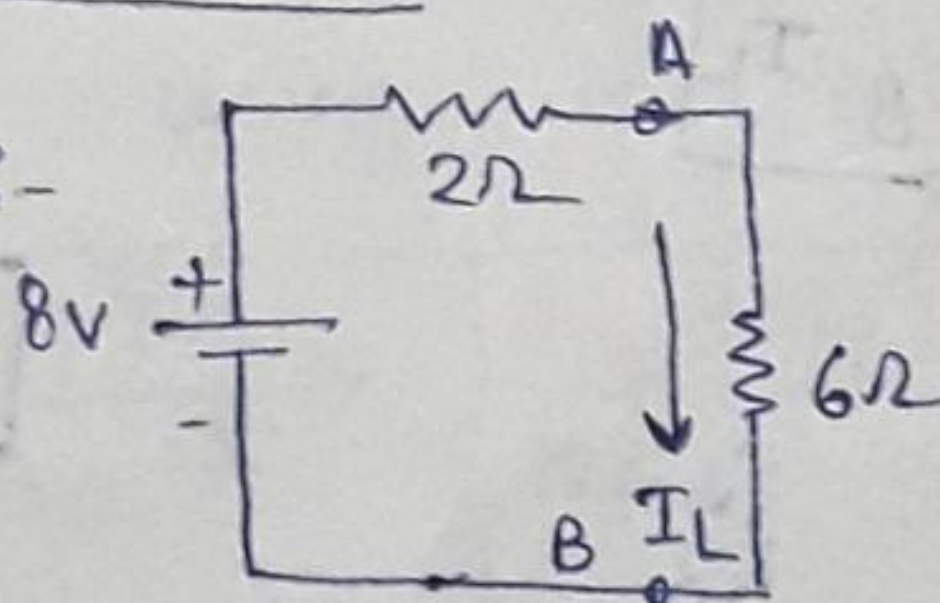
In mesh (I), apply KVL

$$-12 + 4i_1 + 4(i_1 - 2) = 0$$

$$8i_1 - 8 - 12 = 0$$

$$\therefore i_1 = \frac{20}{8} A$$

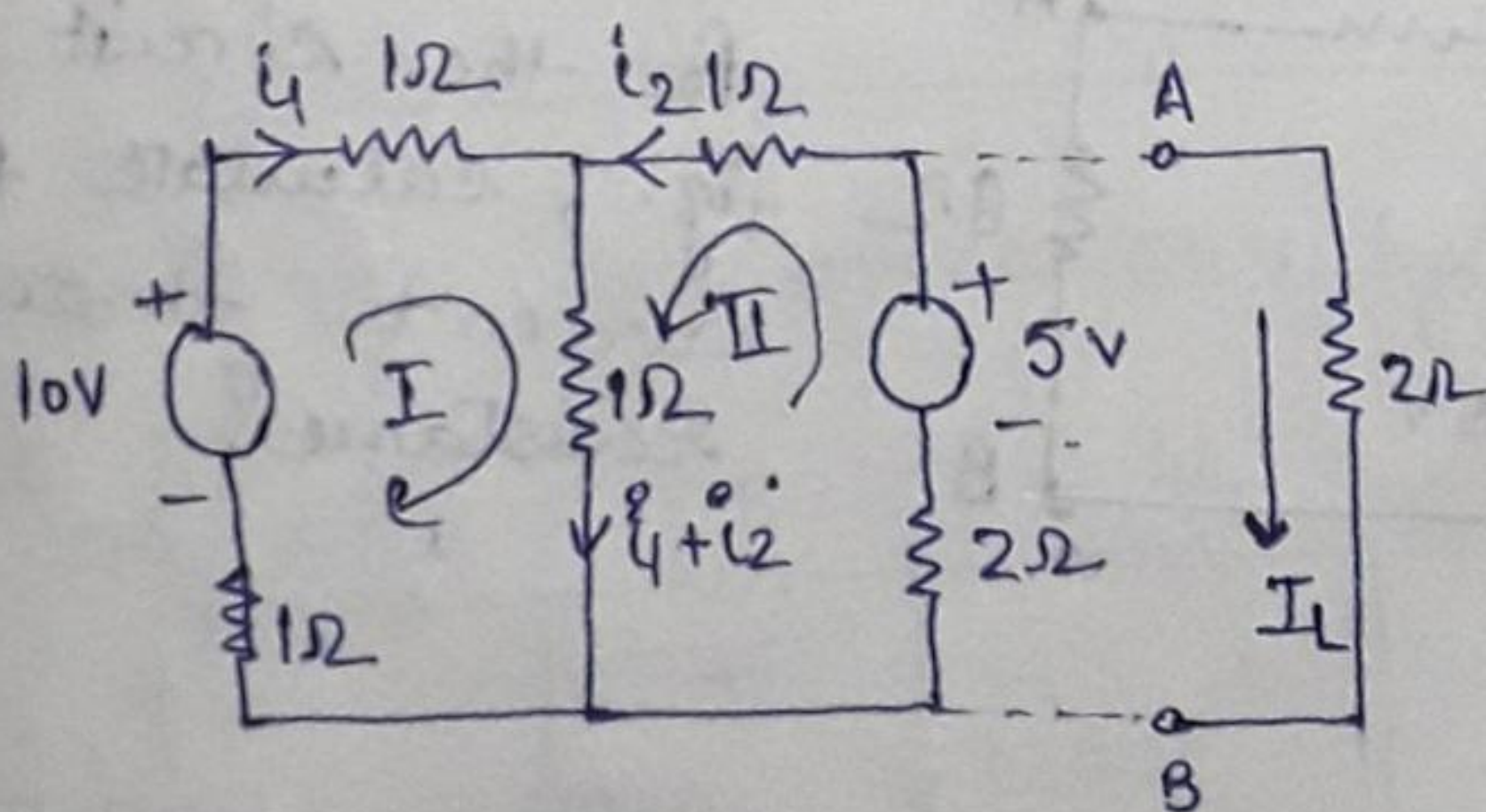
Equivalent circuit :-



Current through 6Ω

$$I_L = \frac{8}{6+2} = 1 A$$

Q6



For the figure shown calculate the current through 2Ω resistance?

Calculation of V_{th} :-

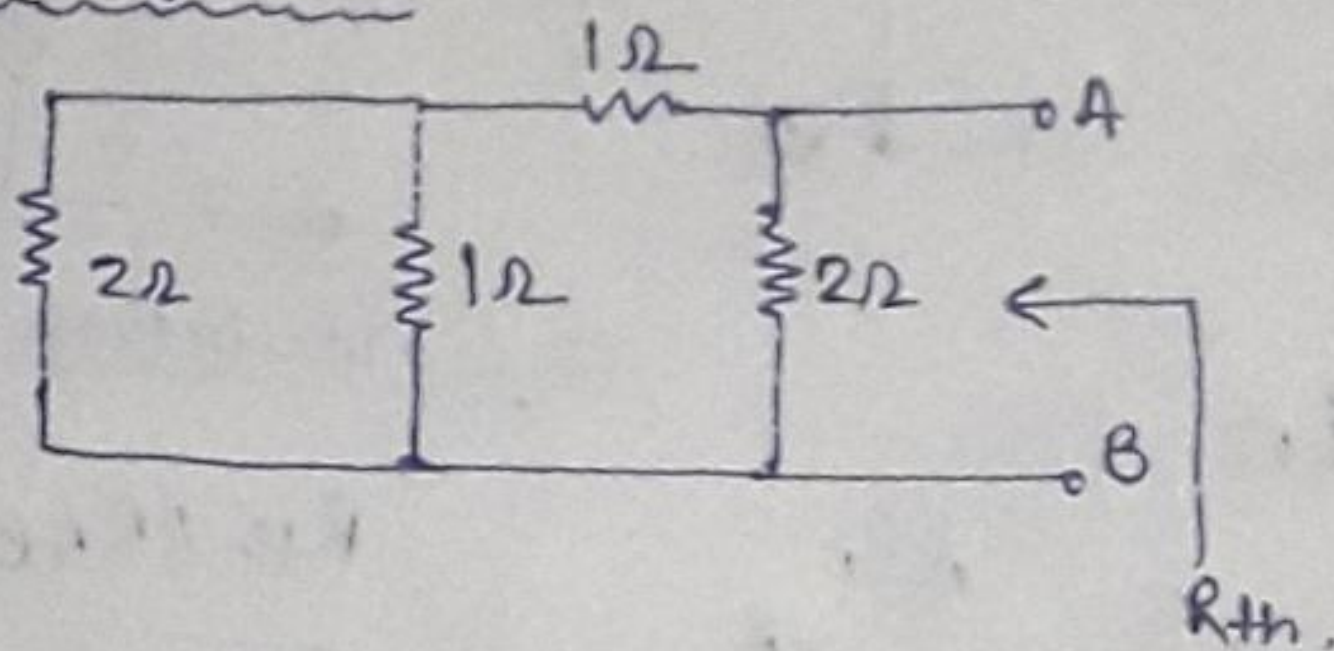
In mesh (I), apply KVL $\Rightarrow -10 + i_1 + i_1 + i_2 + i_1 = 0$
 or $3i_1 + i_2 = 10$ — (1)

In mesh (II), apply KVL $\Rightarrow -5 + i_2 + i_1 + i_2 + 2i_2 = 0$
 or $i_1 + 4i_2 = 5$ — (2)

(22) $\therefore i_1 = 3.18 \text{ A}$ and $i_2 = 0.4545 \text{ A}$

$\therefore V_{th} = 5 - 2i_2 = 5 - 2 \times 0.4545 = \underline{4.091 \text{ V}}$

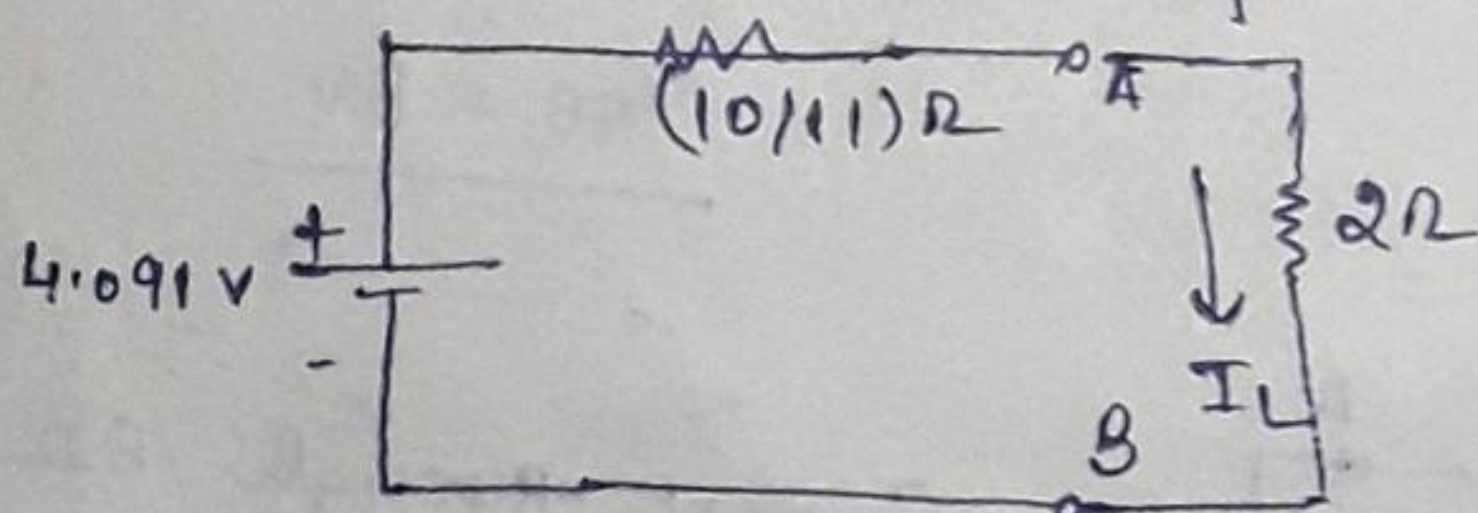
Calculation of R_{th}



$$R_{th} = [(2 \parallel 1) + 1] \parallel 2 = \left[\frac{2}{3} + 1 \right] \parallel 2 = \frac{5}{3} \parallel 2$$

$\therefore R_{th} = \frac{10}{11} \Omega$

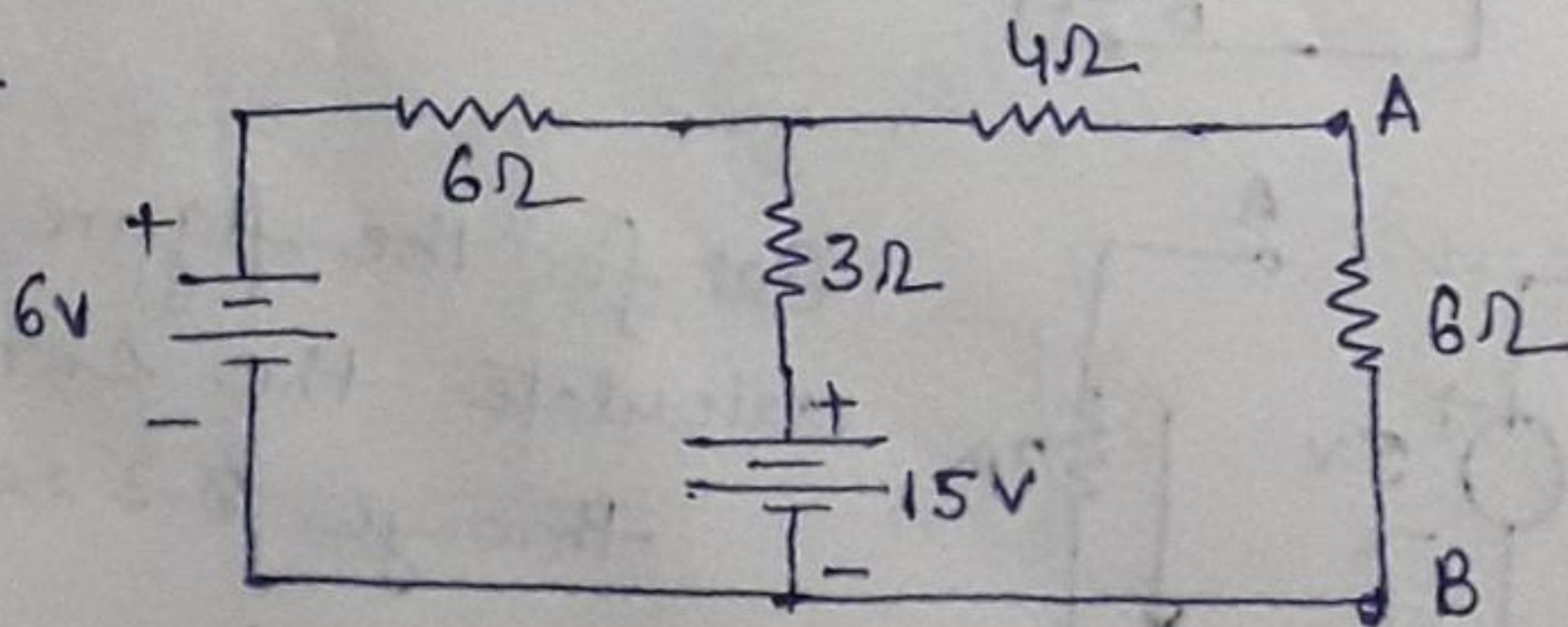
Equivalent circuit is as follows :-



$\therefore$ Current through 2Ω
 $= I_L = \frac{4.091}{\frac{10}{11} + 2}$

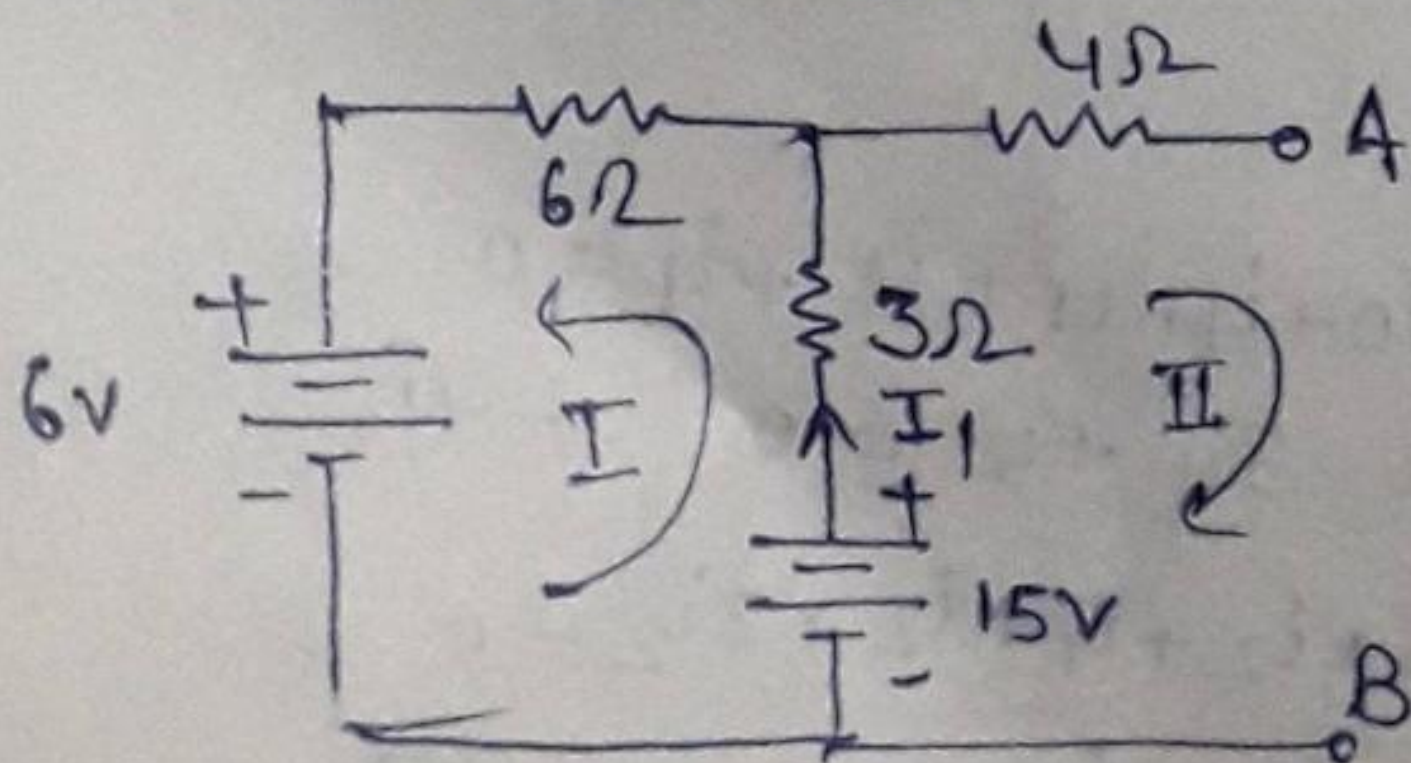
$I_L = 1.4063 \text{ A}$

Q7



For the circuit shown fig., calculate the current through resistance?

Calculation of V_{th}



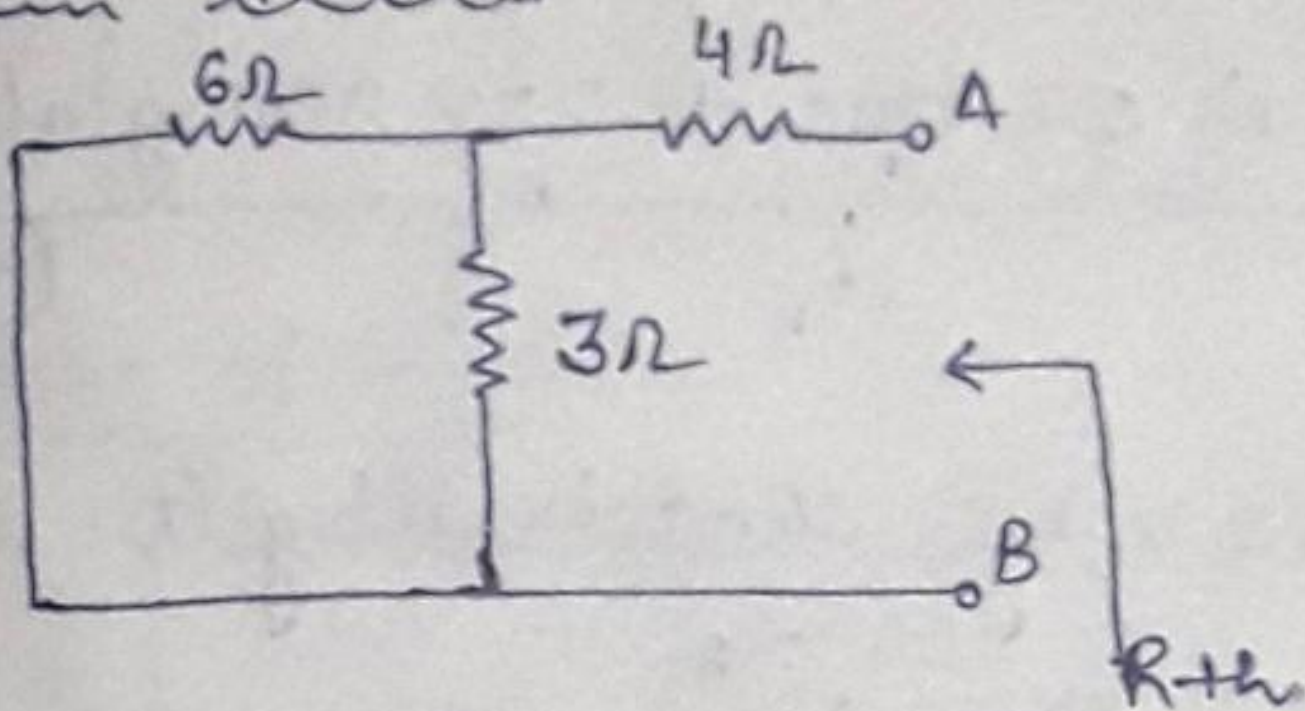
In mesh (I) apply KVL
 $-15 + 3I_1 + 6I_1 + 6 = 0$

$\therefore I_1 = 1 \text{ A}$

In mesh (II), apply KVL [Since AB terminals are open] ⁽²³⁾
 there is a virtual voltage source called V_{AB} .

So, $-15 + 3I_1 + V_{AB} = 0$ or $V_{AB} = 15 - 3I_1 = \underline{12V}$

Calculation of R_{th}

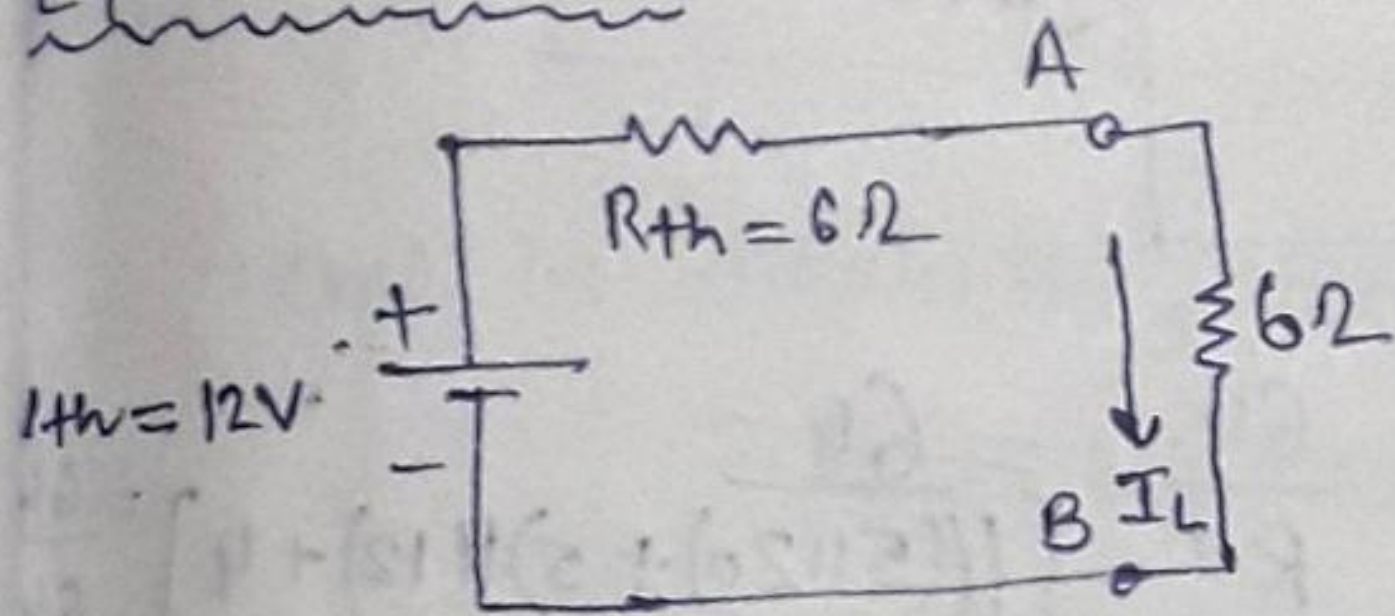


$$R_{th} = 4 + (6 \parallel 3)$$

$$= 4 + \frac{18}{9}$$

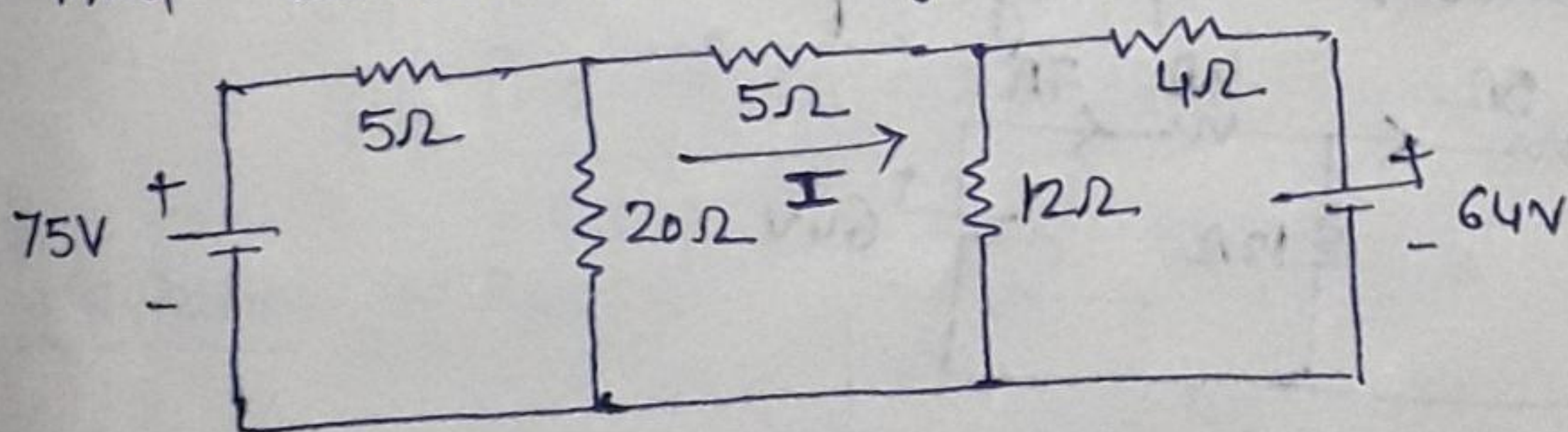
$$\underline{R_{th} = 6\Omega}$$

Equivalent circuit



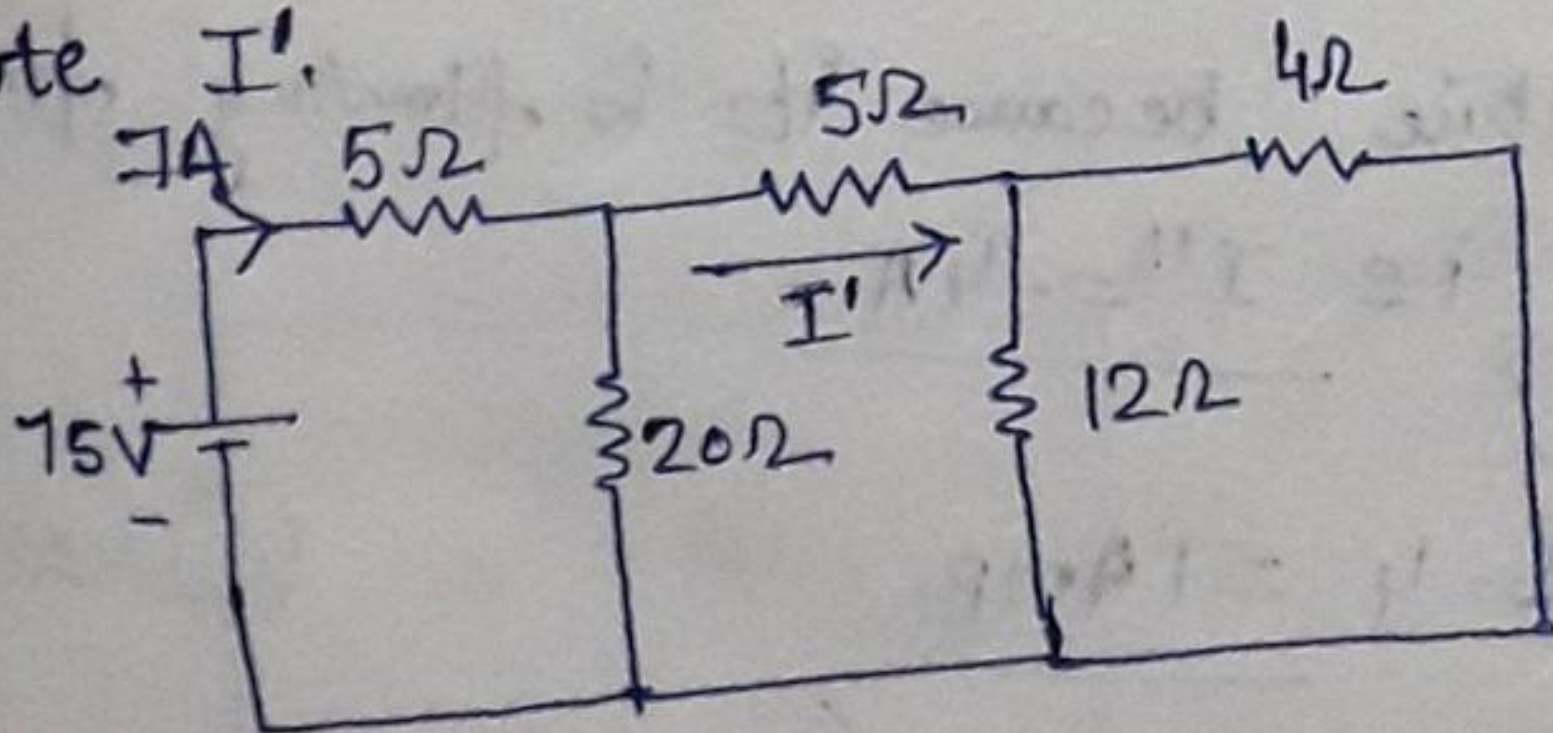
$$\therefore I_L = \frac{12}{6+6} = \underline{1A}$$

Q8 Find current I using superposition theorem?



1) Take 75V and ignore 64V by shortcircuiting it and

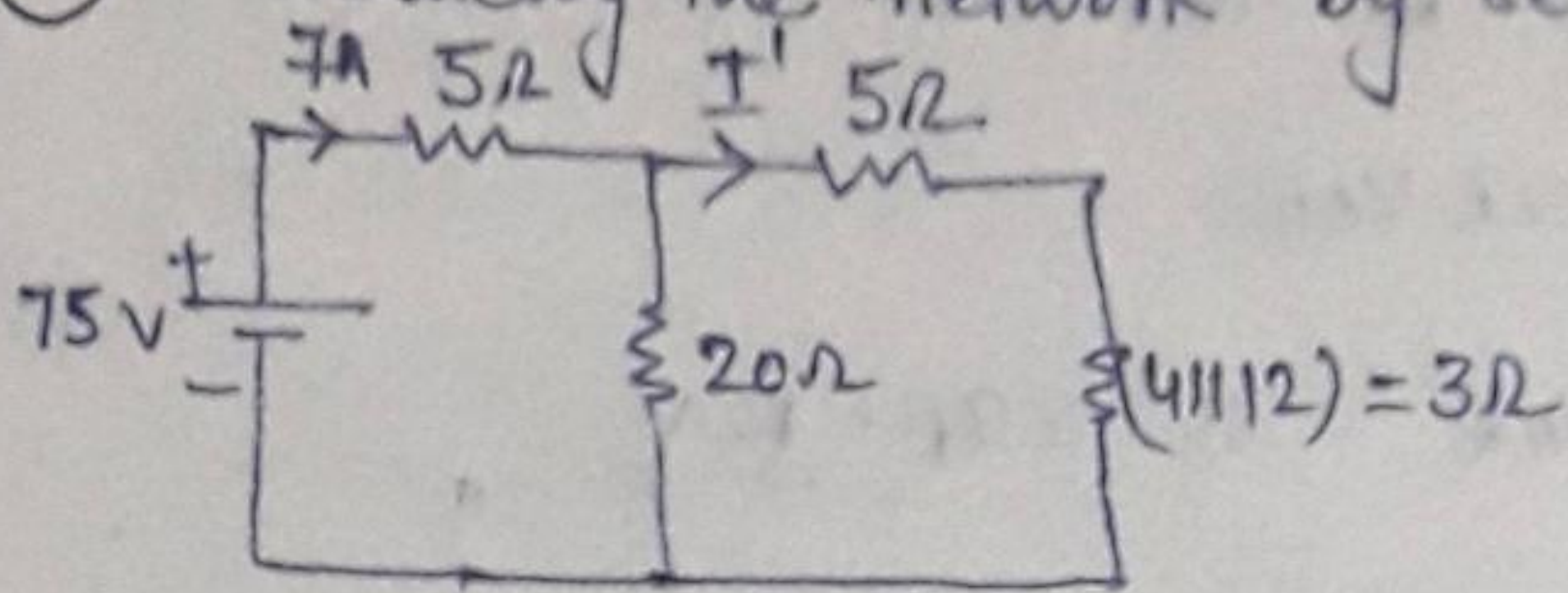
note I' .



Current supplied by battery, $I' = \frac{75}{R_1} = \frac{75}{[(4 \parallel 12) + 5] \parallel 20 + 5} = \frac{75}{75/7} = \underline{7A}$

Q9

Reducing the network by series parallel connection

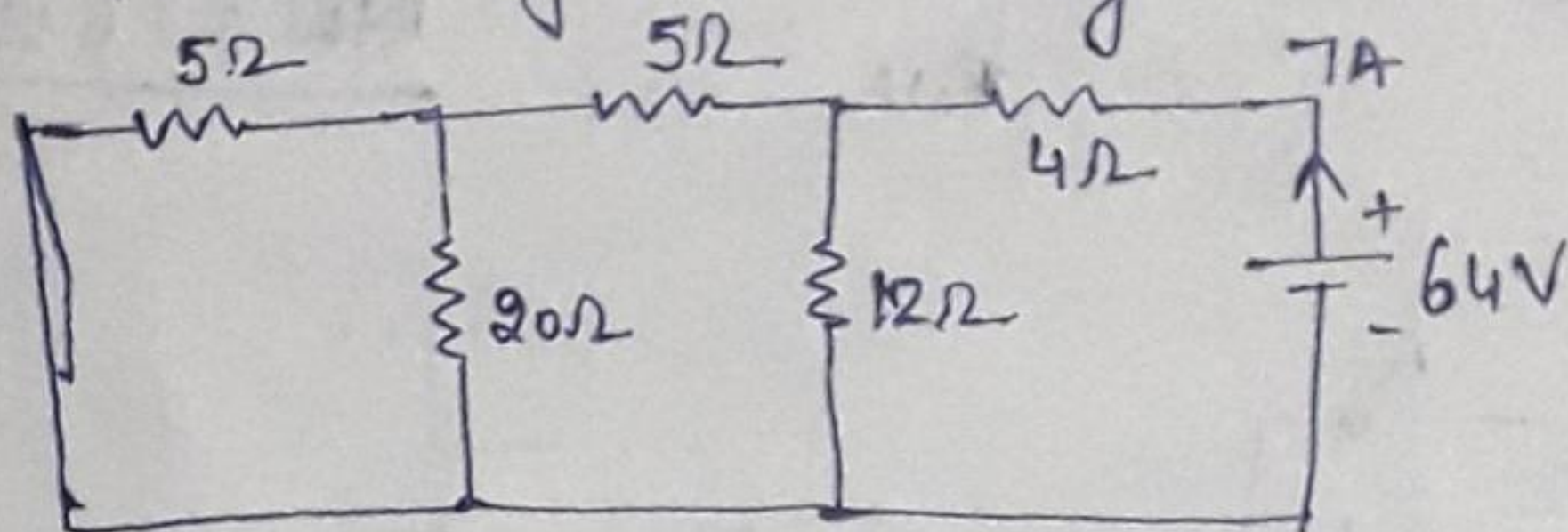


$$\therefore I' = \frac{7 \times 20}{20 + 8} = 5A$$

$\therefore$ Current in 5Ω due to $75V$ acting alone =

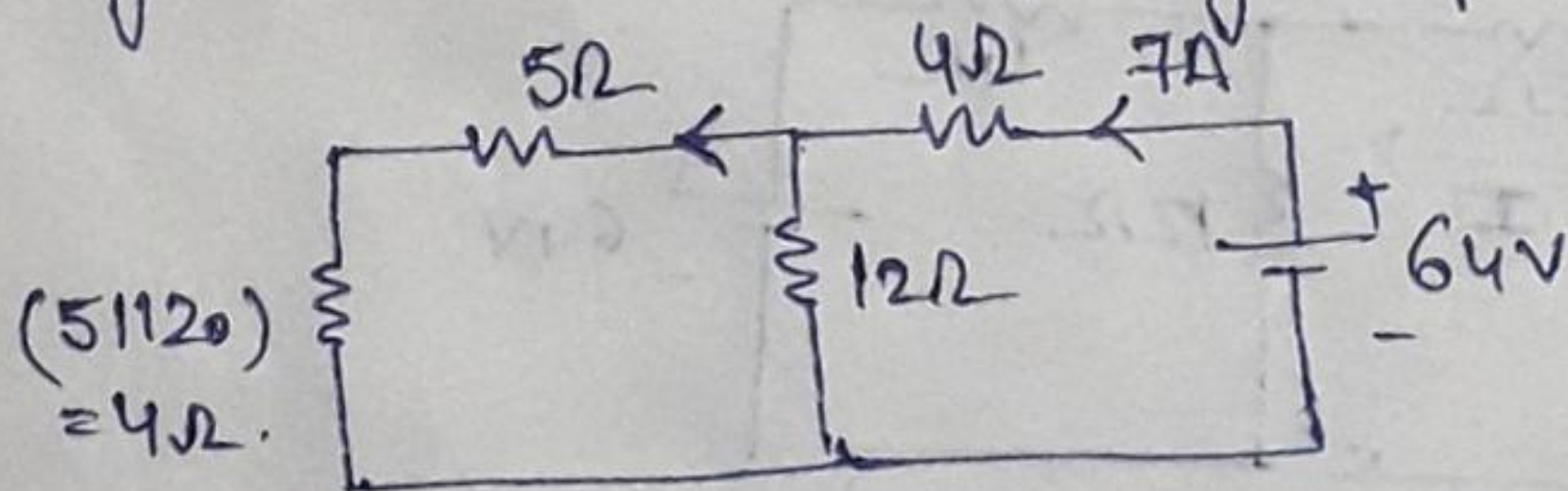
(From Left to Right)

ii) Take $64V$ and ignore $75V$ by shortcircuiting it.



$$\text{Current supplied by battery} = \frac{64}{R''} = \frac{64}{[(5 \parallel 20) + 5] \parallel 12 + 4}$$

Reducing the circuit step by step



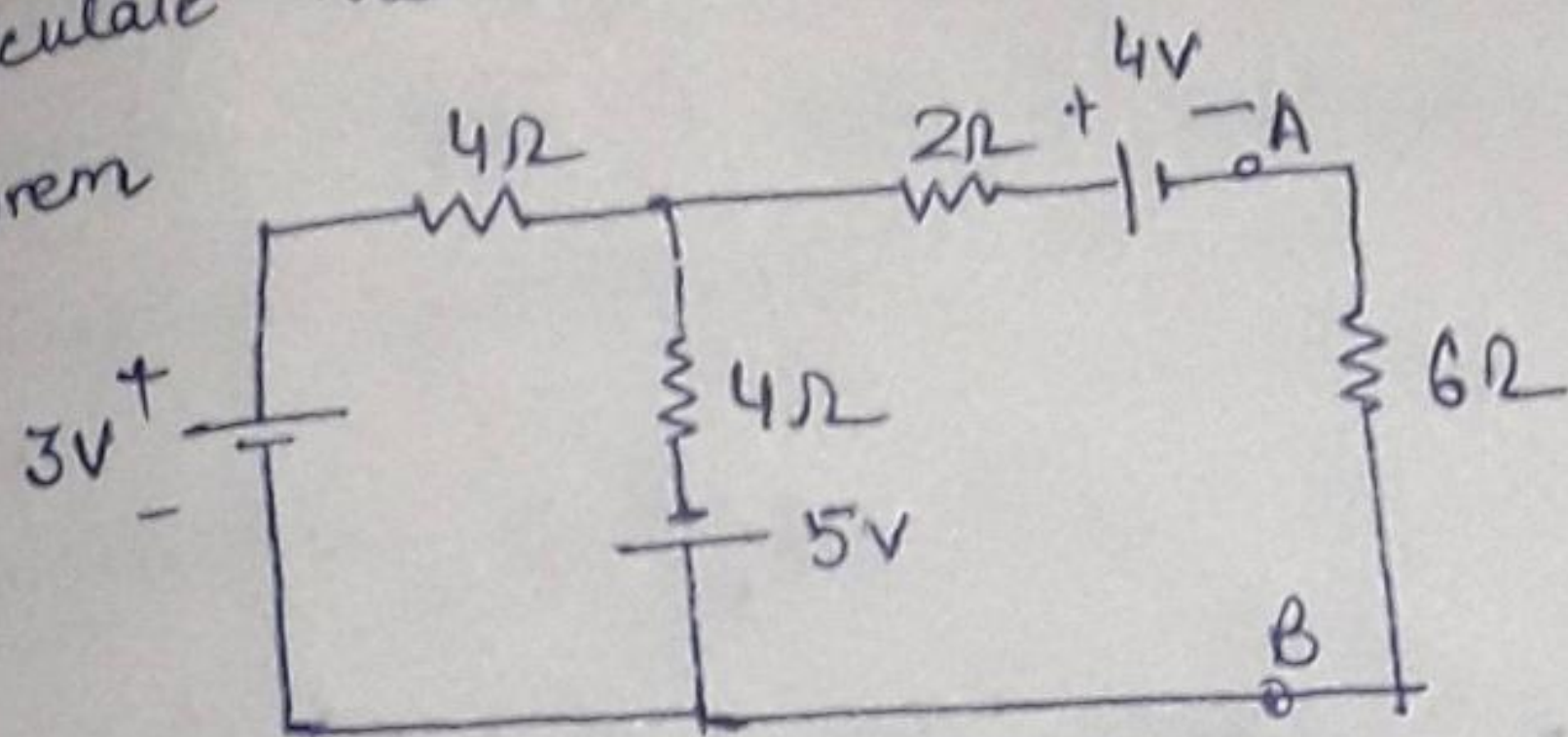
$$\text{Current through } 5\Omega \text{ resistor} = 7 \times \frac{12}{12 + 9} = 4A \text{ and}$$

taken as negative because it is flowing from right to left. i.e. $I'' = -4A$.

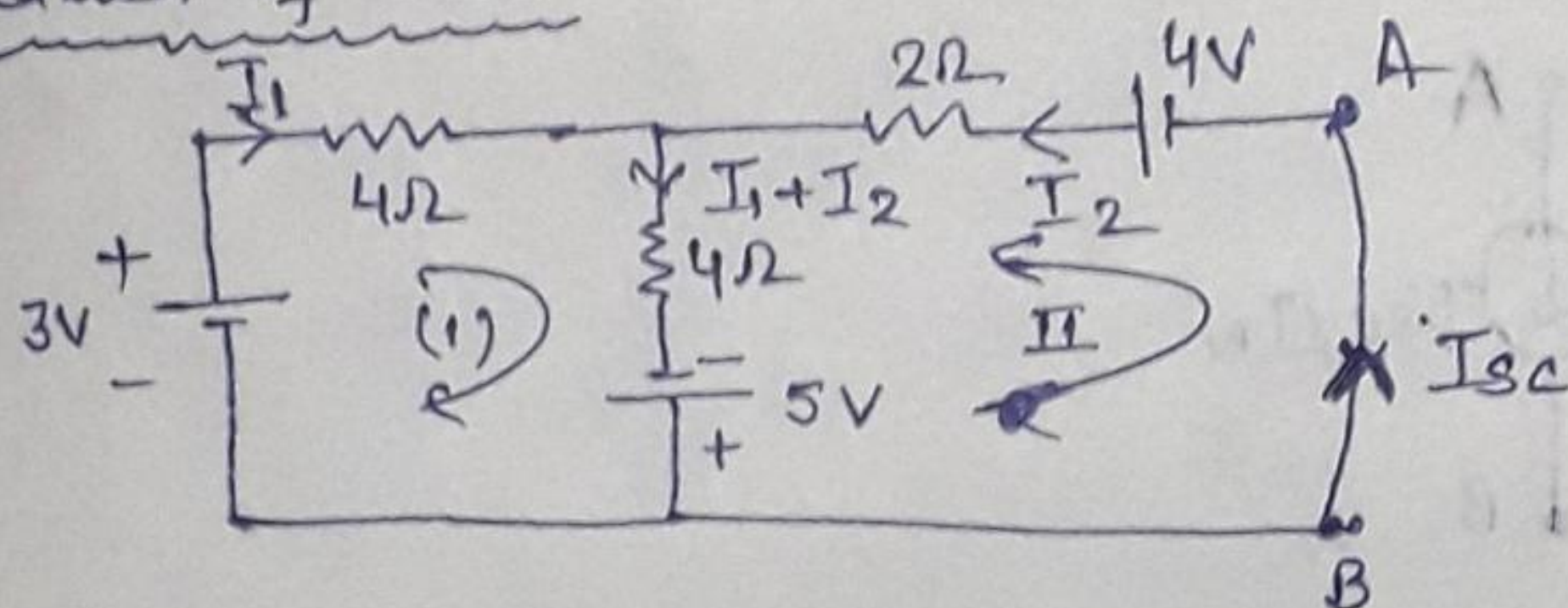
$$\text{Then } I = I' - I'' = 5 - 4 = 1Amp.$$

Find the Norton's equivalent circuit at AB and calculate the current in 6Ω resistance by using Norton's theorem

(25)



Calculation of I_{sc}



In mesh (I), apply KVL

$$4I_1 + 4I_1 + 4I_2 - 5 - 3 = 0$$

$$\text{or } 2I_1 + I_2 = 2 \quad \text{--- (1)}$$

And in mesh (II), apply KVL

$$2I_2 + 4(I_1 + I_2) - 5 - 4 = 0$$

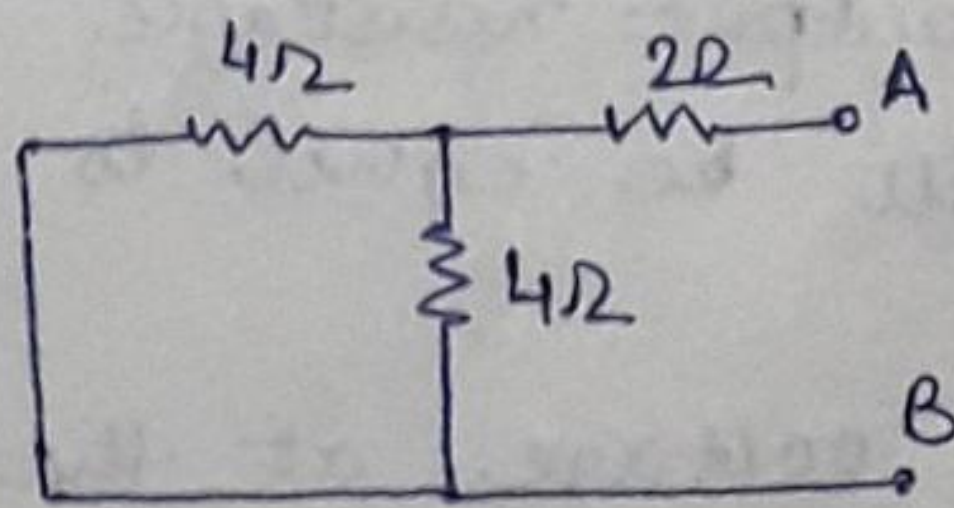
$$2I_1 + 3I_2 = 4.5 \quad \text{--- (2)}$$

From (1) and (II), we have

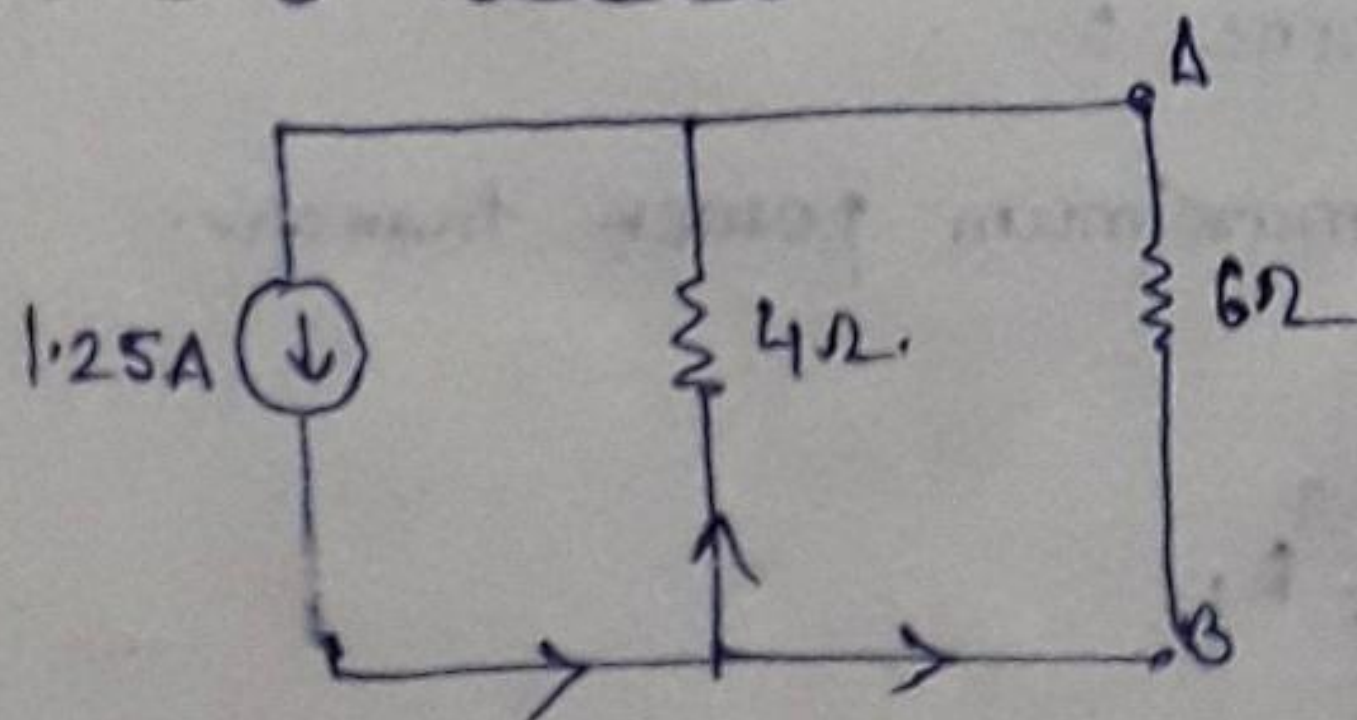
$$\boxed{I_2 = 1.25A = I_{sc}}$$

Calculation of R_N

$$R_N = 2 + (4 \parallel 4) = 4\Omega$$



Equivalent circuit

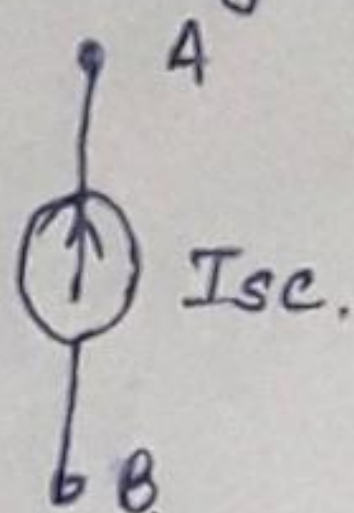


$$I_L = I_{sc} \times \frac{R_N}{R_N + R_L} = 1.25 \times \frac{4}{4 + 6} = 0.5A$$

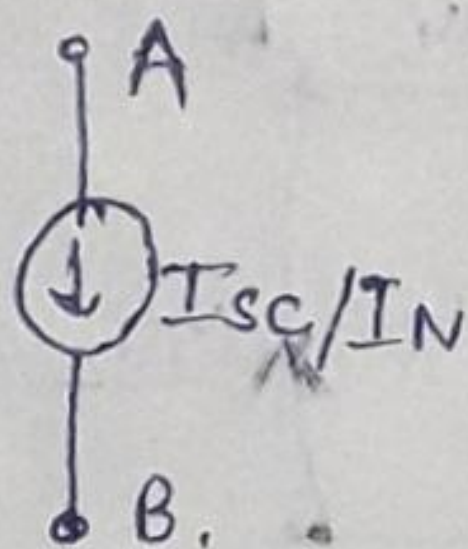
(from B to A)

(26)

Note :- How to decide the direction of morton's current source
if in the circuit diagram, direction of current is from A to B,
then we have



and if in circuit diagram current is flowing from B to A, then the
direction of current is ~~from~~ is I_N is



MAXIMUM POWER TRANSFER THEOREM

In D.C circuits

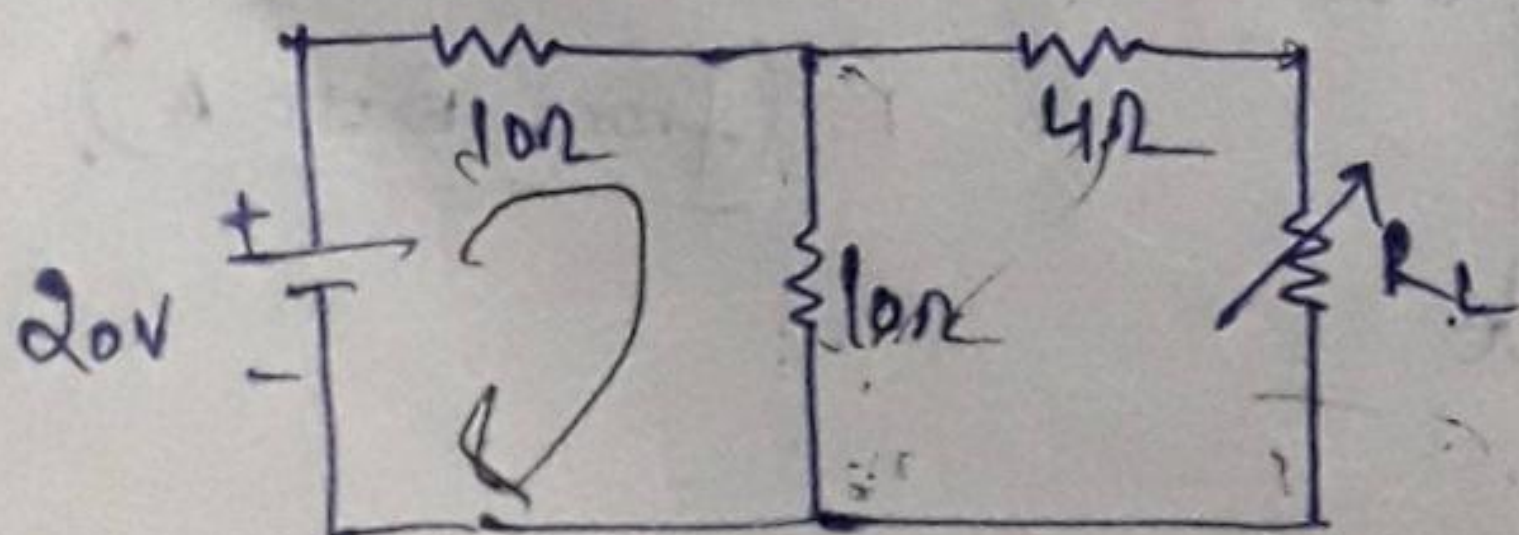
A resistive load will abstract maximum power when
the load resistance value is same as the resistance
viewed by the load as it looks back into the network.

This is called output resistance of network. The maximum
power drawn will be equal to $\frac{V_{oc}^2}{4R_L}$ where V_{oc} is

the open circuit voltage at the terminals from which
has been disconnected.

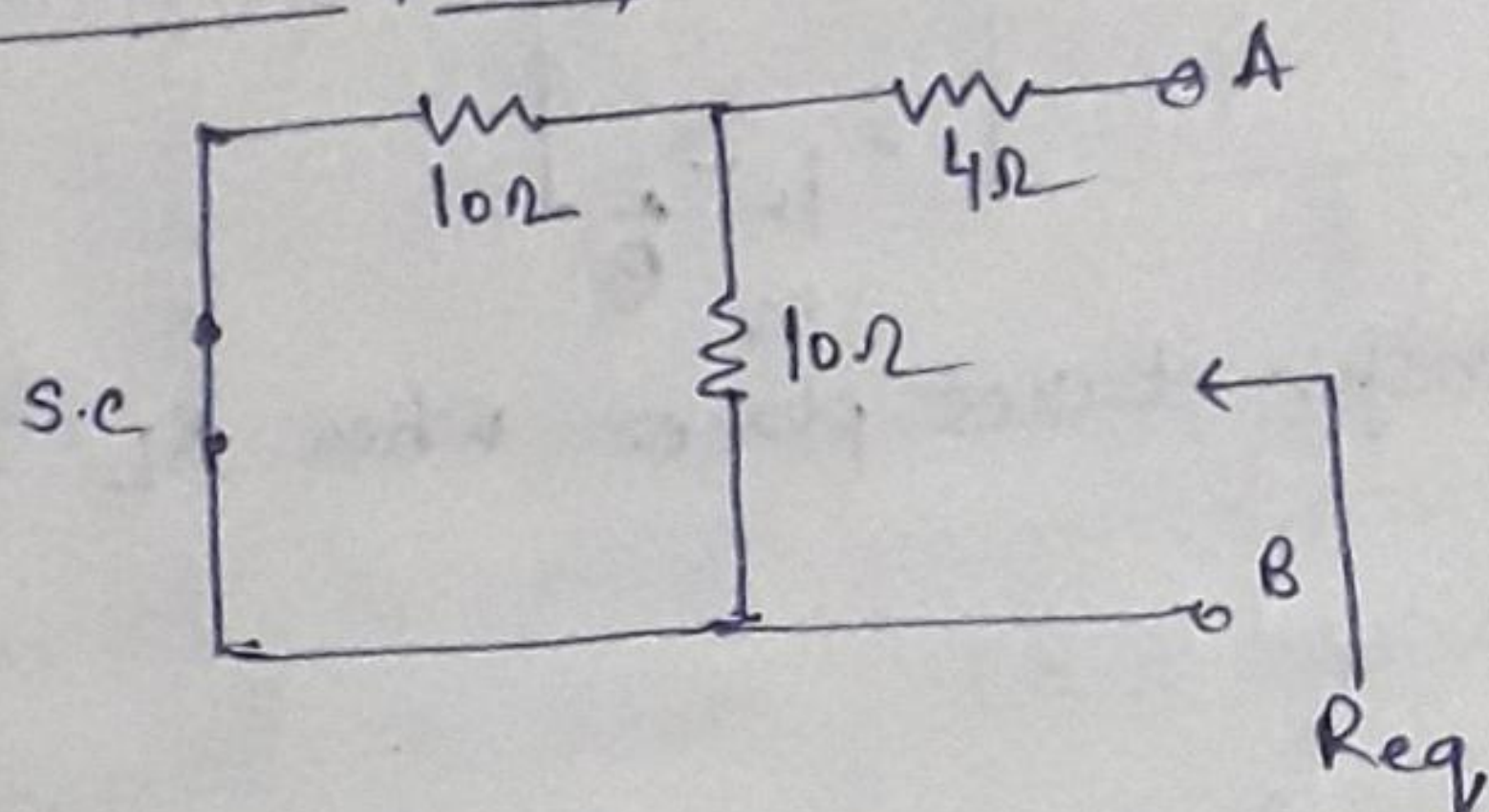
Q1:- In the network, determine :-

(i) The value of R_L to give maximum power transfer.



(27)
 First the load is disconnected ~~and then~~. to calculate the resistance viewed by load as it looks back into the network. This is called output resistance of network or equivalent resistance.

Calculation of R_{eq}



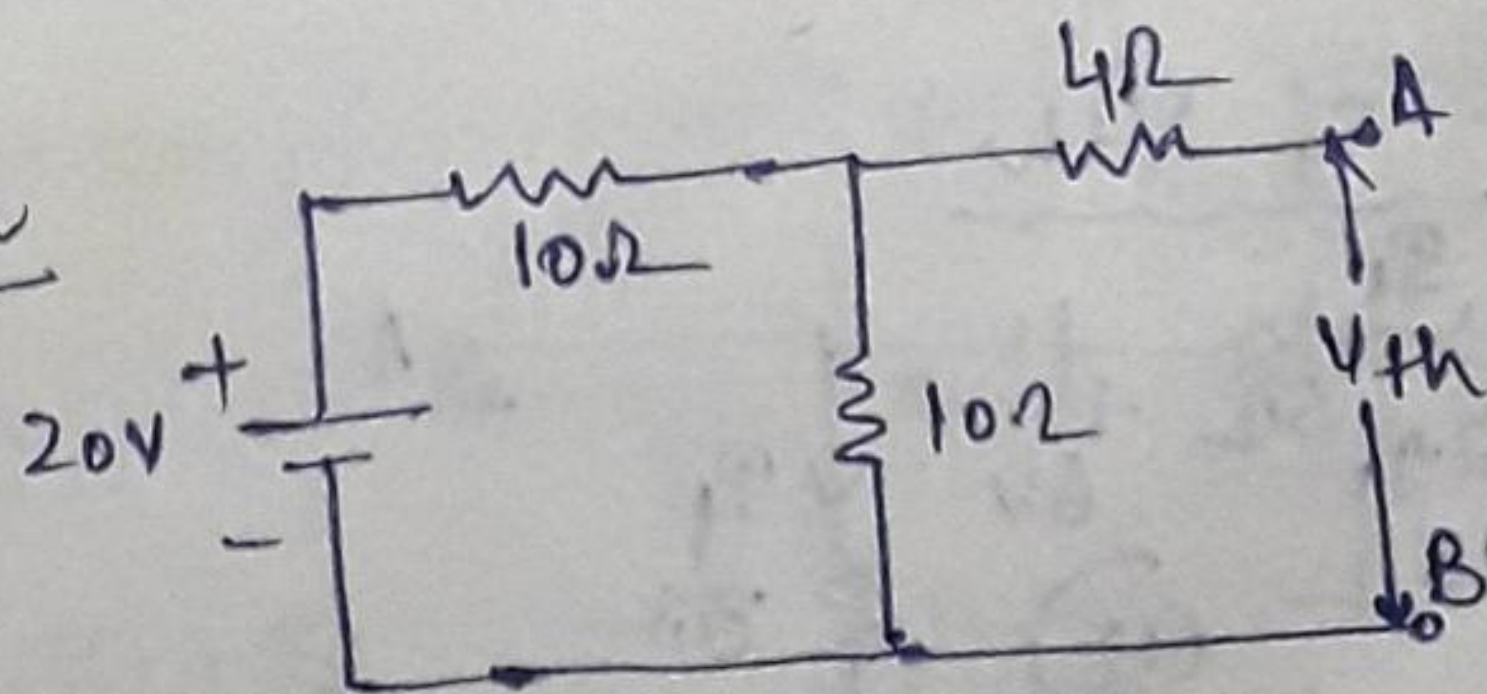
$$R_{eq} = 4 + (10 \parallel 10) \\ = 4 + 5 = \underline{9\Omega}$$

Maximum power will be transferred to R_L when $R_L = R_{eq}$

$$\therefore \boxed{R_L = R_{eq} = 9\Omega}$$

$$(ii) P_{max} = \frac{V_{oc}^2}{4R_L} = \frac{V_{th}^2}{4R_L}$$

Calculation of V_{th}

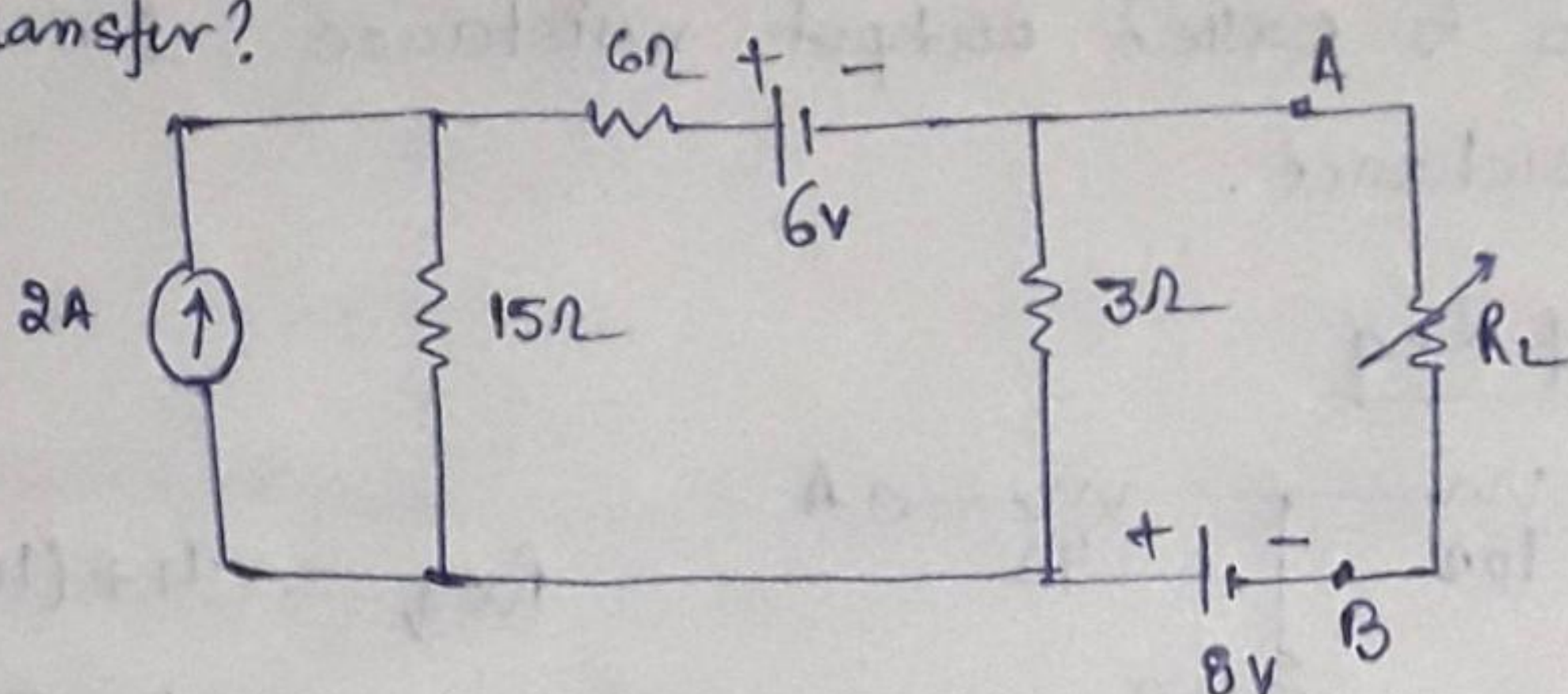


$$\text{Here } V_{AB} = V_{th} = 10 \times I = 10 \times \frac{20}{20} = \underline{10V}$$

$$\therefore P_{max} = \frac{10 \times 10}{4 \times 9} = \frac{100}{4 \times 9} = \frac{25}{9} W = \underline{2.778 W}$$

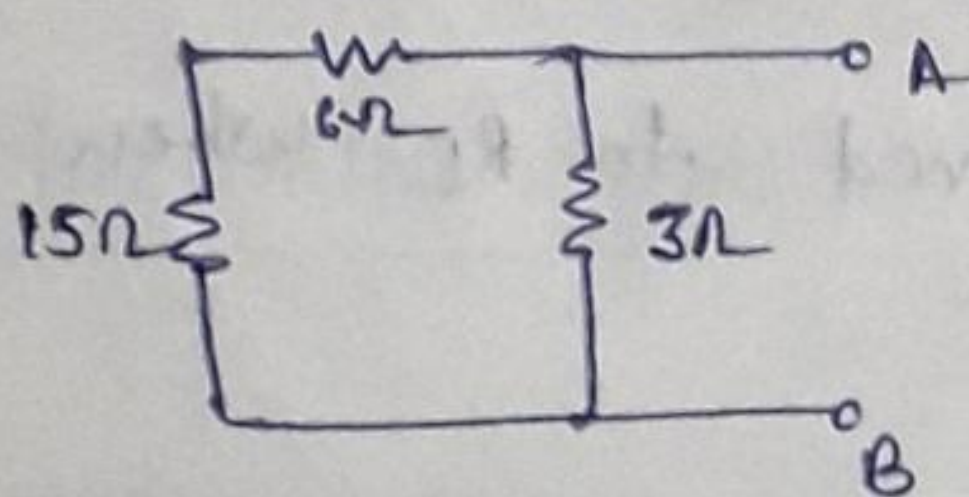
28)

Q2. For the network shown, find the value of R_L to give maximum power transfer and hence find the maximum power transfer?



Maximum power transfer takes place when R_L is eq. to R_{eq} .

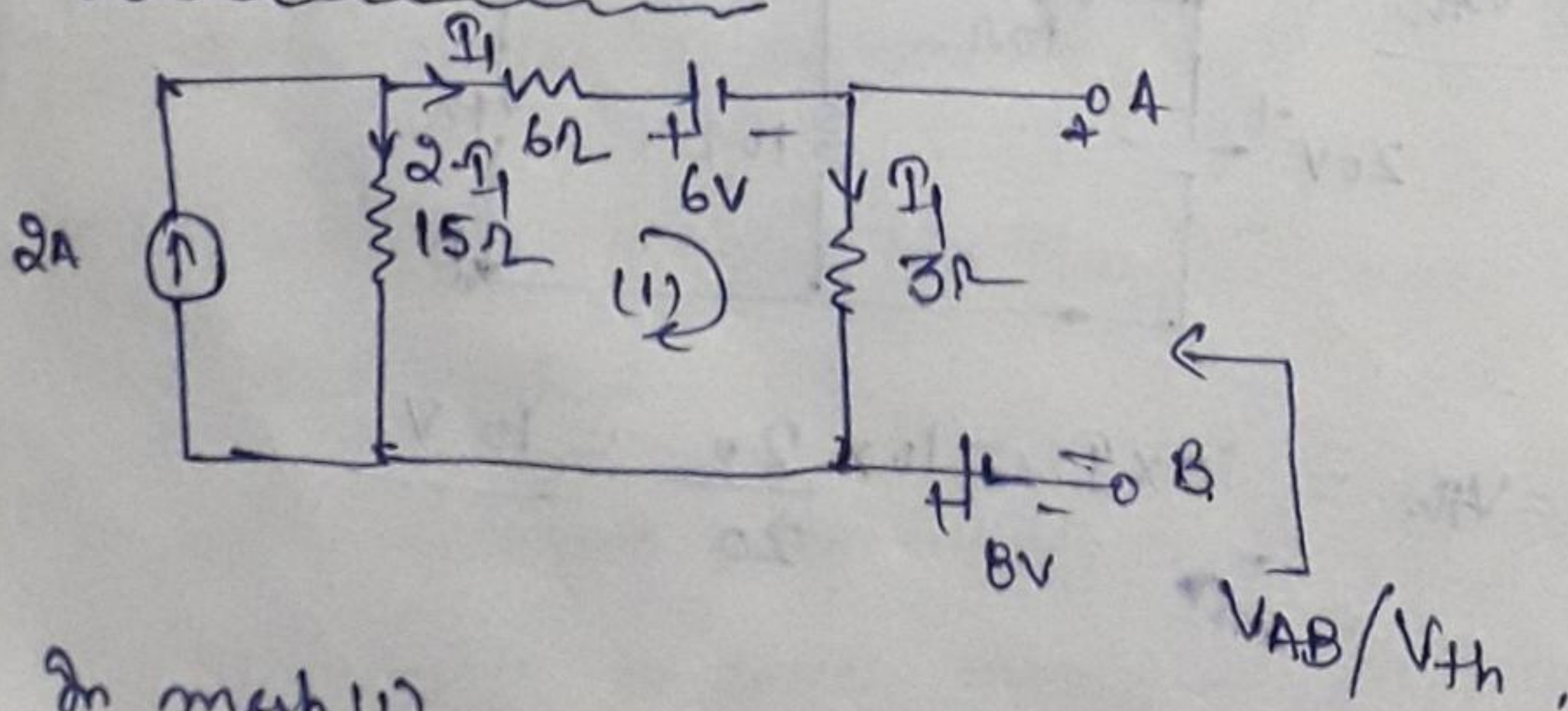
Calculation of R_{eq}



$$\begin{aligned} \therefore R_{eq} &= 3 \parallel (6 + 15) = 3 \parallel 21 \\ &= \frac{3 \times 21}{24} \\ \underline{R_{eq} &= \frac{21}{8} \Omega} \end{aligned}$$

Maximum power, $P_{max} = \frac{V_{oc}^2}{4R_L} = \frac{V_{th}^2}{4R_L}$

Calculation of V_{th}



In mesh (1)

$$\therefore 6I_1 + 6 + 3I_1 - 15(2 - I_1) = 0$$

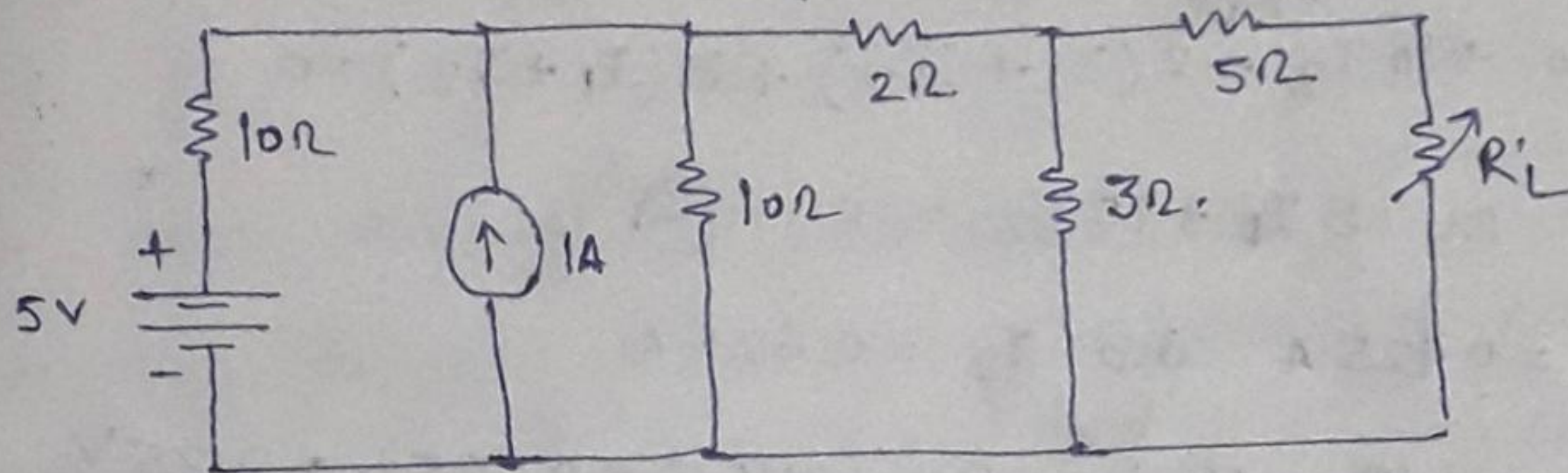
$$\begin{aligned} 24I_1 &= 24 \quad \therefore I_1 = 1A \quad \therefore V_{AB} = 3I_1 + 8 \\ \underline{V_{AB} &= 11V} \end{aligned}$$

$$\therefore P_{\max} = \frac{11 \times 11}{4 \times 21} \times 8 = 11.524 \text{ W}$$

(40)

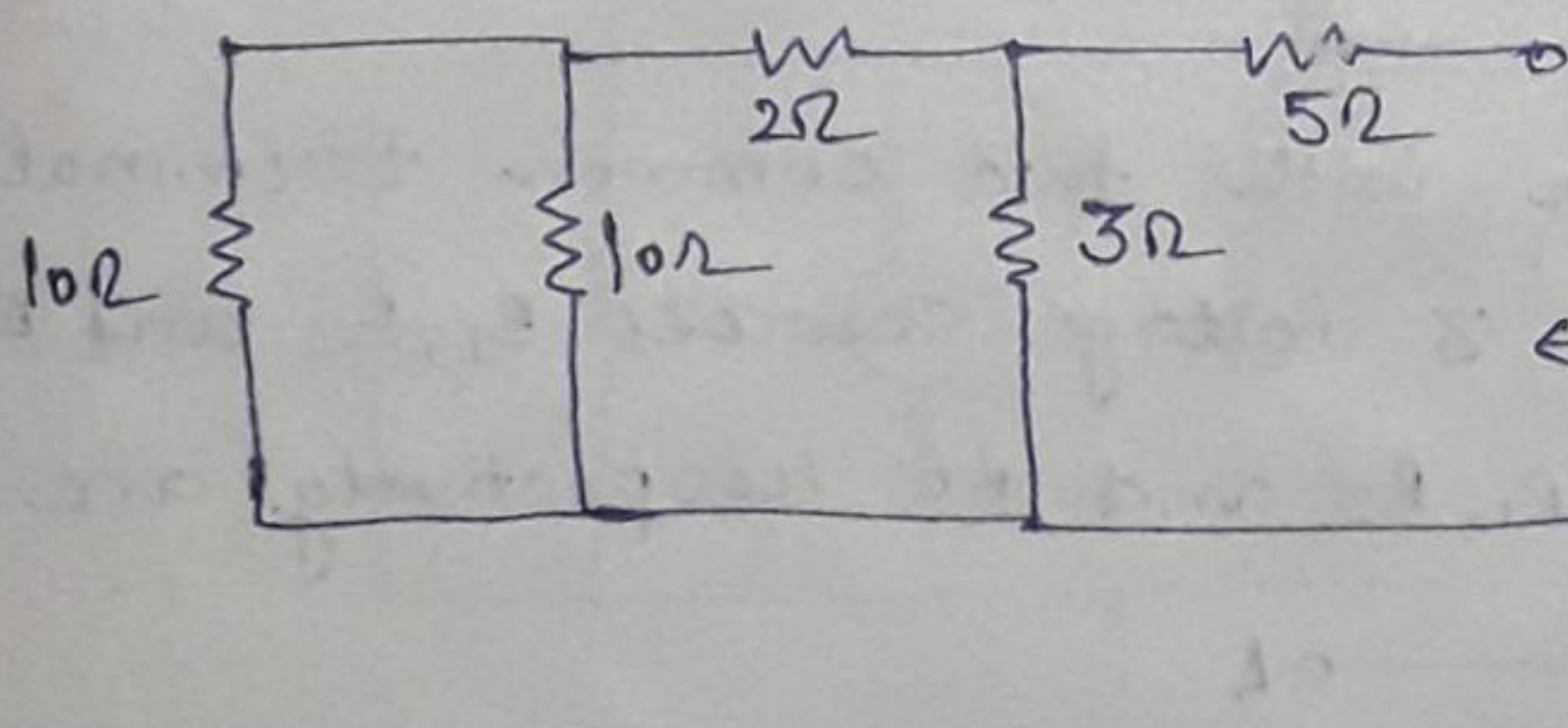
(20)

Q3. Find the condition for maximum power transfer. Hence determine the maximum power transferred?



According to max^m power transfer theorem, R_L must be equal to looking back resistance, R_{eq} .

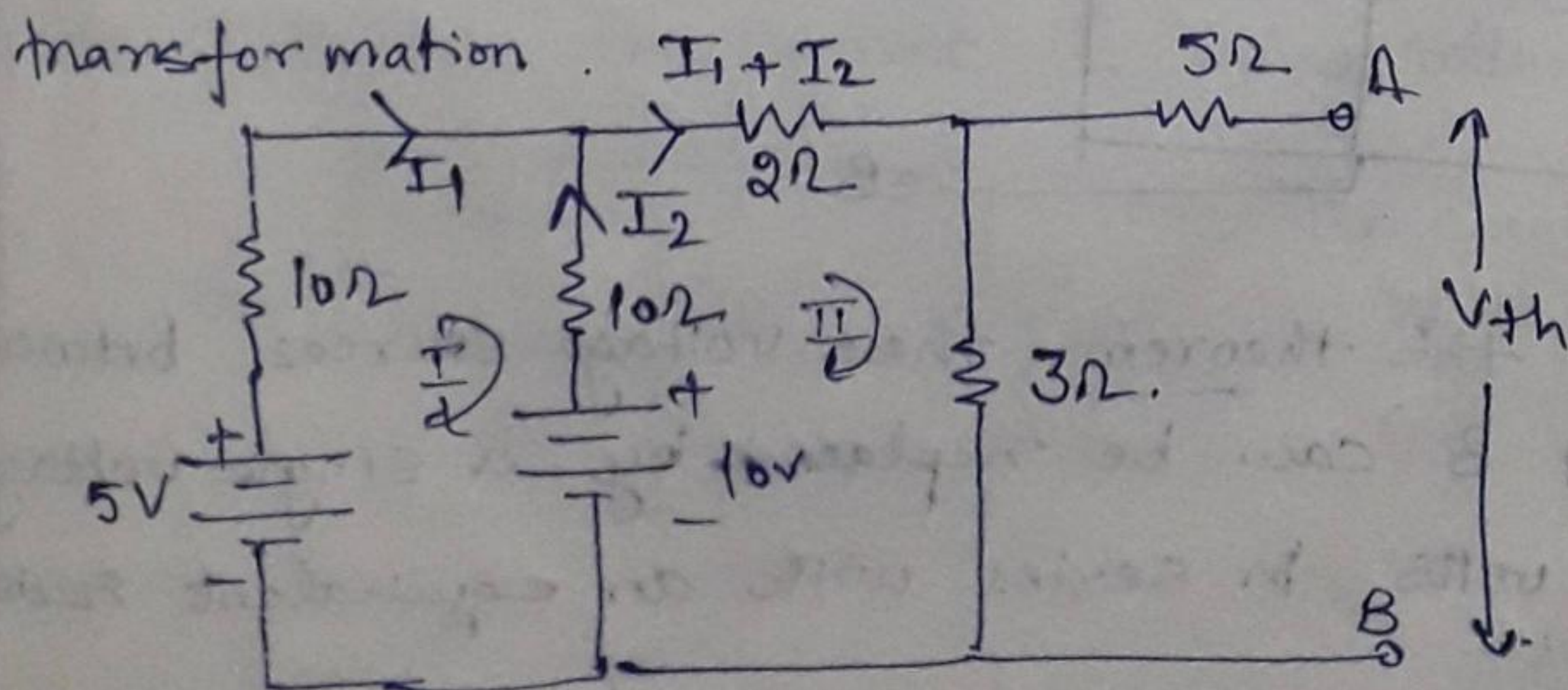
Calculation of R_{eq} (Looking back resistance)



$$\begin{aligned} R_{eq} &= 5 + [3 \parallel (2 + (10 \parallel 10))] \\ &= 5 + [3 \parallel 7] \\ &= 5 + \frac{21}{10} = 7.1 \Omega \end{aligned}$$

$\therefore R_L = 7.1 \Omega$ condition for max^m power Transfer.

Calculation of V_{th} :- Remove load and apply source transformation



Q3) In mesh (I), apply KVL,

$$-5 + 10I_1 - 10I_2 + 10 = 0$$

$$\text{or } 10I_1 - 10I_2 = -5 \quad \text{--- (1)}$$

In mesh (II), apply KVL

$$-10 + 10I_2 + 2(I_1 + I_2) + 3(I_1 + I_2) = 0$$

$$\text{or } 5I_1 + 15I_2 = 10 \quad \text{--- (2)}$$

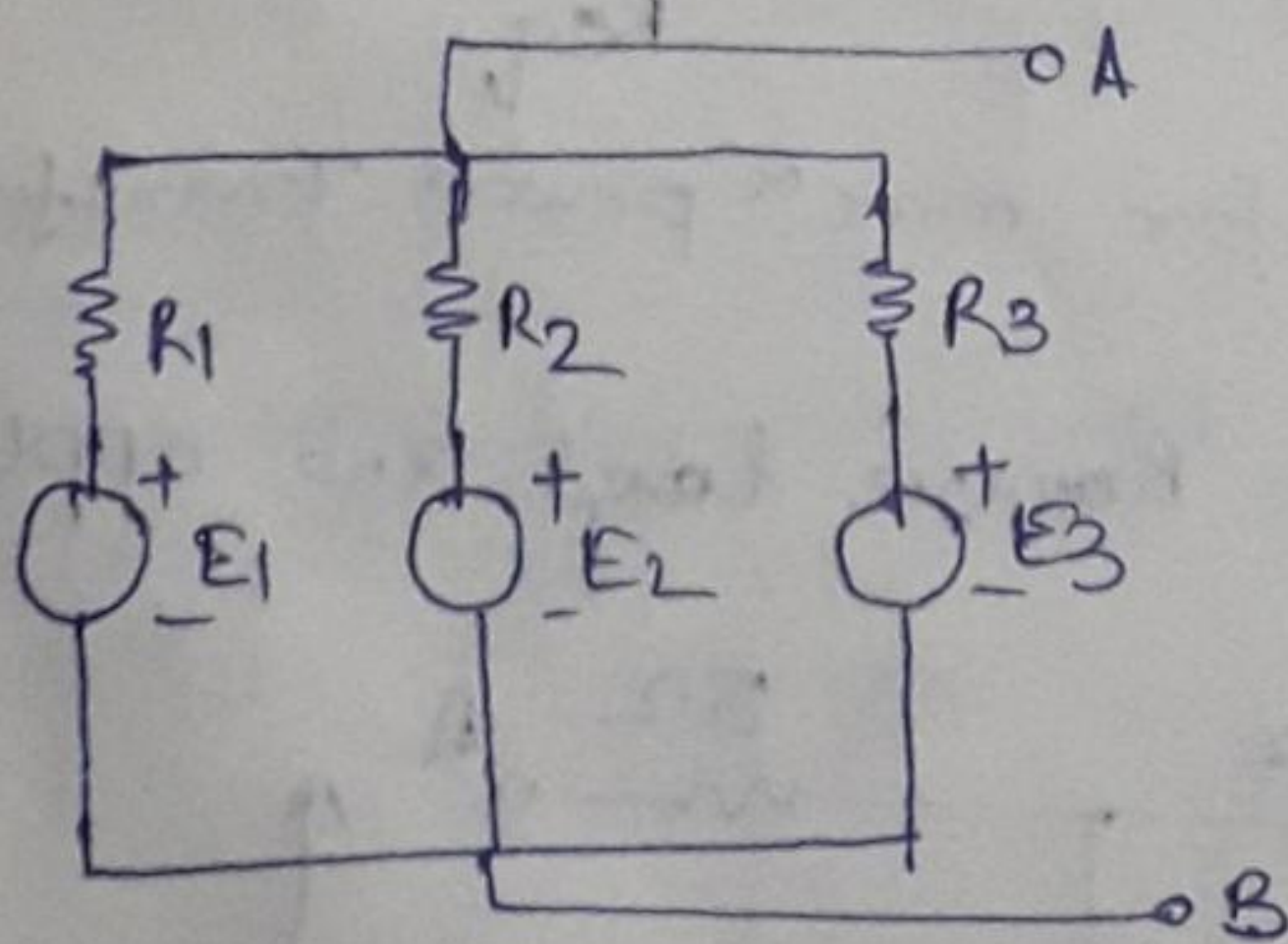
$$\therefore I_1 = 0.125 \text{ A and } I_2 = 0.625 \text{ A}$$

$$\therefore V_{th} = 3(I_1 + I_2) = 3 \times (0.125 + 0.625) = \underline{2.25 \text{ V}}$$

$$\text{and } P_{max} = \frac{V_{th}^2}{4R_L} = \frac{2.25 \times 2.25}{4 \times 7.1} = \underline{0.178 \text{ W}}$$

MILLMAN'S THEOREM

Consider a network shown with two common terminals A and B between which 3 voltage sources E_1, E_2 and E_3 of internal resistances R_1, R_2 and R_3 respectively are connected.



According to this theorem, these voltage sources between terminals A and B can be replaced by a single voltage source of V_{AB} volts in series with an equivalent resistance R_{AB} where

$$V_{AB} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} = \frac{E_1 G_1 + E_2 G_2 + E_3 G_3}{G_1 + G_2 + G_3}$$

$$V_{AB} = \frac{\sum EG}{\sum G}$$

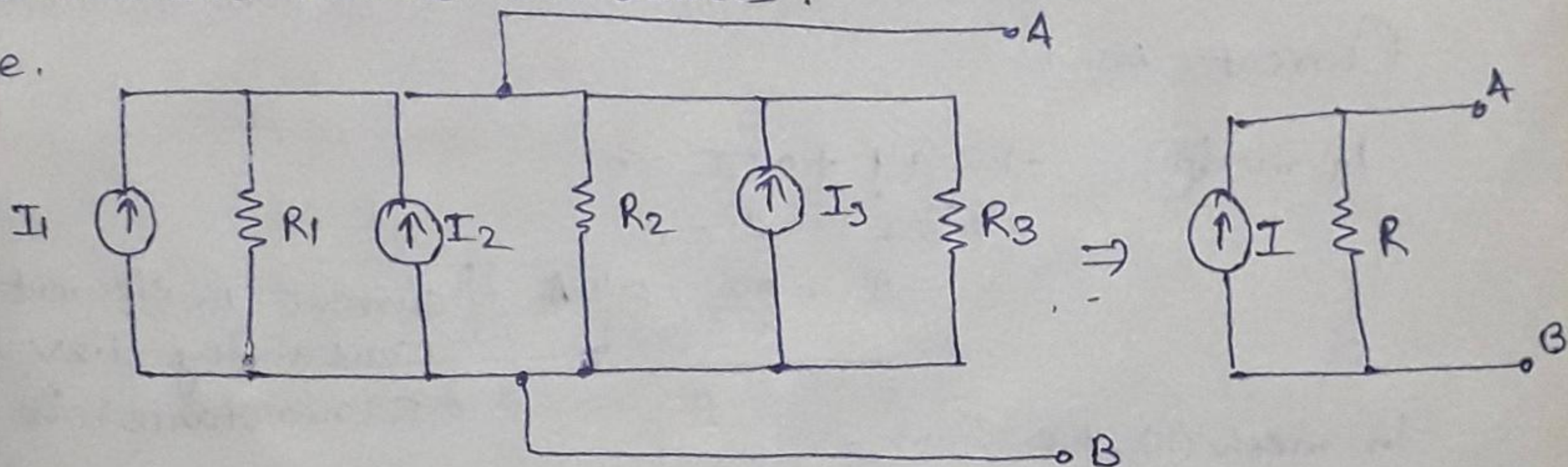
$$\text{and } R_{AB} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \quad (3)$$

$$\Rightarrow R_{AB} = \frac{1}{G_1 + G_2 + G_3}$$

$$R_{AB} = \frac{1}{\sum G}$$

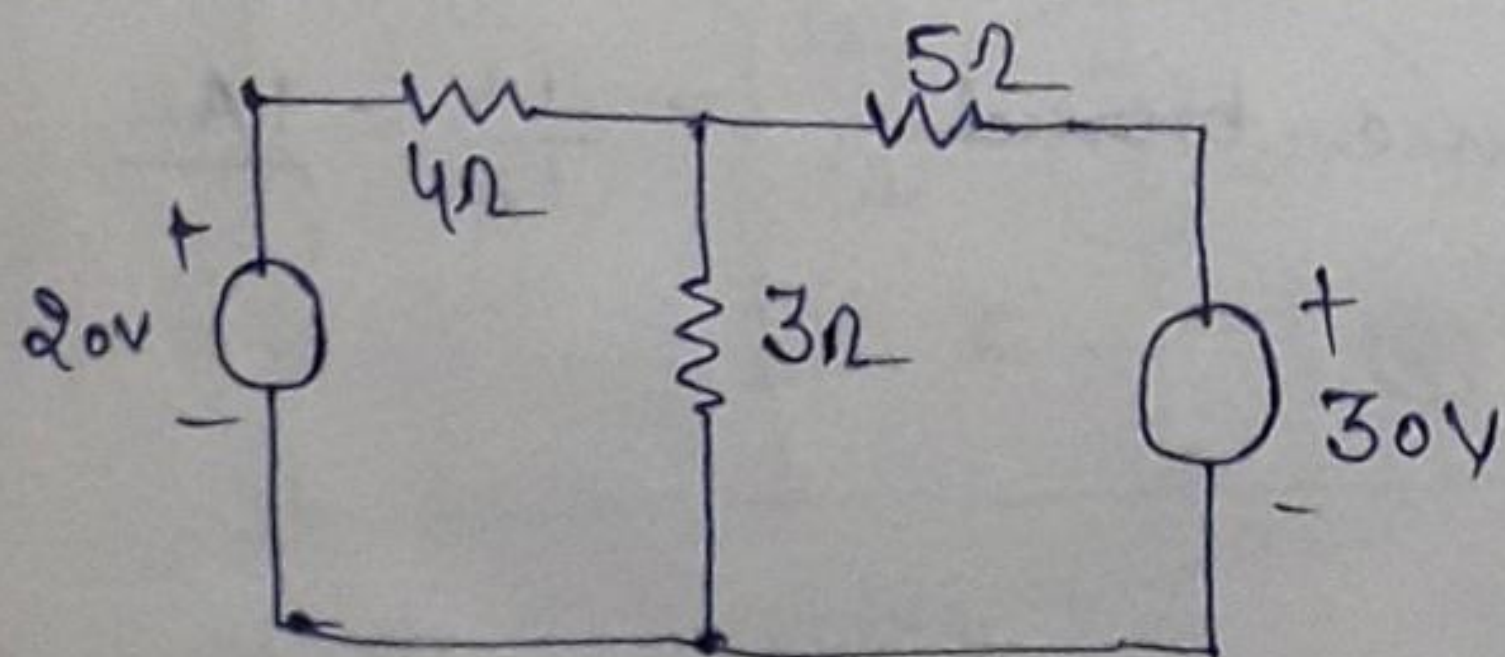
The above theorem is also applicable to current sources. According to this theorem any number of current sources in parallel may be replaced by a single current source, whose current is the algebraic sum of individual source currents and source resistance is the parallel combination of individual source resistances.

i.e.



$$\text{where } I = I_1 + I_2 + I_3 \text{ and } R_{AB} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Q1:- Determine the current I using millman's theorem —



The common terminal voltage

$$V = \frac{\frac{20}{4} + \frac{0}{3} + \frac{30}{5}}{\frac{1}{4} + \frac{1}{3} + \frac{1}{5}}$$

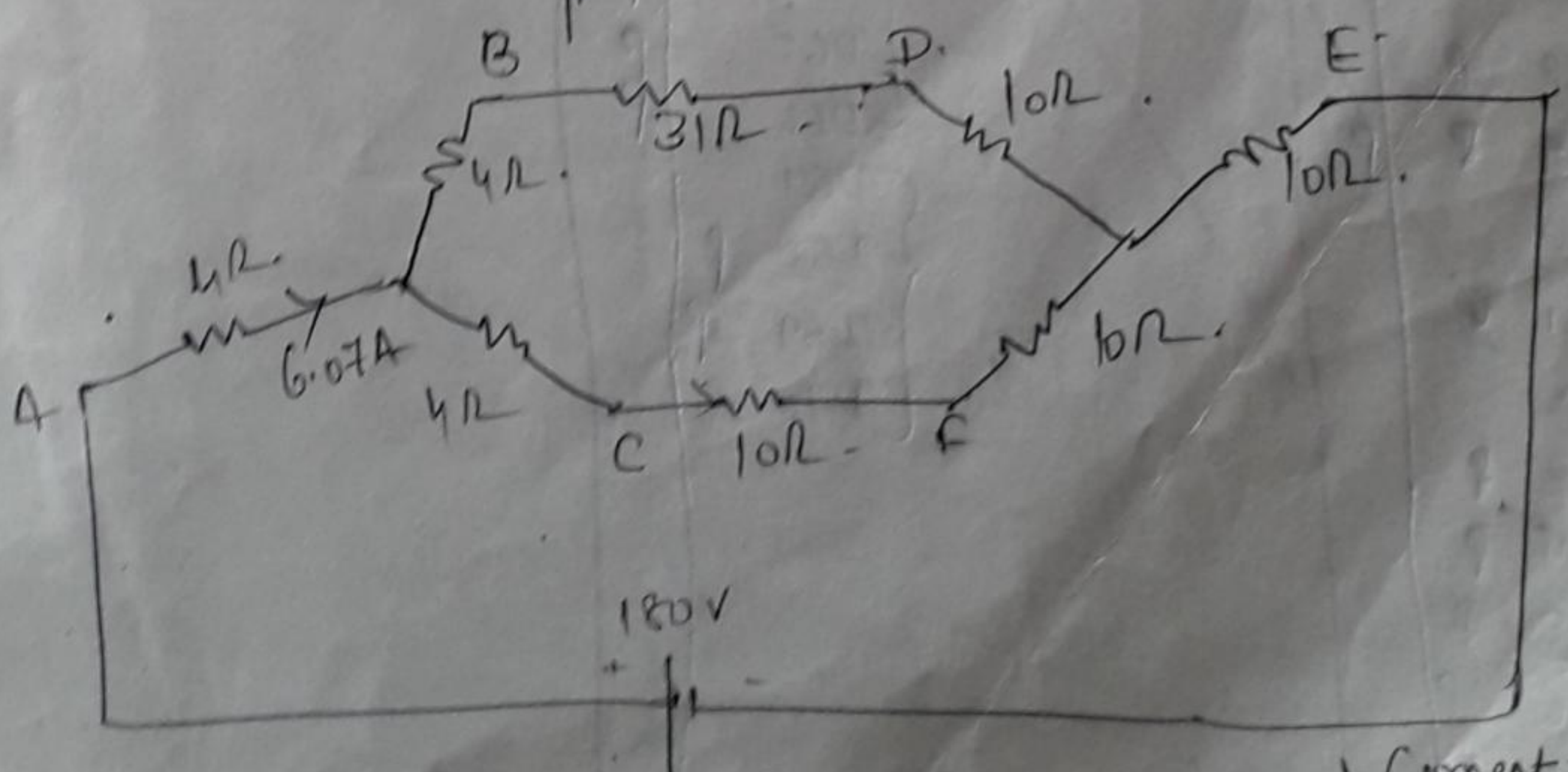
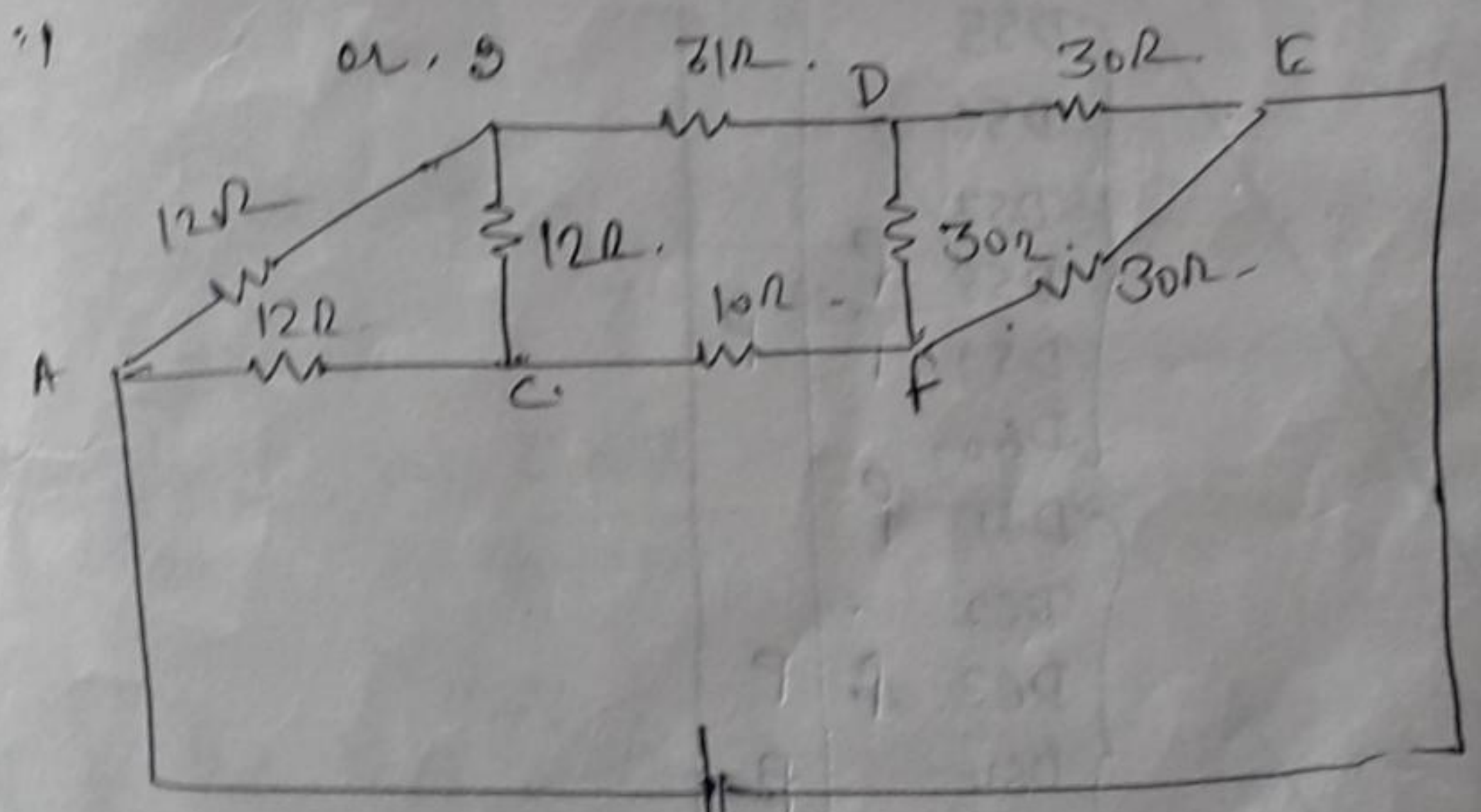
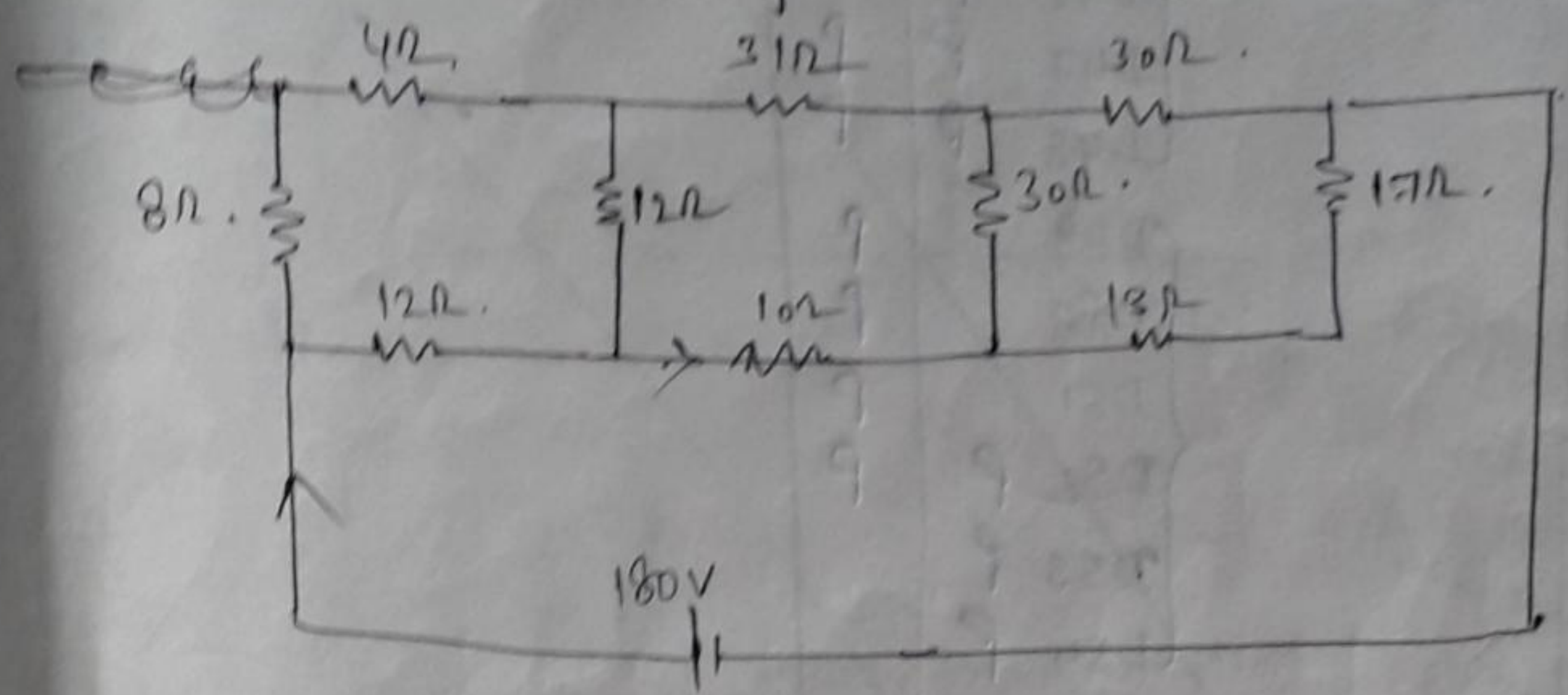
$$V = \frac{660}{47} \text{ V}$$

$$\text{Current in } 3\Omega \text{ resistor} = \frac{V}{3} = \frac{220}{47} \text{ A}$$

Current through 5Ω , $I_5 = \frac{V_A - V_C}{5} = 1.65 \text{ A}$.

Current through 4Ω , $I_6 = \frac{V_C}{4} = 0.95 \text{ A}$.

H.W
~~Q5~~ Find the current in 10Ω resistor in the n/w by star/delta transformation.



$R = 29.65\Omega$ and $I = \frac{180}{29.65} \approx 6.07 \text{ A}$

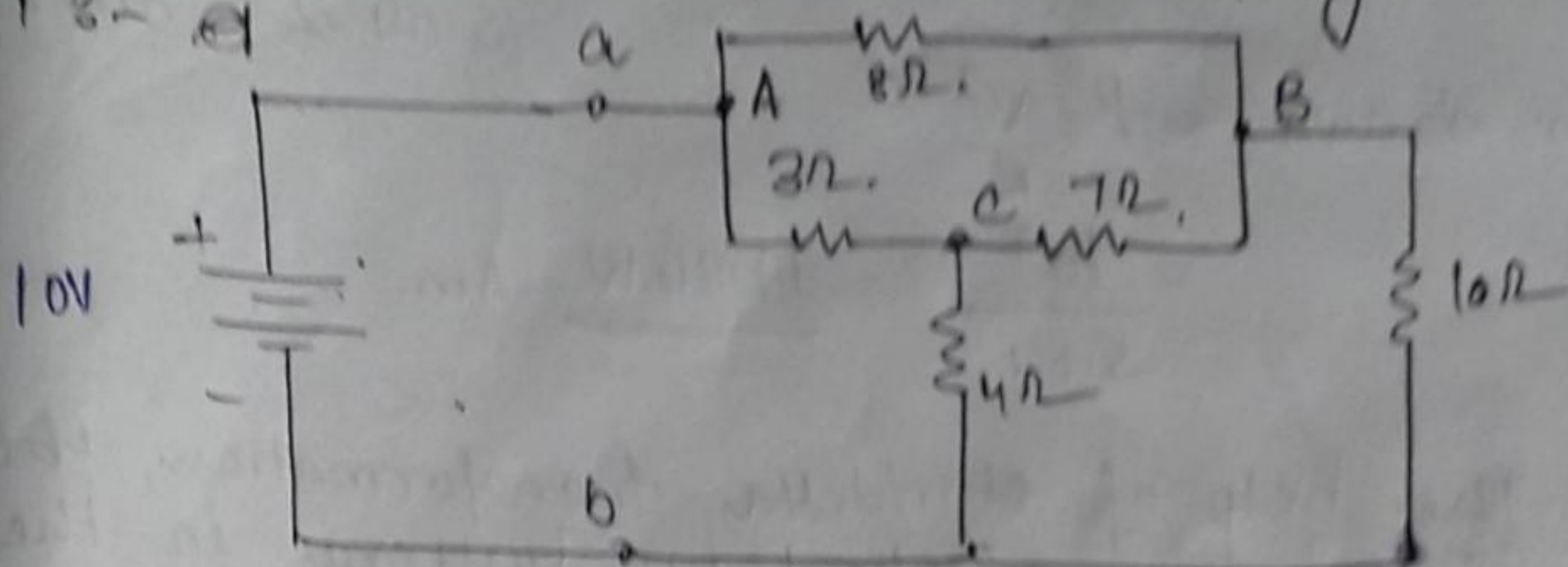
Current through 10Ω .

$$= 6.07 \times \frac{45}{69}$$

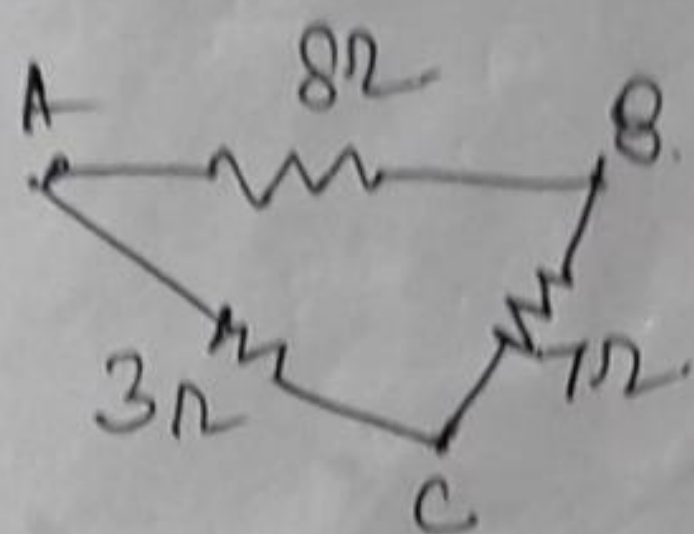
$$= 3.96 \text{ A}$$

Q1 :- Using delta to star transformation determine the resistance between terminals a and b and the total power drawn from the supply.

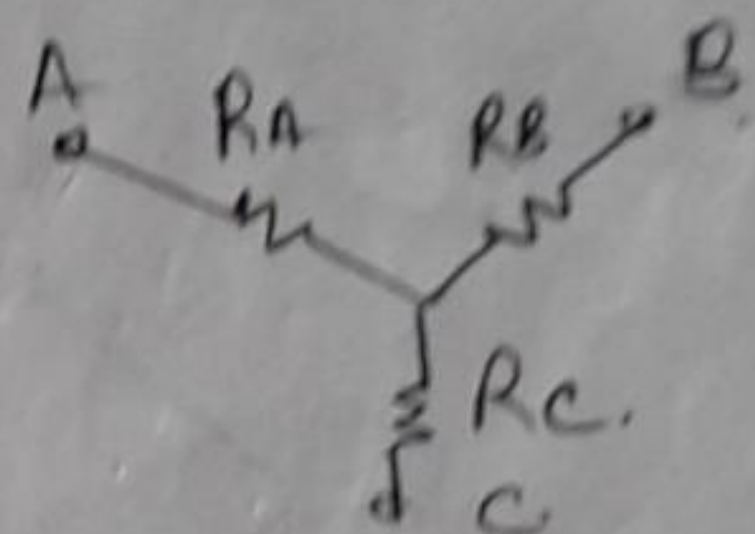
Ans :-



From the given circuit, identify delta network



⇒ apply transformation

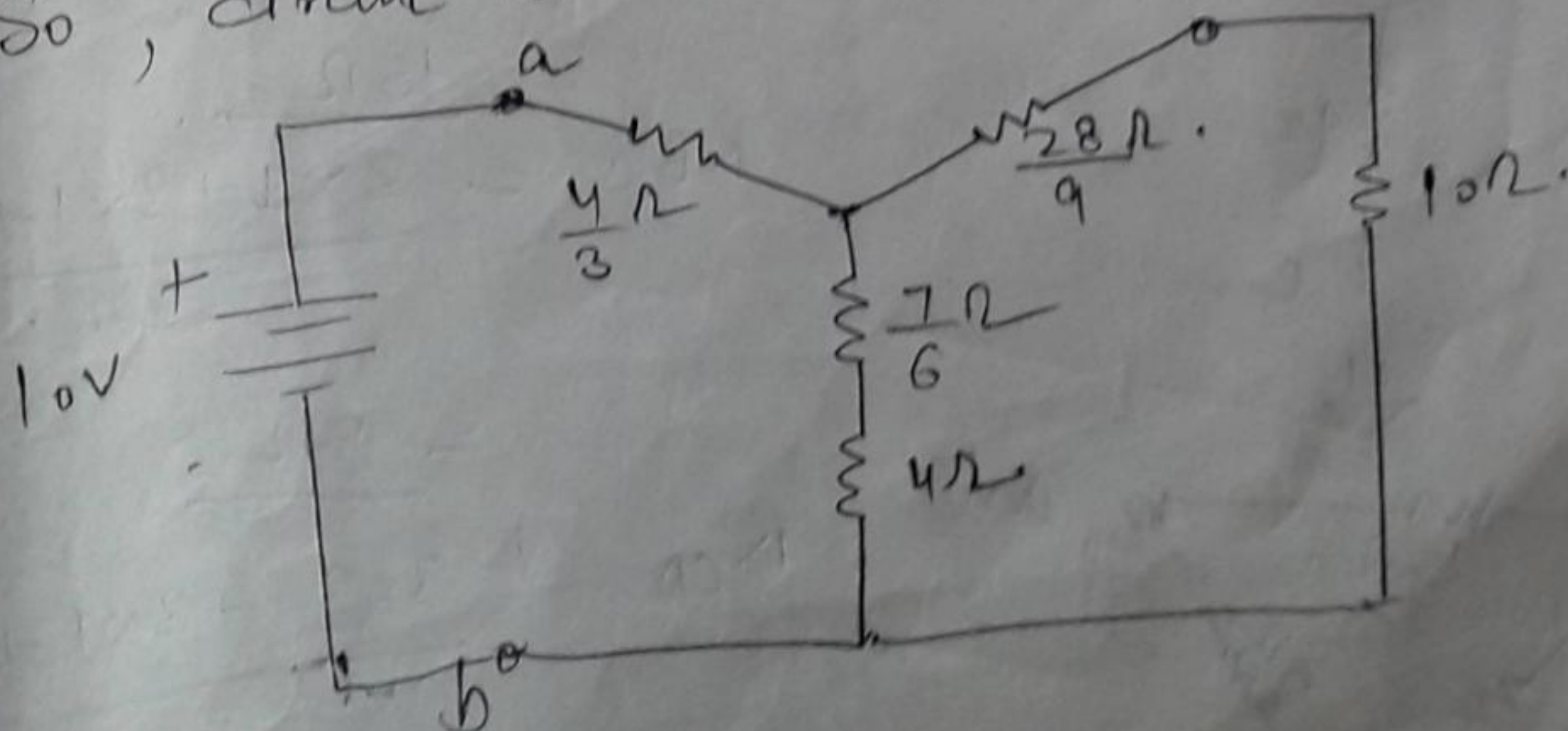


$$\text{where } R_A = \frac{3 \times 8}{3 + 8 + 7} = \frac{4}{3} \Omega$$

$$\text{Similarly } R_B = \frac{8 \times 7}{8 + 7 + 3} = \frac{28}{9} \Omega$$

$$\text{and } R_C = \frac{3 \times 7}{8 + 7 + 3} = \frac{7}{6} \Omega$$

So, circuit is reduced to -



39

$$R = \frac{4}{3} + \left(\frac{7}{6} + 4\right) \parallel \left(\frac{28}{9} + 10\right) = \frac{4}{3} + \frac{31}{7} \parallel \frac{118}{9}$$

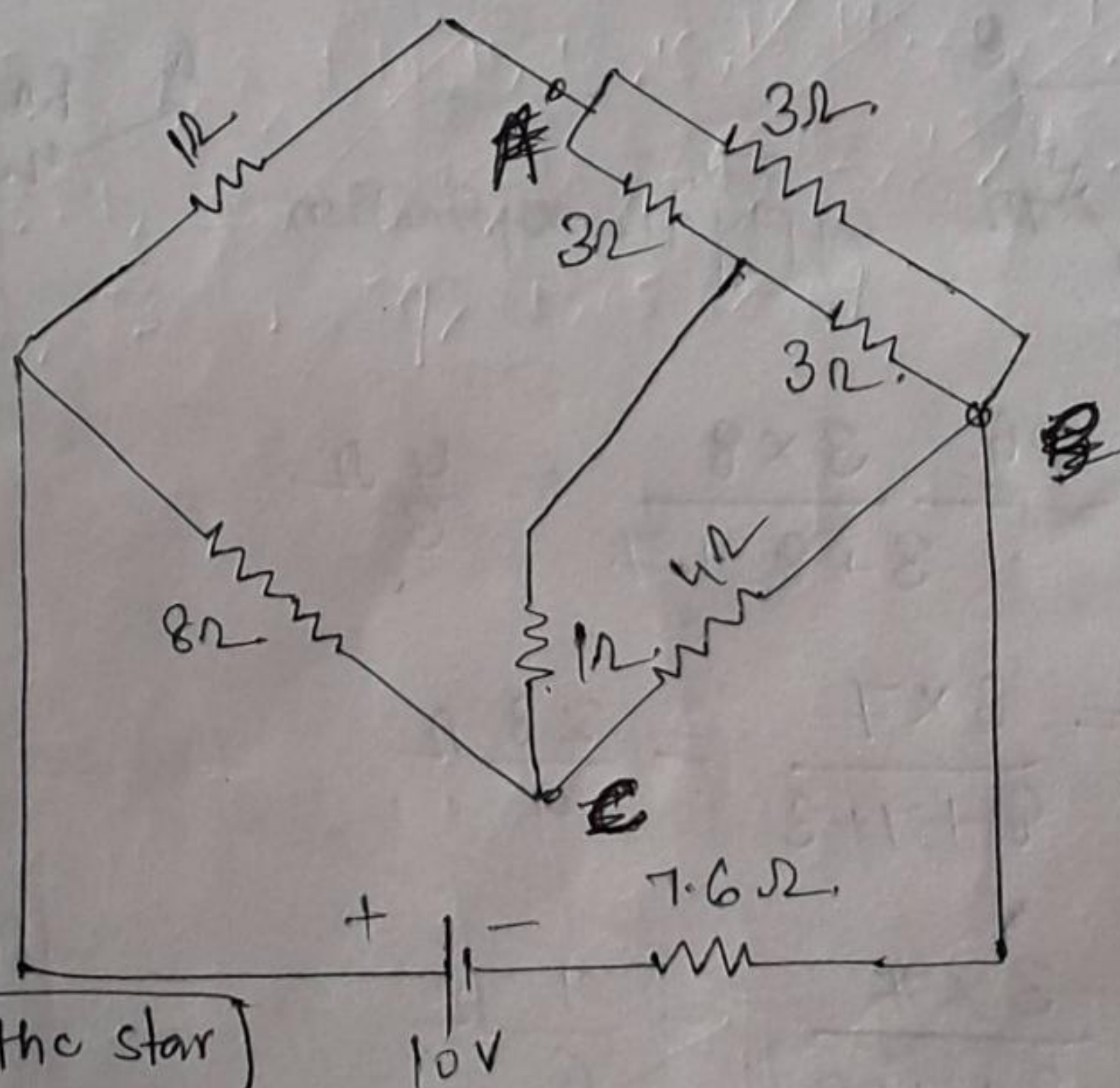
$$= 1.3333 + 3.7062$$

$$= 5.04 \Omega \text{ Ans.}$$

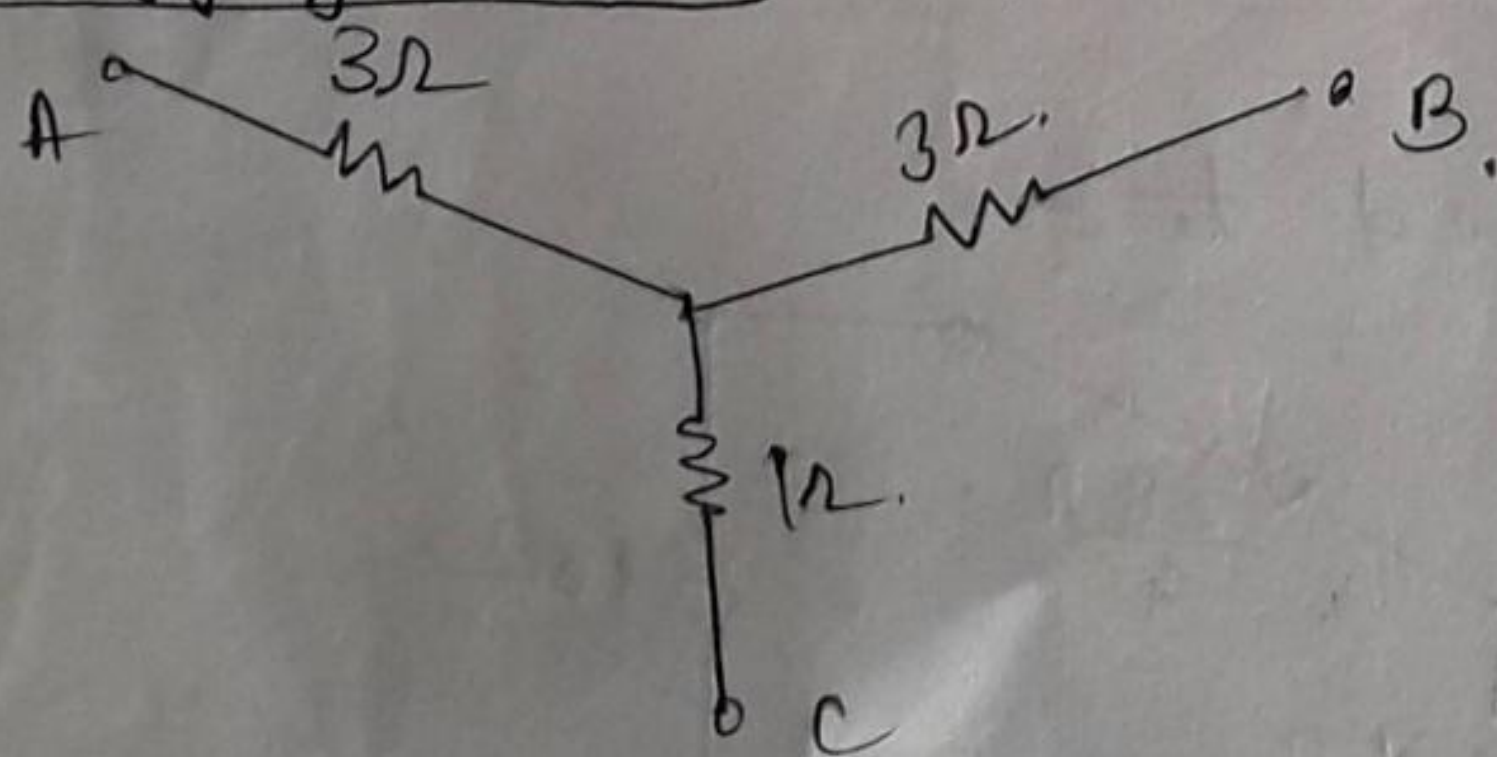
Total power drawn $= P = \frac{V^2}{R}$

$$= \frac{10^2}{5.04} = \underline{19.84 \text{ W}} \text{ Ans.}$$

Q2 H.W. with the help of star/delta transformation, obtain the values of current supplied by battery in the circuit shown.



Identifying the star

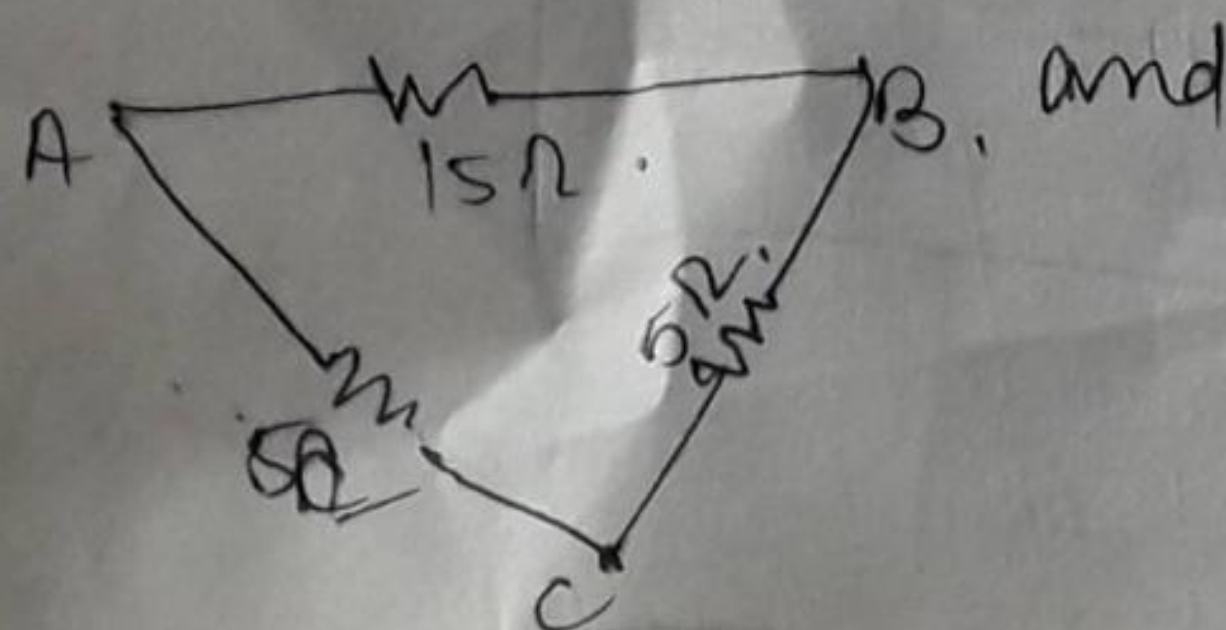


$$R_{AB} = \frac{3 \times 3 + 3 \times 1 + 3 \times 1}{1}$$

$$= \underline{15 \Omega}$$

$$R_{BC} = \frac{3 \times 3 + 3 \times 1 + 3 \times 1}{3 \Omega}$$

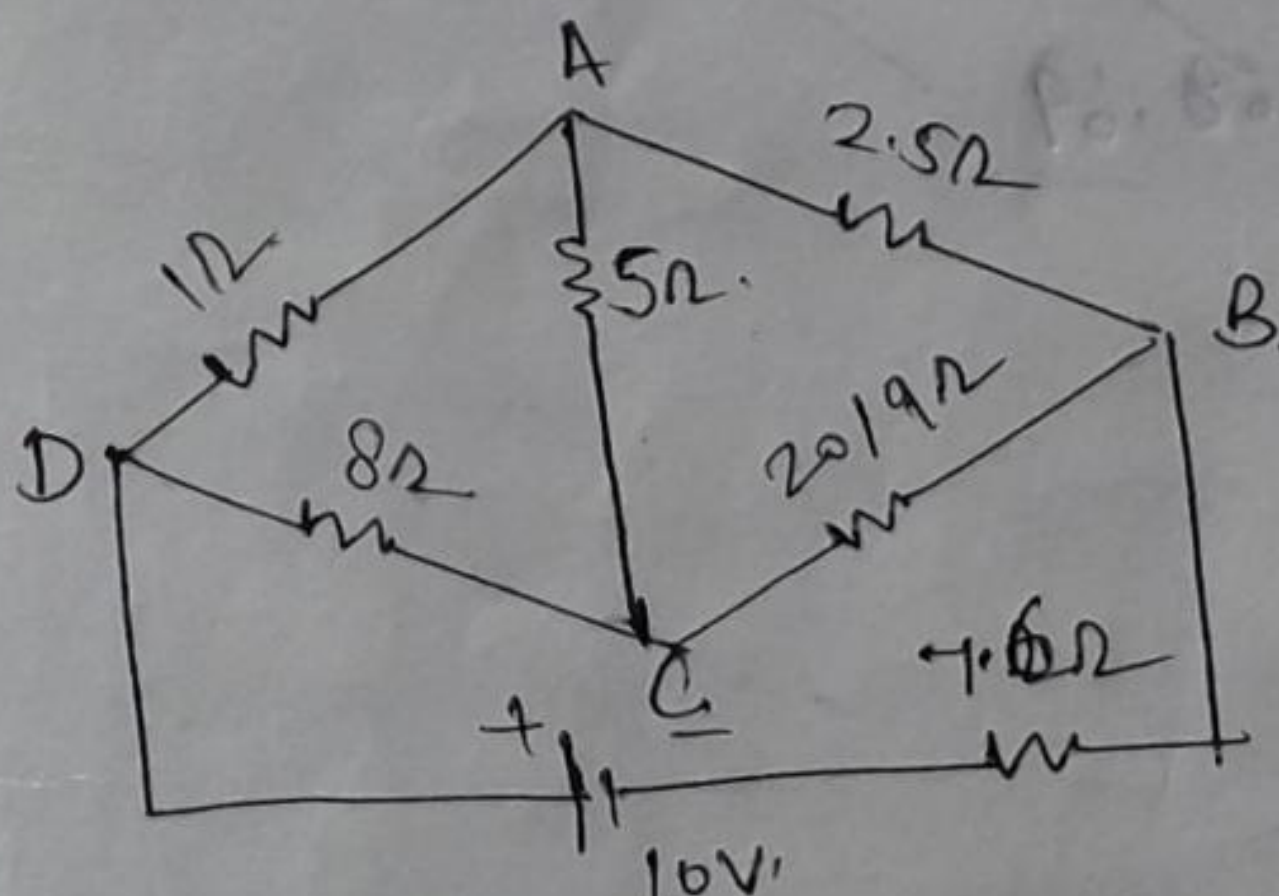
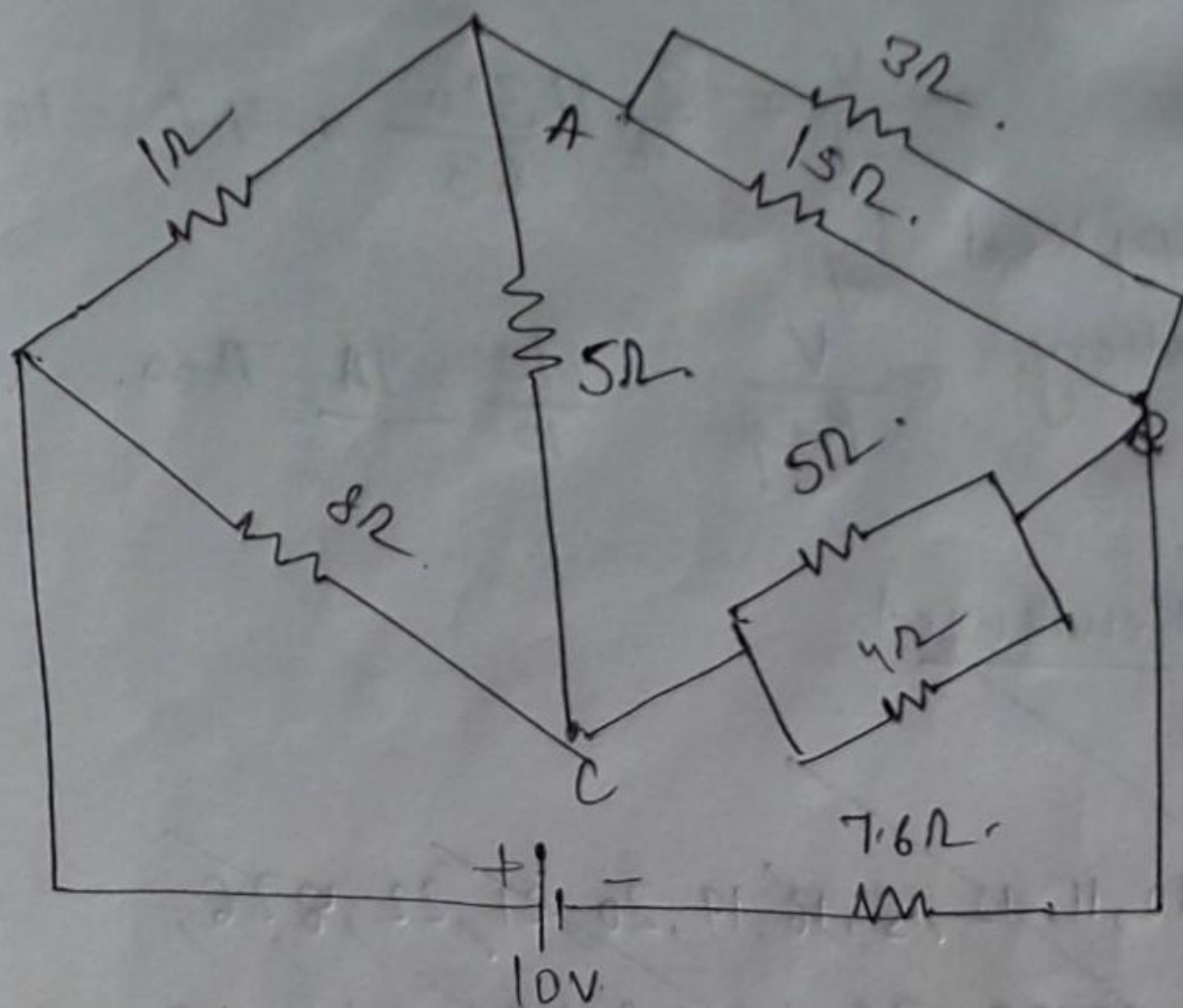
$$= \underline{5 \Omega}$$



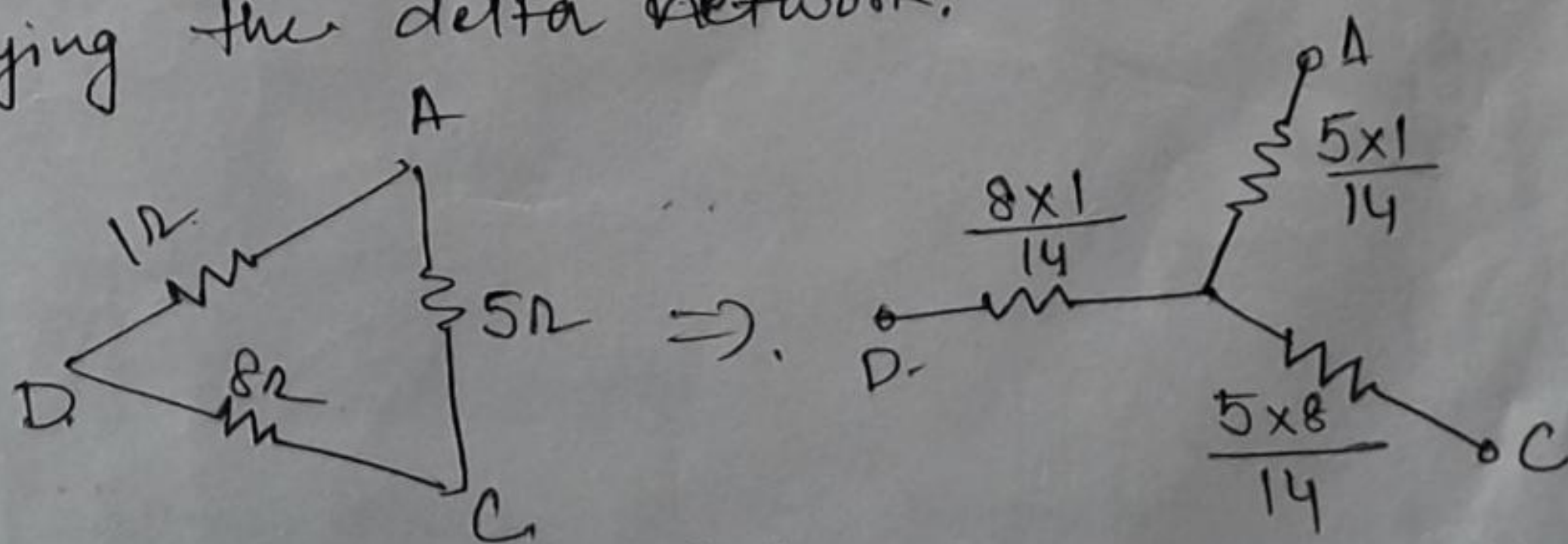
$$R_{CA} = \frac{3 \times 3 + 3 \times 1 + 3 \times 1}{3 \Omega}$$

$$= \underline{5 \Omega}$$

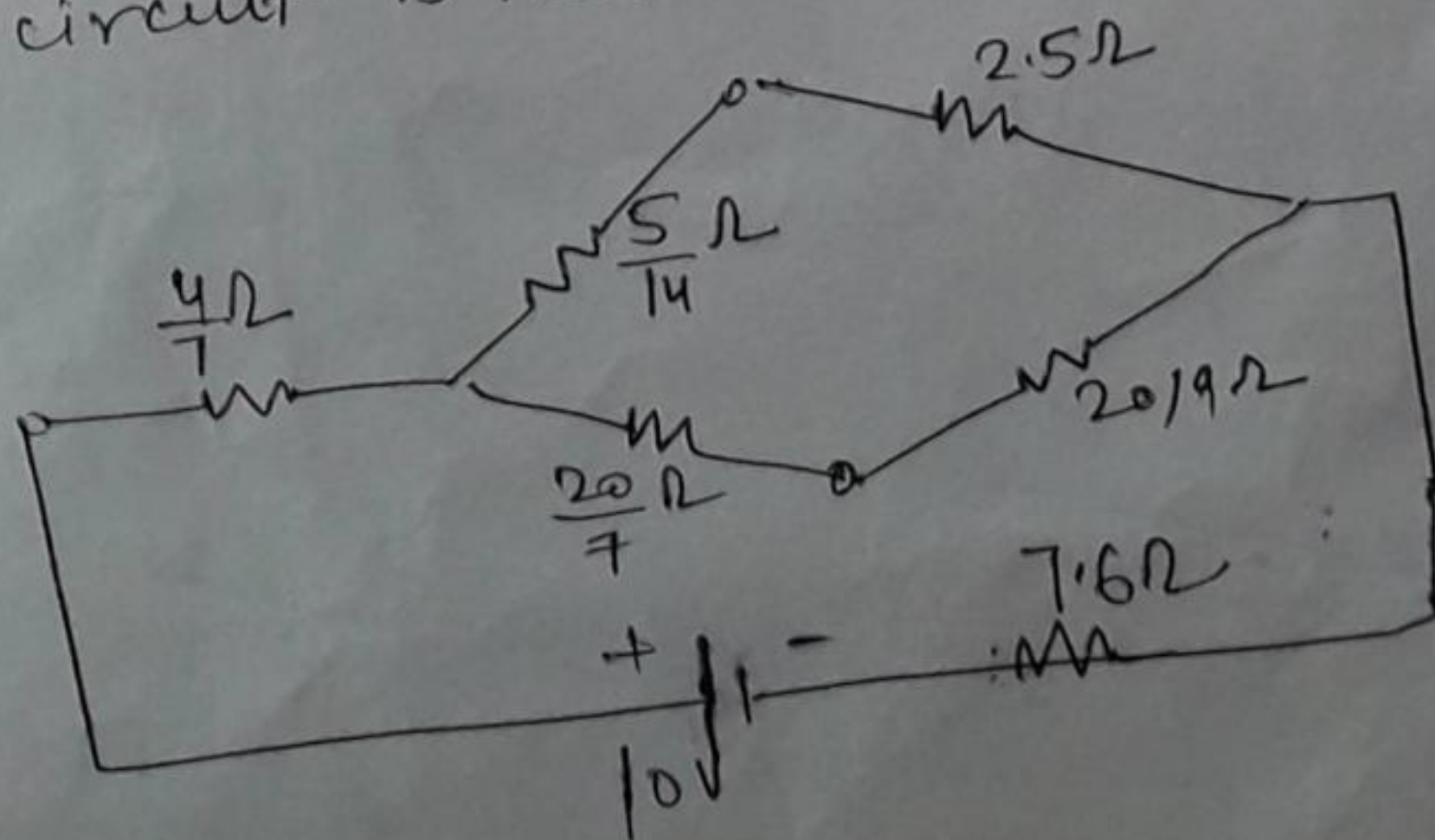
The circuit is reduced to -



Identifying the delta network.



The circuit is reduced to.



(41)

$$R_{eq} = \frac{4}{7} + \left[\frac{5}{14} + 2.5 \right] \parallel \left[\frac{20}{7} + \frac{20}{9} \right] + 7.6$$

$$= \frac{4}{7} + \frac{20}{7} \parallel \frac{320}{63} + 7.6 = 10\Omega$$

So, current supplied by

$$\text{battery} = \frac{V}{R_{eq}} = \frac{10}{10} = 1A \text{ Ans.}$$

~~Attendance~~ (Attendance)

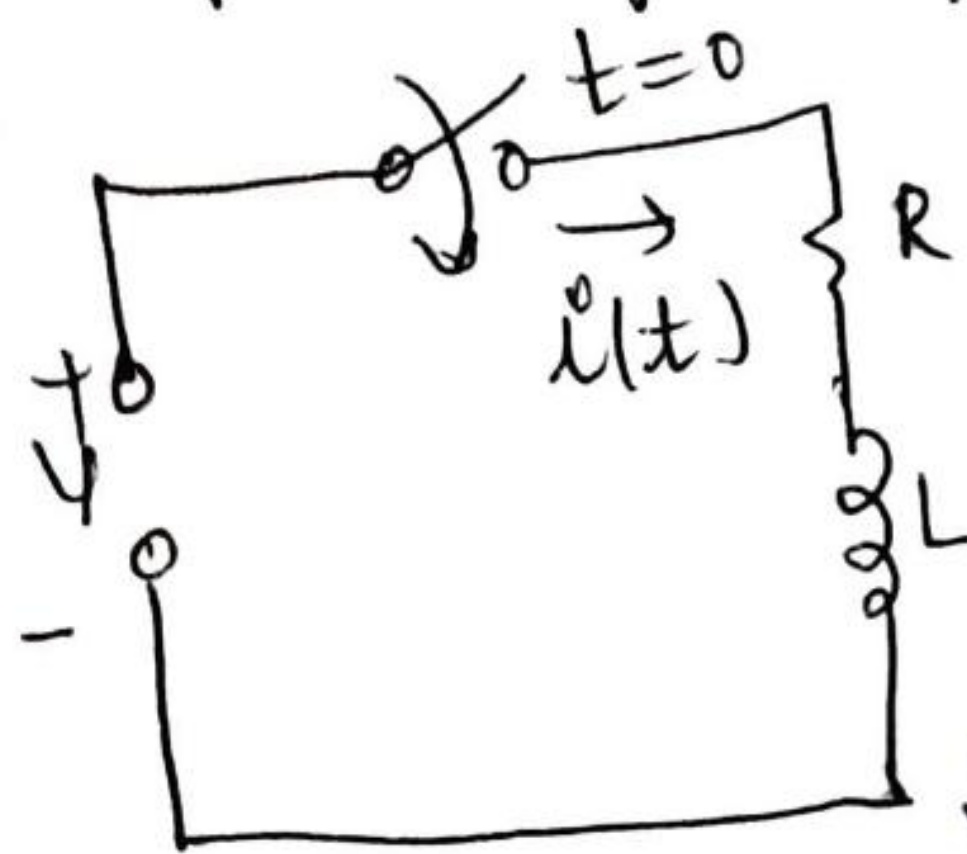
~~12, 52, 28, 68~~

~~3, 5, 6, 7, 8, 9, 10, 11, 12, 15, 16, 19, 20, 21, 22, 26,~~

~~28, 29, 32, 35, 36, 38, 39, 42, 43, 45, 46, 47, 48,~~

~~52, 53, 54, 59, 61, 65, 68, 69~~

1. Step Voltage Response of RL series circuit -



Consider an inductive circuit of Resistance (R) and Inductance L with switch S.

Switch was open since long duration of time, and is closed at $t=0$ just before the applying the step voltage the current through L

$$i_L(0^-) = 0 \quad (1)$$

After closing the switch

$$Ri + L \frac{di}{dt} = V \quad \text{for } t > 0 \quad (2)$$

eqn (2) represents non homogeneous linear differential equation with the excitation term appearing in RHS.

Complete solution of eqn (2) consists of 2 parts

1. Particular Integral (PI) forced Response

2. Complementary functions.

$$\text{i.e. } i_f = \frac{V}{R} \quad (3)$$

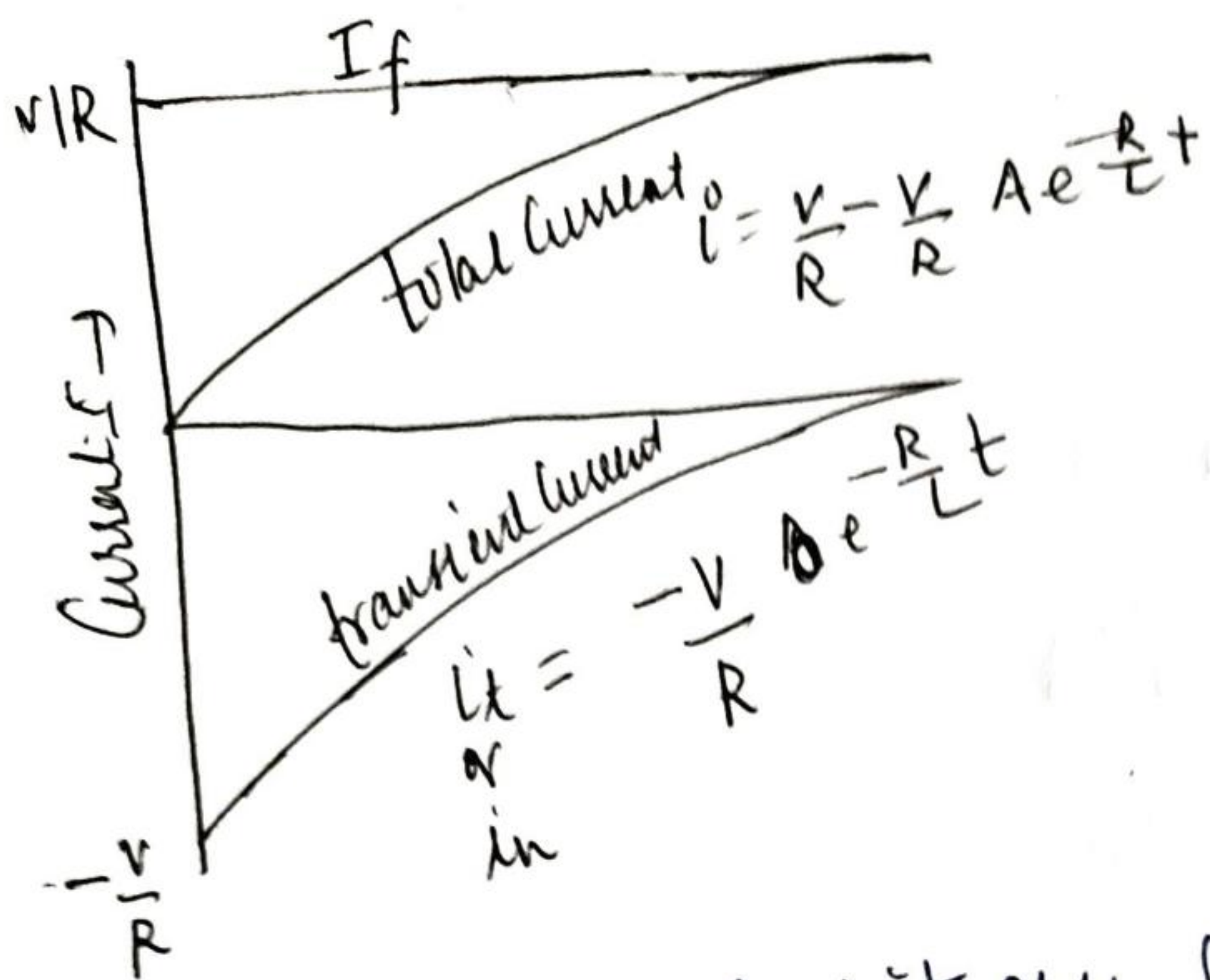
Complementary function represents the initial conditions.

$$\text{Soln for } Ri + L \frac{di}{dt} = 0$$

$$A e^{-\frac{R}{L}t} \quad (4)$$

Complete solution of eq (2) is from eq (3) & (4)

$$i(t) = \frac{V}{R} + Ae^{-\frac{R}{L}t} \quad (5)$$



in eq (5) A = Arbitrary constant
 By using initial condition
 we $i(0^+) = i(0^-) = 0 \quad (6)$

Substitute (6) in (5)

$$0 = \frac{V}{R} + A$$

$$\therefore A = -\frac{V}{R}$$

$$i(t) = \frac{V}{R} - \frac{V}{R} e^{-\frac{R}{L}t}$$

$$= \frac{V}{R} \left[1 - e^{-\frac{R}{L}t} \right] \quad (7)$$

In eq (7) $\frac{V}{R}$ represent steady state condition i.e. forced response i_f and 2nd term $-\frac{V}{R} e^{-\frac{R}{L}t}$ repⁿ transient (Natural, in) response which $\rightarrow 0$ as $t \rightarrow \infty$

2 The way in which transient response dies out depends upon value of R & L .

$$\frac{L}{R} = \text{time constant} = \tau$$

$\frac{R}{L} = \text{damping coefficient of the circuit}$

$$\therefore V_R(t) = i(t) \times R$$

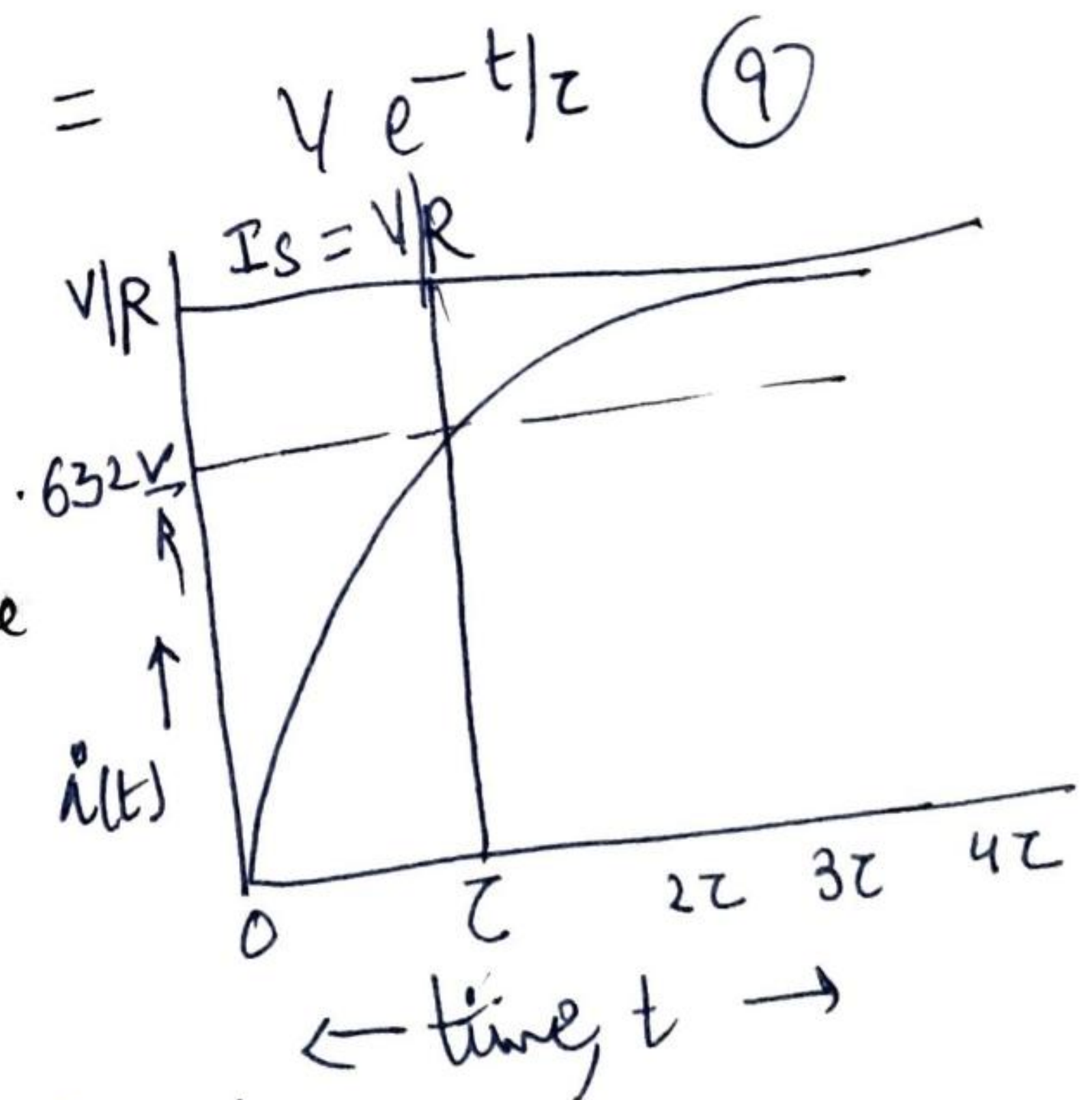
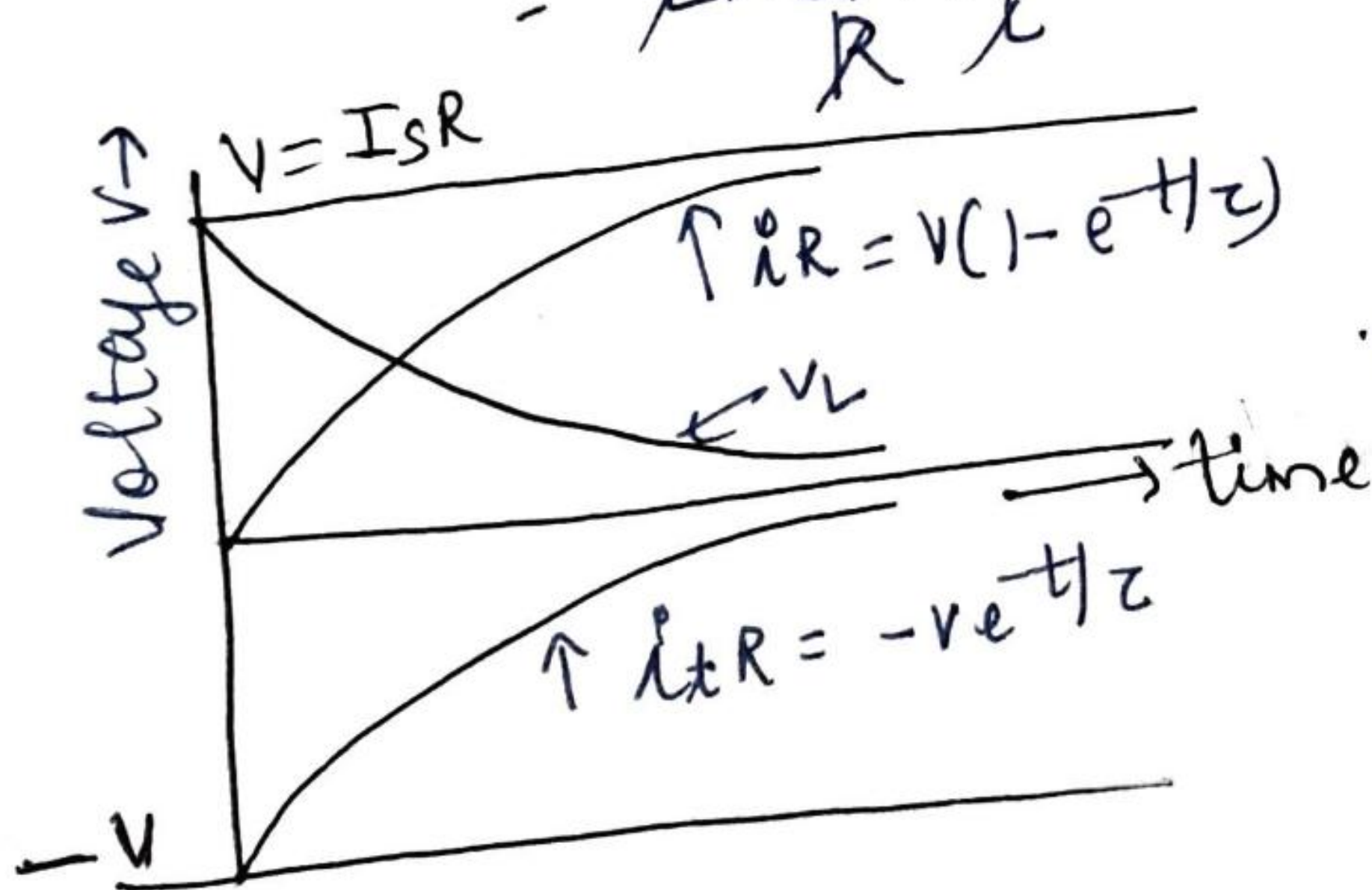
$$= \frac{V}{R} \left[1 - e^{-\frac{t}{\tau}} \right] \times R$$

$$= V \left[1 - e^{-\frac{t}{\tau}} \right] \quad (8)$$

$$V_L(t) = L \frac{di(t)}{dt}$$

$$= L \times \frac{V}{R} \left[0 - \left| -\frac{1}{\tau} \times e^{-\frac{t}{\tau}} \right| \right]$$

$$= \cancel{L} \times \frac{V}{\cancel{R}} \times \frac{R}{\cancel{L}} e^{-t/\tau} = V e^{-t/\tau} \quad (9)$$

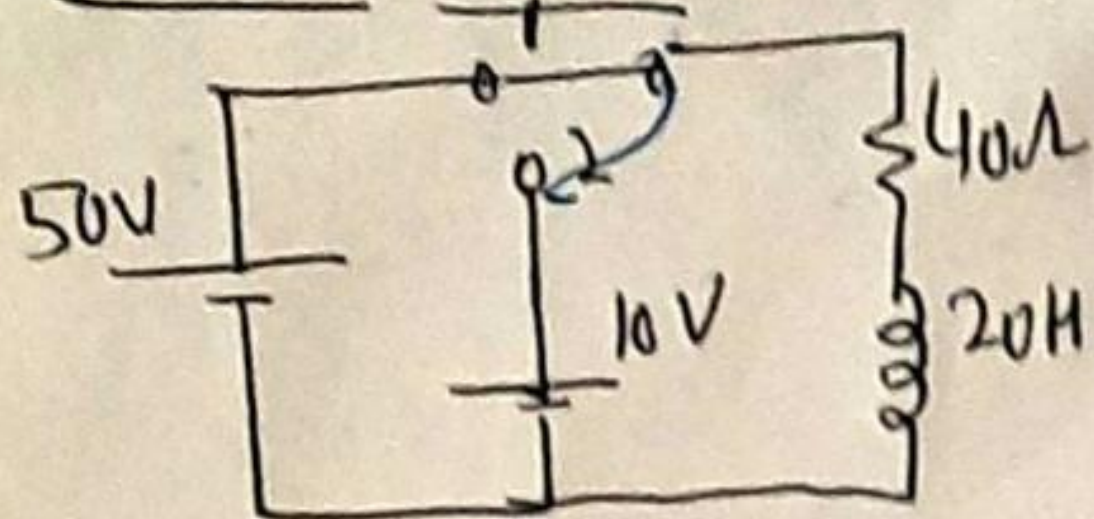


Immediately after switch is closed,
Rate of Rise of current i.e. initial rate of Rise

Can be determined by w.r.t $t=0$

$$\frac{di(t)}{dt} = \frac{V}{R \times \frac{L}{R}} e^{-\frac{R}{L}t} = \frac{V}{L} e^0 = \frac{V}{L}$$

Numerical -



Switch is in 1st posn

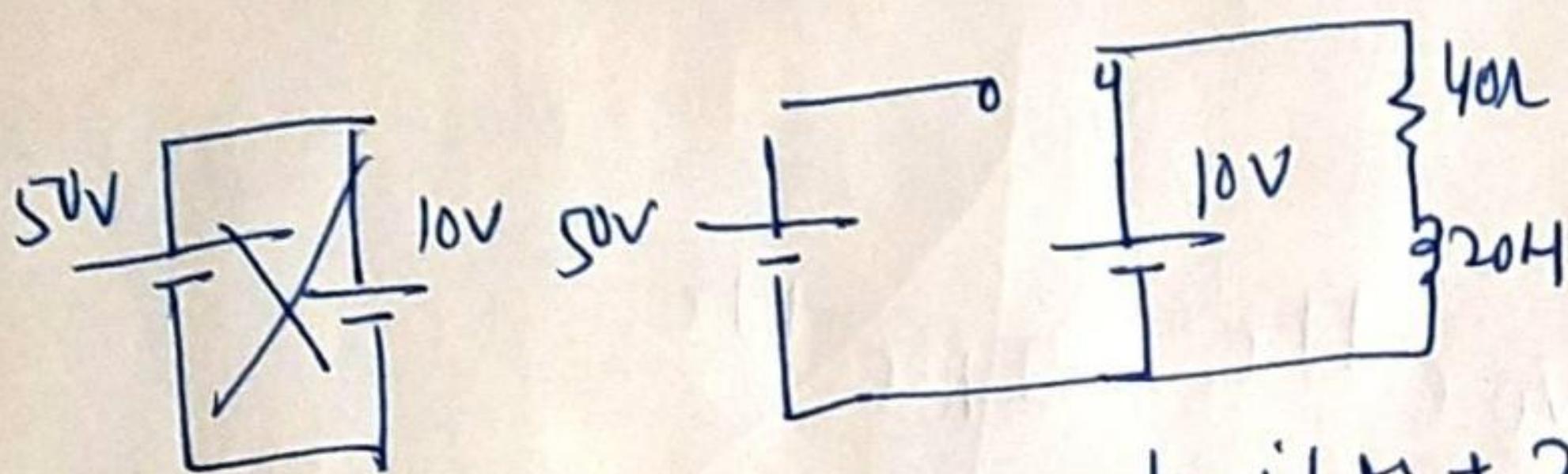
moved to 2 at $t=0$, obtain the Expⁿ for $t > 0$

in 1st posn

$\frac{50}{40} = I$ at steady state as [L as short ckt]

$$I = 1.25 \text{ A}$$

So when ckt comes to 2nd posn - $I = 1.25 \text{ A}$



$$10 = 40 i(t) + 20 \frac{d i(t)}{dt}$$

$$\frac{d i(t)}{dt} + 2 i(t) = 0.5$$

$$i(t) = \frac{0.5}{2} + A e^{-2t}$$

$$\text{at } i(0^+) = 1.25 \text{ A}$$

$$1.25 = 0.25 + A$$

$$A = 1 \text{ Amp}$$

$$\therefore i(t) = (0.25 + 1 \times e^{-2t}) \text{ Amp}$$

Step Voltage Response of RC series ckt -

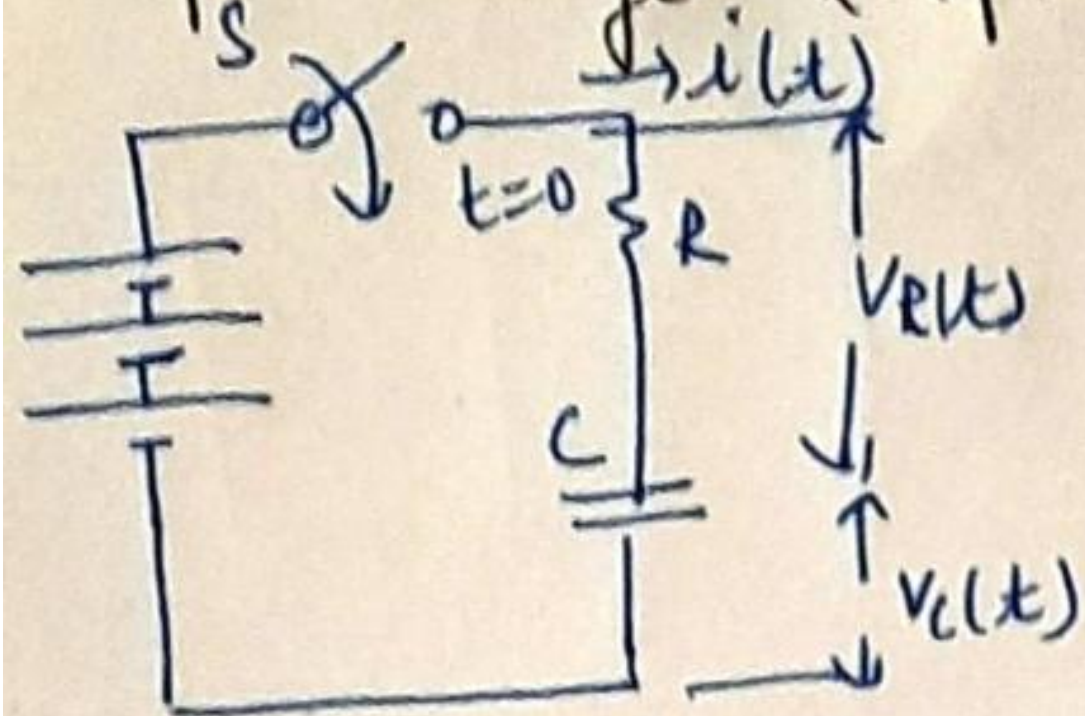


fig (1) repⁿ Series Combn of R-C.

at any instant PD across the capacitor be $V_C(t)$ and charging current be $i(t)$ amp

voltage drop across Resistor $V_R(t) = Ri(t)$ (1)

voltage drop across Capacitor $V_C(t) = \frac{q(t)}{C}$
 $= \frac{1}{C} \int_{-\infty}^t i(t) dt$

As applied voltage

$$V = V_R(t) + V_C(t)$$

$$V = Ri(t) + \frac{1}{C} \int_{-\infty}^t i(t) dt$$

$$= Ri(t) + \frac{1}{C} \int_0^t i(t) dt + V_C(0^+) \quad (1)$$

Assuming initial condⁿ to be zero
 i.e. Capacitor is initially Discharged

$$V = Ri(t) + \frac{1}{C} \int_0^t i(t) dt \quad (2)$$

Differentiate (2) w.r.t t

$$R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

$$\frac{di(t)}{dt} + \frac{1}{RC} i(t) = 0$$

③ General Eqⁿ of above differential Eqⁿ

$$i(t) = A e^{-\frac{1}{RC}t} \quad (1)$$

at $t = 0^+$, $i(0^+) = \frac{V}{R}$ as capⁿ act as short ckt

$$\therefore \frac{V}{R} = A$$

$$\therefore i(t) = \frac{V}{R} e^{-\frac{1}{RC}t} \quad (2)$$

$$\begin{aligned} \text{Voltage across Resistor } V_R(t) &= R i(t) \\ &= \frac{V}{R} e^{-\frac{1}{RC}t} \times R \\ &= V e^{-t/\tau} \end{aligned}$$

$$\text{where } \tau = \frac{RC}{1}$$

voltage across Capacitor $V_C(t) =$

$$\frac{1}{C} \int_0^t i(t) dt$$

$$= \frac{1}{C} \int_0^t \frac{V}{R} e^{-\frac{1}{RC}t} dt$$

$$= \frac{V}{RC} \left[\frac{e^{-\frac{1}{RC}t} \times -RC}{-1} \right]_0^t$$

$$= \frac{V \times RC}{RC} \left[-e^{-\frac{t}{RC}} + 1 \right]$$

$$= V \left[1 - e^{-t/RC} \right]$$

$$= V \left[1 - e^{-t/\tau} \right]$$

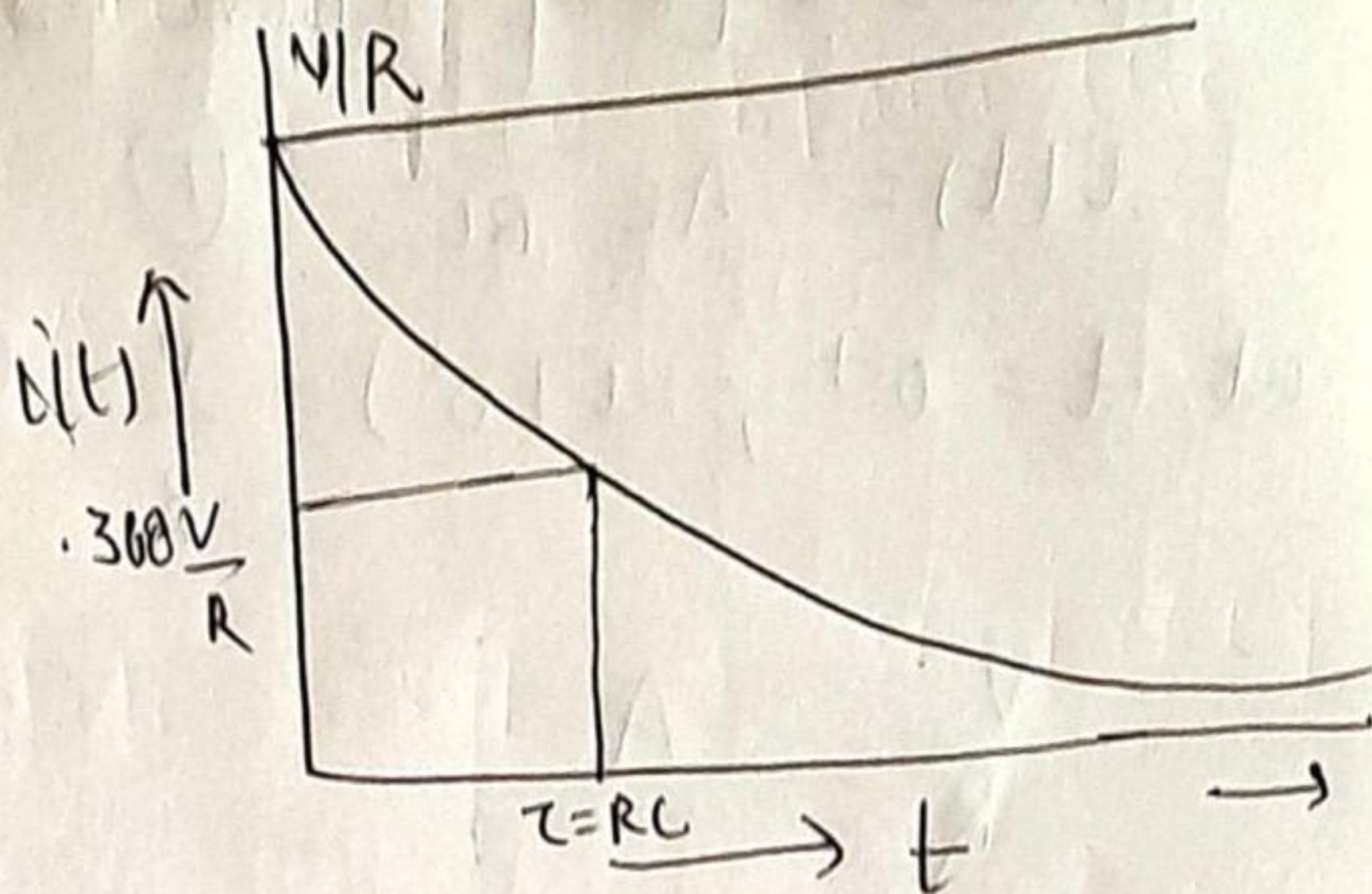
at $t=0^+$ $i(t) = \frac{V}{R}$

$V_R(t) = V$

$V_C(t) = 0$

at $t = \infty$ $V_R(t) = 0$

$V_C(t) = V$



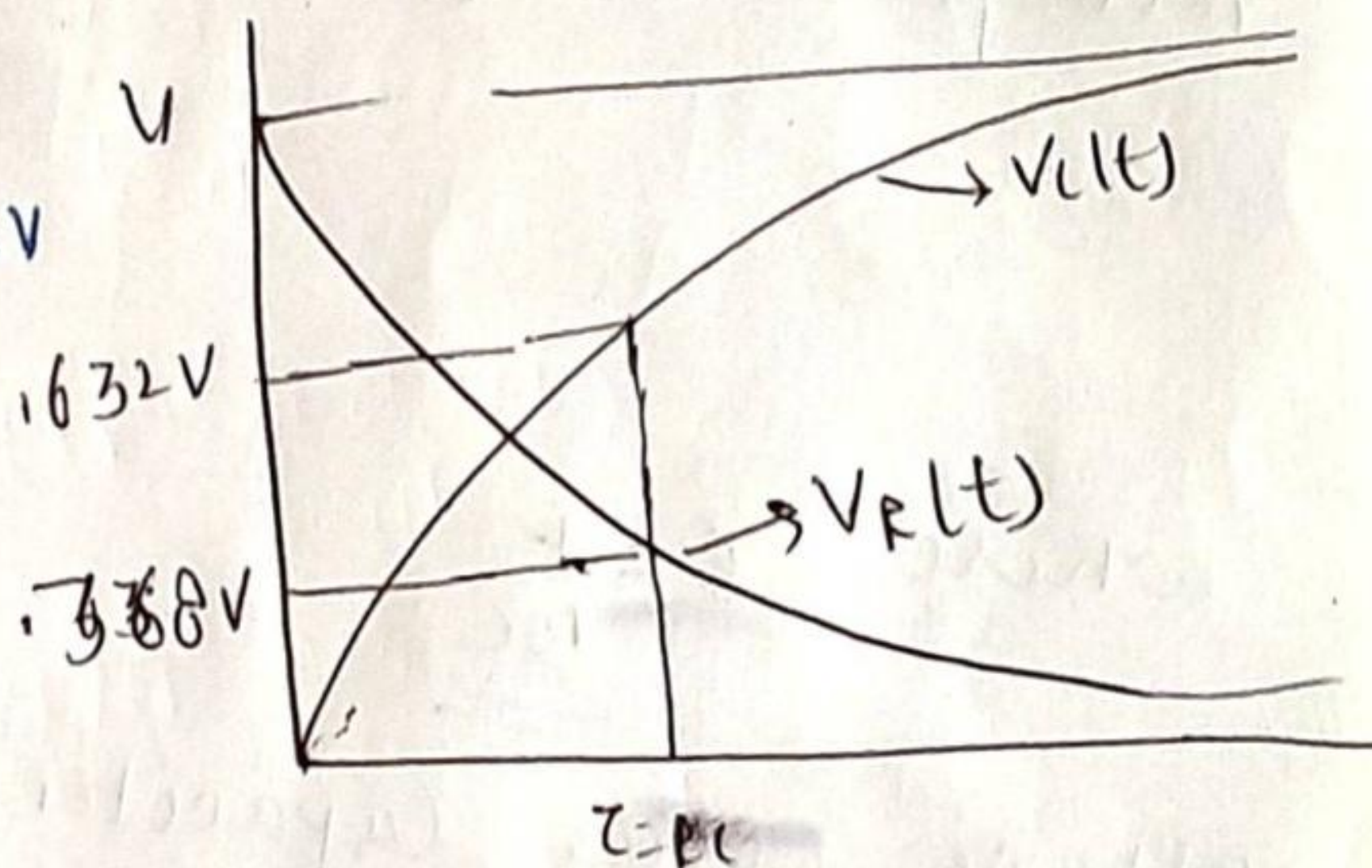
at $t = RC$

$i(t) = \frac{V}{R} e^{-1} = 0.368 \frac{V}{R}$

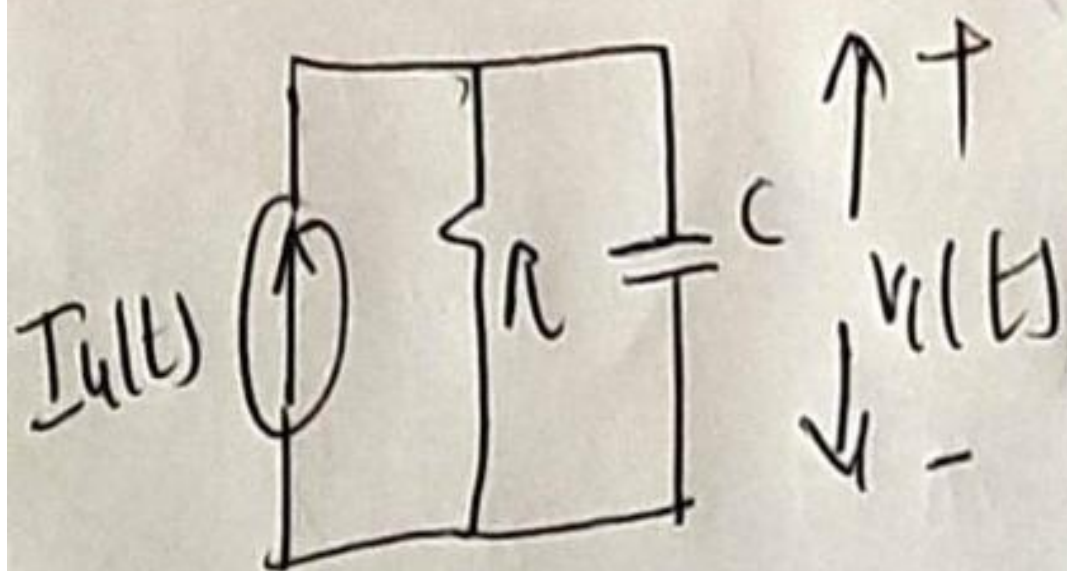
$V_R(t) = V e^{-1} = 0.368 V$

$V_C(t) = V [1 - e^{-1}]$

$V [1 - 0.368] = 0.632 V$



Step Current Response of RC Parallel circuit—



time constant of circuit with
current source O.C
 $\tau = RC$ (1)

Transient Response of Voltage across Capacitor

$V_{cr} = A e^{-t/\tau}$ (2)

In steady state, capacitance act as open circuited
So all current pass through R

$V_{cf} = V_R = R I$ (3)

$$V_C = IR + Ae^{-t/\tau} \text{ for } t > 0$$

~~the~~ initially $V_C = 0$ i.e. $t = 0$

$$0 = IR + A$$

$$A = -IR$$

$$\therefore V_C = IR(1 - e^{-t/\tau}) \quad (3)$$

if capⁿ initially have some voltages.
consider $V_C = V_0$

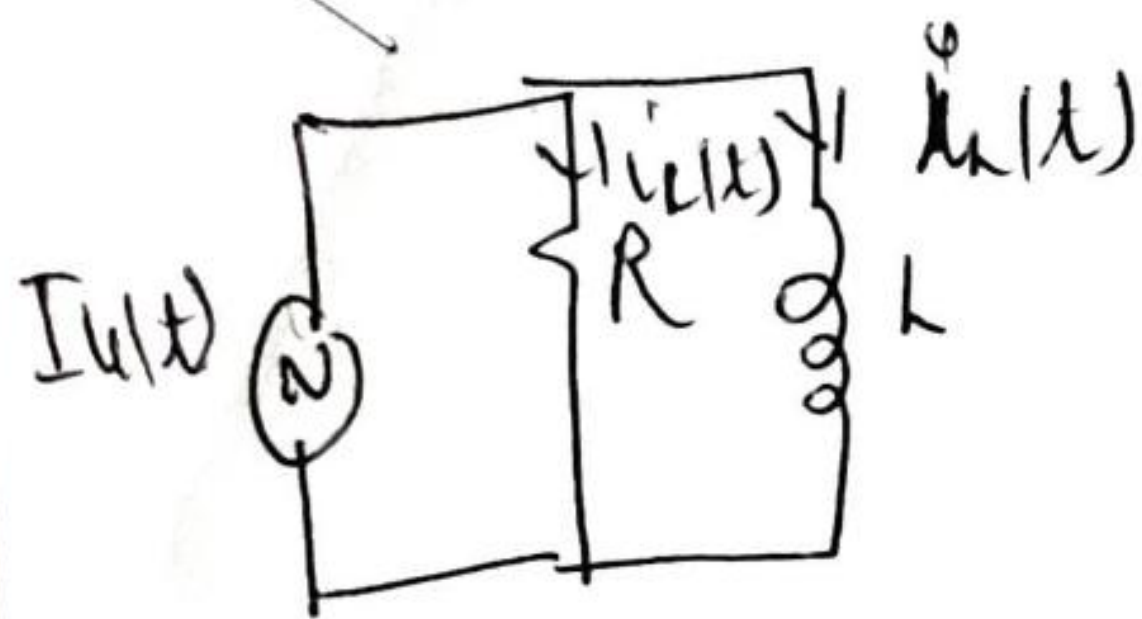
$$\therefore V_0 = IR + A$$

$$A = V_0 - IR$$

$$\therefore V_C = IR + V_0 e^{-t/\tau} - IR e^{-t/\tau}$$

$$\therefore V_0 e^{-t/\tau} + IR(1 - e^{-t/\tau}). \quad (4)$$

② Step Current Response of R-L parallel ckt-



Consider a parallel combination of R & L.

$I_s(t)$ is the current source.

time constant $\tau = \frac{L}{R}$ when current source open circuited.

Natural or transient compⁿ of current through inductance

$$= i_{LN} = A e^{-t/\tau}$$

at $t \rightarrow \infty$, inductance behaves as a short circuited

So steady state component

$$i_{sf} = I$$

$$\begin{aligned} \text{hence } i_L(t) &= i_{LN} + i_{sf} \\ &= \underline{A} + A e^{-t/\tau} \quad (1) \end{aligned}$$

$$\text{at } t = 0^+ \quad i_L(t) = 0$$

$$\therefore 0 = I + A$$

$$\therefore A = -I$$

By putting the value in (1)

$$\begin{aligned} i_L(t) &= I - I e^{-t/\tau} \\ &= I(1 - e^{-t/\tau}) \quad \text{for } t \geq 0 \end{aligned}$$